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ISAAC DOUGHAN

REFRACTIVE INDEX SENSORS USING HYBRID WAVEGUIDES:

DESIGN, FABRICATION, AND CHARACTERIZATION

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DESIGN, FABRICATION, AND CHARACTERIZATION

ACADEMIC DISSERTATION

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Isaac Doughan
Refractive index sensors using hybrid waveguides: Design, fabrication, and characterization
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ABSTRACT

In this thesis, we investigate the design, fabrication, and characterization of hybrid waveguide structures, with a focus on polymer-titania (TiO_2) strip composites aimed at enhancing photonic sensing applications. Initially, silicon nitride strip waveguides were used to analyze mode behavior, for which a custom algorithm was developed to accurately retrieve mode strengths within the waveguide. Experimental results validated the algorithm's effectiveness, showing a strong correlation between retrieved and measured fields. These insights were foundational for the development of hybrid waveguide structures, as a clear understanding of mode dynamics is necessary for successful waveguide design.

The second part of the work focuses on the development of polymer-titania hybrid waveguides that demonstrate low propagation losses of 1.75 ± 0.32 dB/cm while maintaining the advantages of polymers, such as affordability and fabrication ease. Techniques like atomic layer deposition (ALD) and electron beam lithography (EBL) were employed to precisely control waveguide dimensions, resulting in hybrid structures that enhance light confinement. By integrating TiO_2 with polymers, the study addresses light confinement limitations typically associated with polymer waveguides, making them better suited for sensing applications.

Finally, the thesis explores the application of these hybrid waveguides in a strip-loaded Mach-Zehnder interferometer (MZI) designed for ethanol-water refractive index sensing. To overcome challenges in using strip-loaded waveguides for evanescent field sensing, a large opening was introduced in the MZI's sensing arm, allowing the evanescent field to interact with the analyte. The MZI sensor was characterized by measuring output power as a function of the analyte's refractive index and determining the corresponding effective index change. The device demonstrated a homogeneous sensitivity of 0.128 and a detection range of 1.1×10^{-3} RIU. These results showcase the practical effectiveness of the hybrid waveguides in achieving precise and sensitive measurements for photonic sensing applications.

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Keywords: Sensors, waveguide, Mach–Zehnder interferometer, refractive index sensing, Strip loaded waveguide, hybrid polymer waveguide, nanostructures

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"When the time is right, I, the Lord, will make it happen." – Isaiah 60:22

I want to begin by thanking God for how far He has brought me. This journey, which once seemed impossible to complete, has been made possible by His grace. Truly, My God is not dead.

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Joensuu, November 30, 2024

Isaac Doughan

LIST OF PUBLICATIONS

This thesis consists of the present review of the author's work in the field of optical phenomena and the following selection of the author's publications:

I **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, 'Low Loss Vertical TiO_2/Polymer Hybrid Nano-Waveguides," *Nanomaterials*, **13**, no. 469, 2023.

II **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing," *Scientific Reports*, **14**, no. 3064, 2024.

III **I. Doughan**, A. Halder, I. Reduto, M. Koivurova, T. Aalto, M. Roussey, and J. Turunen, "Determination of mode strengths in channel waveguide from the complex electric field," *Scientific Reports*, **14**, no. 28665, 2024.

Throughout the overview, these papers will be referred to by Roman numerals.

AUTHOR'S CONTRIBUTION

The publications selected in this dissertation are original research papers. In Paper **I**, the author contributed to the design, participated in the characterization, contributed to the data analysis, and contributed to the writing of the manuscript. In Paper **II**, the author was responsible for the design, contributed to the fabrication and characterization, contributed to the data analysis, and co-wrote the manuscript. In Paper **III**, the author contributed to performing the numerical calculations and simulations, built the experimental setup, contributed to running the experiments and the data analysis. The author was also involved in manuscript preparation and contributed to scientific discussions across all three papers.

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1 Introduction

The rapid advancement in photonics and integrated optics has revolutionized various technological fields, from telecommunications to medical diagnostics [1–4]. At the heart of these advancements lie waveguides, one of the fundamental components required for guiding light in a controlled manner [5]. Waveguides serve as the backbone of integrated optical circuits, which are designed to perform complex tasks by manipulating light rather than electrical signals, thereby overcoming some of the inherent limitations of traditional electronics, such as bandwidth constraints and speed limitations [6,7].

Waveguides are essential for miniaturizing optical devices, which aim to create compact, efficient, and highly functional photonic circuits. Several materials can be used in fabricating waveguides, yet only a few exhibit the necessary characteristics for integrated optics input/output (I/O) device development. Among these materials are lithium niobate ($LiNbO_3$) [8–10], valued for its excellent piezoelectric, nonlinear optical, and electro-optic properties. Glass-based waveguides have been widely used, especially in optical fiber communications, where silica glass plays a key role mainly due to its low optical losses. Moreover, silica glass is also the standard material for optical fibers, offering excellent transmission properties across various wavelengths [11]. Additionally, III–V semiconductor compounds [12, 13] rank high in impact on integrated optics technology, alongside silicon-based materials [14–17] which stands out for their moderately low propagation losses, affordability, and compatibility with conventional complementary metal-oxide semiconductor (CMOS) technology. Each material type boasts distinct advantages and disadvantages, with the choice depending on the specific application of the integrated device. However, these materials often involve complex and costly fabrication processes. Polymers [18–20] have emerged as a promising alternative in recent years due to their ease of fabrication, low cost, and compatibility with various substrates. Despite their advantages, polymers have lower refractive indices compared to conventional materials, which limits their ability to confine light tightly.

To address this limitation, researchers have explored hybrid waveguide structures that combine polymers with high-refractive-index materials such as titanium dioxide (TiO_2) [21]. These hybrid structures leverage the beneficial properties of both materials, enhancing light confinement and reducing propagation losses while maintaining the fabrication simplicity and cost-effectiveness of polymers. Advanced fabrication techniques, such as atomic layer deposition (ALD) and nanoimprint lithography (NIL) have been pivotal in realizing these sophisticated waveguide structures. These methods enable precise control over the waveguide dimensions and material composition, crucial for optimizing optical performances.

This thesis deals with the design, fabrication, and characterization of hybrid waveguide structures, aiming to translate theoretical designs into practical, high-performance devices mainly targeting sensing applications. The research emphasizes the development and optimization of waveguides that integrate advanced materials and fabrication techniques to enhance performance and functionality. Chapter 2 provides a theoretical framework for photonic waveguides, covering optical

slab waveguides, channel waveguides, and polymer-titanium hybrid waveguides. Chapter 3 details the fabrication techniques such as ALD of TiO_2 and spin coating of polymers, including thin-film characterization and electron beam lithography. Understanding the complexity of the modes in a waveguide is crucial for sensing applications aiming at very low limit of detection (LOD). Therefore, Chapter 4 presents the design and characterization of waveguides, focusing on mode retrieval, mode visualization, and propagation losses, with a comprehensive study of polymer-titania hybrid waveguides. The final chapter introduces the design and application of a strip-loaded Mach-Zehnder interferometer (MZI) for ethanol concentration measurement, demonstrating the practical utility of the developed waveguides.

In summary, this thesis aims to bridge the gap between theoretical designs and practical applications of complex-cross-section waveguide structures. By developing and optimizing hybrid waveguide structures that combine polymers with high-refractive-index materials (TiO_2), to enhance light confinement and reduces propagation losses. Practical applications are demonstrated by designing and applying a strip-loaded Mach-Zehnder interferometer (MZI) for ethanol concentration measurement, showcasing the utility of the developed waveguides. Advanced fabrication techniques, namely, ALD and electron beam lithography (EBL), are employed to achieve precise control over waveguide dimensions and material composition. The optical performances are characterized through detailed analyses of mode retrieval, mode visualization, and propagation losses, evaluating the efficiency and effectiveness of the fabricated waveguides.

2 Design and simulations of waveguides

Waveguides are structures designed to guide electromagnetic waves, typically light, with minimal energy loss by confining the waves along a specific path. The design process involves selecting appropriate materials and geometries to achieve the desired guiding properties such as speed and confinement. In a waveguide, light propagates in the form of modes, which are distinct patterns of electromagnetic fields along the transverse plane. Each mode has specific propagation characteristics determined by its propagation constant β. In this chapter, we explore photonics waveguides, focusing on slab waveguides and channel waveguides specifically strip waveguides, hybrid polymer waveguides, and strip-loaded waveguides. Next we continue to introduce a technique for analysing and retrieving modes in waveguides and finally describes the concept for integrated Mach-Zehnder interferometers.

2.1 SLAB WAVEGUIDES

A slab waveguide consists of a core film, which is sandwiched between a cladding and a substrate, confines light propagation in one dimension. Consider a slab waveguide as shown in figure 2.1. In this structure, the refractive indices of the cladding (n_c), core (n_f), and substrate (n_s) must satisfy the condition $n_f > n_s \geq n_c$ to ensure confinement of light mostly within the core region. This arrangement ensures that light remains guided within the core layer, primarily due to total internal reflection at the interfaces between the core and its surrounding layers. The core has a thickness, h_f, and light propagates along the z-axis, while the electric field ($\mathbf{E}$) is assumed to be invariant in the y-direction, simplifying the problem to a two-dimensional system with variations only in the x and z directions.

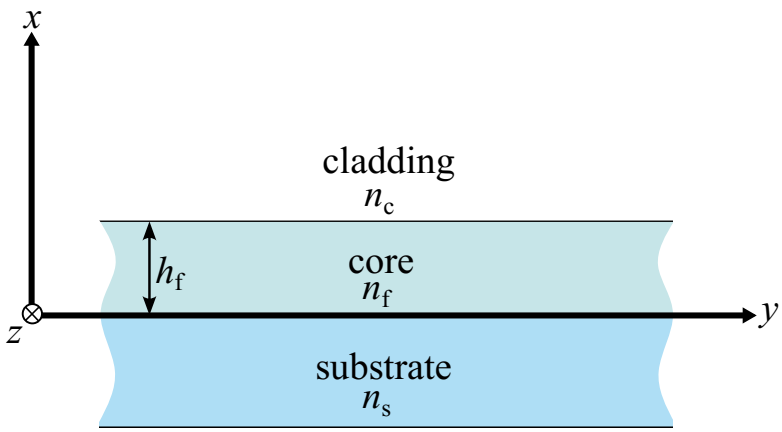

Figure 2.1: Cross section of an asymmetric 3-layer slab waveguide with film thickness h_f. The film with refractive index n_f, is sandwiched between the cladding (n_c) and the substrate (n_s). The stack extends infinitely in the y-direction, with light propagating in the z-direction.

In general, the propagation of electromanetic waves in any medium is governed by the Maxwell's equations, which are Eqs. (2.1 - 2.4) [22]. These fundamental equations describe the relationships between the electric field, the magnetic field, the electric displacement field, and the magnetic flux density.

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{2.1}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}, \tag{2.2}$$

$$\nabla \cdot \mathbf{D} = \rho, \tag{2.3}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{2.4}$$

Here, $\mathbf{E}$ and $\mathbf{H}$ represent the electric and magnetic fields, respectively, while $\mathbf{D}$ is the electric displacement field, related to the electric field by $\mathbf{D} = \varepsilon\mathbf{E}$, where ε is the permittivity of a medium. Similarly, $\mathbf{B}$ is the magnetic flux density, related to the magnetic field by $\mathbf{B} = \mu\mathbf{H}$, where μ is the magnetic permeability of a medium. In Eq. (2.2), $\mathbf{J}$ denotes the current density, and in Eq. (2.3) ρ represents the charge density.

In the specific case of a slab waveguide, where the materials are assumed to be non-magnetic ($\mathbf{B} = \mu\mathbf{H}$, with $\mu \approx \mu_0$ in non-magnetic media) and isotropic, and where no free charges or currents are present ($\rho = 0$, $\mathbf{J} = 0$), Eqs. (2.1) and (2.2) simplify to

$$\nabla \times \mathbf{E} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t}, \tag{2.5}$$

$$\nabla \times \mathbf{H} = \varepsilon_0 n^2 \frac{\partial \mathbf{E}}{\partial t}, \tag{2.6}$$

In Eqs. (2.5) and (2.6), the term n refers to the refractive index of the medium in the respective region, whether it is the core (n_f), the cladding (n_c), or the substrate (n_s). The refractive index is connected with the relative permittivity $\varepsilon_r = \varepsilon/\varepsilon_0$ via $n = \sqrt{\varepsilon_r}$ relation. In general, permittivity can be complex valued, however, in this thesis we mostly deal with real valued permittivity (meaning no losses). Moreover, the free space permeability $\mu_0 = 4\pi \times 10^{-7}\,\mathrm{H/m}$ and the free space permittivity $\varepsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F/m}$.

To find solutions to Maxwell's equations, trial solutions of the following forms

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \exp^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \tag{2.7}$$

$$\mathbf{H}(\mathbf{r}, t) = \mathbf{H}_0 \exp^{i(\omega t - \mathbf{k}\cdot\mathbf{r})}, \tag{2.8}$$

are often used. In Eqs. (2.7) and (2.8), $\mathbf{E}_0$ and $\mathbf{H}_0$ represent the amplitudes of the electric and magnetic fields, respectively, $\mathbf{k}$ is the wave vector, ω is the angular frequency of the electromagnetic wave, and $\mathbf{r}$ denotes the position vector (x, y, z) with respect to the origin. By substituting Eqs. (2.7) and (2.8) into Eqs. (2.5) and (2.6), it can be shown that they satisfy the system of differential equations, thus confirming their validity as solutions in free space. The Equations (2.7) and (2.7) denote plane waves, the simplest solutions of Maxwell's equations for/in isotropic medium. However, one needs to solve the Maxwell's equations under certain boundary conditions (provided by the geometry and mediums) while dealing with waveguides.

In simple geometries, such as the slab waveguide illustrated in Fig. 2.1, an analytical solution exist. In this case, we consider light propagation along the z-axis, and the electric and magnetic fields can be expressed as functions of x and z

$$\mathbf{E}(x,z,t) = \mathbf{E}_0(x)\exp^{i(\omega t-\beta z)}, \tag{2.9}$$

$$\mathbf{H}(x,z,t) = \mathbf{H}_0(x)\exp^{i(\omega t-\beta z)}. \tag{2.10}$$

Here, β represents the propagation constant along the z-axis. For this geometry, boundary conditions at the core-cladding interface lead to discrete allowed values of β, corresponding to the supported modes of the waveguide.

For figure 2.1, we specifically consider Transverse Electric (TE) polarization, in which the electric field is oriented perpendicular to the plane of incidence (x-z plane) and lies along the y-axis. This means the electric field only has a y-component (E_y), while the magnetic field lies in the x-z plane. Under these conditions, Maxwell's equations reduce to

$$i\beta E_y = -i\omega\mu_0 H_x, \tag{2.11}$$

$$\frac{\partial E_y}{\partial x} = -i\omega\mu_0 H_z. \tag{2.12}$$

By combining these equations, we arrive at the Helmholtz equation, which governs the distribution of the transverse electric field component $E_y(x)$ across the waveguide also expressed as

$$\frac{\partial^2 E_y}{\partial x^2} + (n_i^2 k_0^2 - \beta^2)E_y = 0, \tag{2.13}$$

where n_i represents the refractive index in the i-th region (either the core, cladding, or substrate), and $k_0 = \frac{\omega}{c}$ is the free-space wavenumber, with c being the speed of light in vacuum.

The solution to Eq. (2.13) describes the transverse profile of the electric field within the waveguide which is described further in Section 2.3.2. For the waveguide to support guided modes, the term $(n_i^2 k_0^2 - \beta^2)$ must be positive. This condition ensures that the light remains confined within the core of the waveguide through total internal reflection. In the core, where the refractive index is highest, the electric field varies sinusoidally, leading to the formation of standing wave patterns. These standing waves represent the guided modes of the waveguide, which are determined by the waveguide's dimensions, particularly the core thickness h_f, as well as the refractive indices of the surrounding layers (n_c, n_f, and n_s).

2.2 CHANNEL WAVEGUIDE

Unlike planar waveguides, channel waveguides confine light in two dimensional plane and their mode determination is not trivial. They are typically rigid and can follow curved paths. Some channel waveguides are surface-based, where light guidance is influenced by the high refractive index contrast between the waveguide material and air. Buried waveguides are other forms of channel waveguides which

involves covering the waveguide core with an additional material which can minimise scattering losses. The mode simulations in this thesis were conducted using the finite difference eigenmode (FDE) solver in Ansys Optics MODE software [23]. For complex geometries, we used finite-difference time-domain (FDTD) [24–28], 2.5D variational FDTD (varFDTD) [29, 30], and the eigenmode expansion (EME) solver [31], all are modules integrated within Ansys Optics. Initial FDTD simulations were done with varFDTD, while only final ones used 3D-FDTD due to its higher memory and computational needs. MATLAB was used to analyze and plot the simulation results.

2.2.1 Strip waveguide

A strip waveguide typically comprises a high-refractive-index core material surrounded by a lower refractive index cladding [32–34]. This is such that $n_{sw} > n_s > n_c$. sw-subscripts refer to the strip material to avoid confusion with the s-subscript of the substrate. The refractive index configuration facilitates the confinement of light within the core through total internal reflection, allowing it to propagate with minimal loss. Figure 2.2 shows a strip waveguide with the width of the strip w_{sw} and height of the strip as h_{sw}.

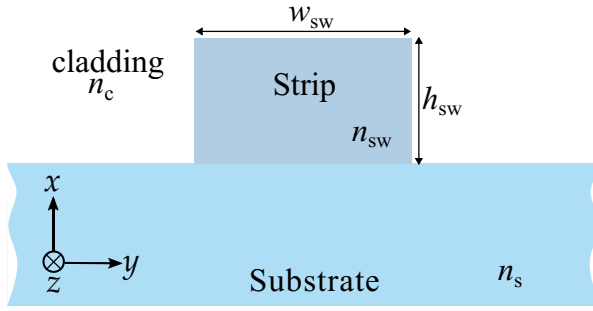

Figure 2.2: Cross section geometry of a strip waveguide with width and height as w_{sw} and h_{sw} respectively

For a strip waveguide the mode field is confined in the high index core (strip) with a little percentage in the substrate and the cladding. Figure 2.3 shows the field distribution of a mode of a silicon nitride (Si_3N_4) on top of a silicon dioxide (SiO_2) substrate and air as a cladding material.

Figure 2.3(a) shows the mode field distribution for the TE fundamental mode. The refractive index silicon nitride core n_{sw} = 2.04 and refractive index of the SiO_2 = 1.46 at λ = 520 nm wavelength. Figure 2.3(b) shows the fundamental mode profile $|E_y|$ along x in the center of the loading strip h_{sw} = 165 nm. It shows that as w_{sw} increases at a fixed h_{sw}, the mode is more laterally confined inside the core and the boundaries have less influence on the field. For most applications, strip waveguides are designed to support a single mode in a particular polarisation. There are other cases where multi-mode operation may be required or prefered.

2.2.2 Polymer-titanium hybrid waveguides

The basis of the designed hybrid polymer waveguide consists of AZ nLoF 2070 (nLoF) polymer waveguide. The refractive index of the polymer is 1.601 at a wave-

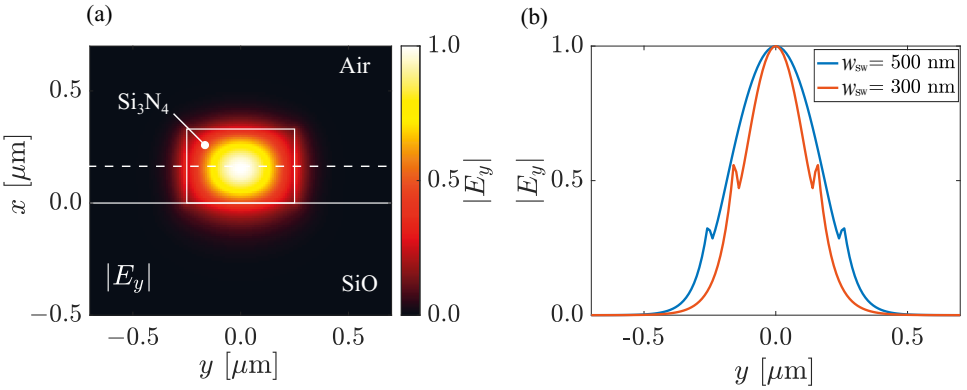

Figure 2.3: (a) Mode distribution in a Silicon nitride strip waveguide with w_{sw} = 500 nm and h_{sw} = 330 nm. (b) Comparison of the fundamental mode profiles $|E_y|$ for different strip width w_{sw} = 500 nm and w_{sw} = 300 nm. The profile is extracted from (a) along x (white dash line.)

length of 1.55 μm. In between the polymer is a trench positioned at y_t which varies from $-a$ to $+a$. The trench position $y_t = 0$ indicates that the trench is in the center of the waveguide. The field is confined in the central core by filling the trench with a material of high refractive index [35]. Figure 2.4 is a schematic cross-section of the hybrid polymer waveguide with the trench filled with TiO_2 at $n_{TiO_2} = 2.247$ at a wavelength of 1.55 μm. The TiO_2 is deposited using atomic layer deposition (ALD). This forms a conformal layer around the nLoF polymer channel waveguide. This leads to a TiO_2 thickness of t around the polymer with the trench filling as $2t$ as shown in Figure 2.4.

The substrate is an oxidized silicon wafer and the cladding is air. The design of the trench as $2t$ is dependent on the fabrication technique used. One benefit of the hybrid polymer waveguide that the trench can be position at any relative position within the nLoF-strip which is a great advantage for development of optical bends.

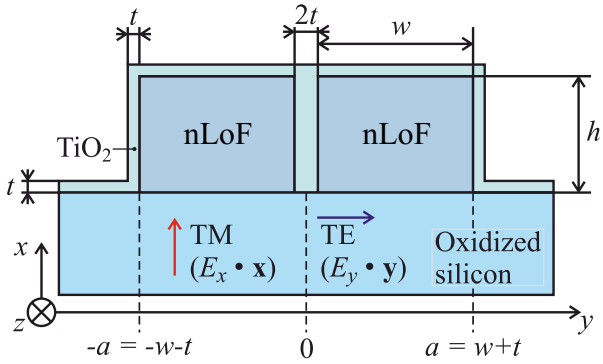

Figure 2.4: Hybrid waveguide cross section geometry. TiO_2 is deposited on an nLoF polymer strip within a trench. The trench width is $2t$. The trench is positioned in the center corresponding to $y_t = 0$.

Considering xz plane as our plane of incidence, figure 2.5 is the mode field

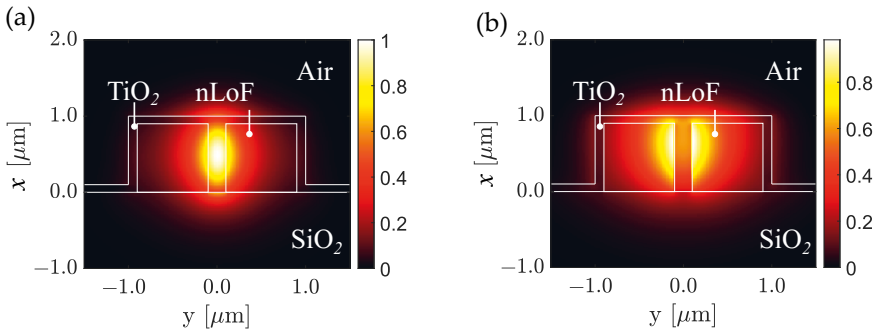

Figure 2.5: Mode distribution of the fundamental quasi- (a) TM mode ($|E_x|$) and (b) TE mode ($|E_y|$). TiO_2 width $2t$ = 200 nm, w = 800 nm, and h = 900 nm.

distribution for the fundamental quasi-TM and TE modes inside the waveguide. The height h of the nLoF polymer is 900 nm, the width w is 800 nm and the high refractive index TiO_2 width is 200 nm. It can be observed in figure 2.5 that the fundamental TM mode is vertically confined with a high percentage of the electric field confined mainly in the TiO_2 high index region. Contrarily to the fundamental TM mode, the TE mode has the electric field more confined to the nLoF low index polymer. This waveguides acts as a vertical strip nano-waveguide.

2.2.3 Strip-loaded waveguides

A strip-loaded waveguide comprises of a high refractive index slab (guiding layer) with a lower refractive index strip on top. The strip changes the effective index locally, weakly confining the light underneath [36,37]. This structure is easier to fabricate because it does not involve a patterning of the high refractive index material, which can be challeneging in some cases. It results a reduction of the sensitivity to the strip's quality and sidewall roughness. It is suitable for applications that require minimal sensitivity to sidewall imperfections and can be implemented using various materials. The optical mode is mainly confined in the high refractive index material. Figure 2.6 shows the geometry of a strip-loaded waveguide. The loading strip increases the effective index in the strip section, which yields the mode to be localised under the strip and therefore creates a channel. The design of this waveguide, requires optimisation of the film thickness, the strip width and the strip height.

Figure 2.7 illustrates the electric field distribution for both the fundamental quasi-TE and quasi-TM modes with TiO_2 as the film and nLoF as the strip at 1550 nm wavelength. Figure 2.7(a) displays the electric field distribution for the fundamental TE mode, while Figure 2.7(b) shows the distribution for the TM mode. In the case of the TE mode, the field is predominantly confined within the high refractive index material, i.e., TiO_2 film. For the TM mode, the field is mainly localized in the nLoF and the SiO_2 substrate.

When designing waveguides, several critical factors must be considered to ensure optimal performances. These include mesh size, convergence testing, and boundary conditions (such as perfectly matched layers). The mesh size determines the resolution of the computational model used in simulations. Using finer meshes provides more accurate results by capturing detailed behaviors of light within the

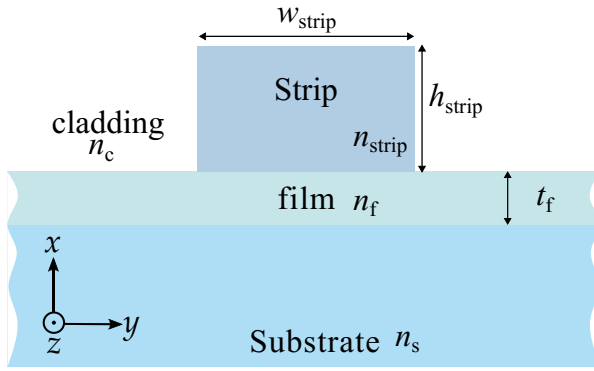

Figure 2.6: Cross-sectional view of a strip-loaded waveguide platform. The field is mainly within the high refractive index film and a fraction is laterally confined beneath the strip. w_{stip}, h_{strip}, and n_{stip} represents the strip width, the strip height, and the strip refractive index respectively. n_{f} and t_{f} represents the refractive index of high index film and its thickness respectively.

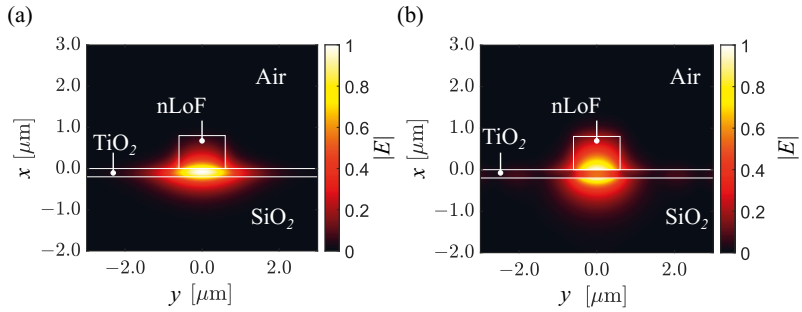

Figure 2.7: Electric field distribution of the strip-loaded waveguide for the fundamental (a) TE mode and (b) TM mode

waveguide and preserve the geometrical tiny variations that we may consider. However, finer meshes also require greater computational resources.

Convergence testing verifies that the solution is stable and accurate, ensuring it does not change significantly with further mesh refinement. Proper boundary conditions are essential for accurately simulating the waveguide environment, preventing artificial reflections and ensuring correct propagation of light waves. It is important to note that simulations are approximations of real-world phenomena, limited by the assumptions and simplifications of the model. Therefore, while optimizing these factors enhances simulation accuracy and reliability, results should be validated against experimental data or practical observations whenever possible.

2.3 MODAL STRENGTHS DETERMINATION

The design of photonics devices requires the understanding of the modes inside the waveguide. The design of on-chip sensors, for instance, requires the knowledge of the amount of power carried by each mode inside the waveguide. In this section,

we explore how these modes are represented within a waveguide and discuss a technique for analysing and retrieving modes using their complex field profiles.

2.3.1 Modal representations in waveguides

In waveguides, modes are described by the spatial distribution of the electromagnetic field, including amplitude and phase of the electric and magnetic fields. Solving Maxwell's equations with boundary conditions results in a set of modes that define how electromagnetic energy is guided, characterized by parameters like mode number, polarization, and frequency.

For a slab waveguide, the total guided field E_{out} at the exit plane can be expressed as a combination of established modal fields $\phi_p(x)$

$$E_{\text{out}} = \sum_{p=0}^{P-1} \alpha_p \phi_p(x), \tag{2.14}$$

where p is the mode number, P is the total number of modes the waveguide can support, and α_p are complex-valued modal weights. The modes are orthogonal and satisfy

$$\int_{-\infty}^{\infty} \phi_p^*(x)\phi_q(x)\,\mathrm{d}x = \delta_{pq}, \tag{2.15}$$

where δ_{pq} is the Kronecker delta. The total modal weights are obtained by

$$\alpha_p = \int_{-\infty}^{\infty} E_{\text{out}}\phi_p(x)\,\mathrm{d}x. \tag{2.16}$$

The intensity weights, which are real-valued, are given by the expression [38,39]

$$c_p = \left|\alpha_p\right|^2. \tag{2.17}$$

The far-field representation of the modes is obtained through the Fourier transform, which converts the spatial domain functions into their corresponding frequency domain representations. The resulting expression for the total exiting field in the spatial frequency domain is expressed as

$$\tilde{E}_{\text{out}}(k_x) = \sum_{p=0}^{P-1} \alpha_p \tilde{\phi}_p(k_x), \tag{2.18}$$

In Eq. (2.18), α_p are the weights corresponding to each mode, obtained through near-field and far-field measurements. This summation indicates that the total exiting field is a combination of the individual mode contributions in the spatial frequency domain. This formulation can also be extended to handle two-dimensional (2D) modes, where the same principles apply, but the integrals and summations would consider both k_x and k_y spatial frequency components.

To accurately determine the contribution of each mode, we employ the overlap integral method, which calculates the modal weights based on the overlap between

the input and mode fields. In the overlap integral method, the modal weights of an output field are computed by evaluating how much of the input field overlaps with the specific modes of the waveguide. This technique, similar to coupling a coherent incident field $E_{\text{in}}(x)$ into a waveguide, calculates modal weights α_p through

$$\alpha_p = \int_{-\infty}^{\infty} E_{\text{in}}(x)\phi_p(x)\,\mathrm{d}x. \tag{2.19}$$

These modal weights represent projections of the incident field onto the waveguide-mode basis, influenced by the polarization (TE or TM). The incident field can be decomposed into waveguide modes and uncoupled radiation modes expressed as

$$E_{\text{in}}(x) = \sum_{p=0}^{P-1} \alpha_p\phi_p(x) + E_{\text{r}}(x), \tag{2.20}$$

where $E_{\text{r}}(x)$ represents uncoupled radiation modes, leading to energy loss during coupling. The coupled field $E_{\text{c}}(x)$ is therefore expressed as

$$E_{\text{c}}(x) = E_{\text{in}}(x) - E_{\text{r}}(x). \tag{2.21}$$

Eq. (2.21) describes the coupled field that propagates through the waveguide, potentially transforming into the output field $E_{\text{out}}(x)$, though interactions between modes can occur due to waveguide imperfections.

The overlap integral method assesses the alignment of an incident field with a waveguide mode, indicating coupling efficiency. Perfect alignment excites only the matched mode, while deviations can excite multiple modes, reducing efficiency. Although the overlap integral method provides a general measure of coupling efficiency, it is less accurate when applied to end-coupling scenarios. This inaccuracy arises because the method typically neglects phenomena such as reflections at the waveguide boundaries and the influence of evanescent fields that decay exponentially outside the waveguide. These effects can play a significant role in determining the actual coupling efficiency at the waveguide's end facets. Despite this limitation, the method remains highly effective for analyzing the total output fields in waveguides and is adaptable to coupling between waveguides or fibers, where reflections and evanescent fields are less dominant [40, 41].

Typically, measuring total output field characteristics involves complex phase measurements with interferometers, e.g., the Twyman–Green interferometer [42]. Alternatively, we present in Section 2.3.2 a mirror-based technique [43, 44] can measure both amplitude and phase profiles accurately, offering a self-referencing solution.

2.3.2 Complex field and modal strength retrieval

In the case of a general scalar field $E(x)$, the cross-spectral density (CSD) is defined as [43].

$$W(x_1, x_2) = \langle E^*(x_1)E(x_2)\rangle, \tag{2.22}$$

In Eq. (2.22), the electric field is denoted as $E(x_i)$, the ensemble average is also represented as angle brackets, and the complex conjugate is denoted by asterisk. The CSD can be expressed in factored form for a fully coherent field $E_{\text{out}}(x)$ as [45,46]

$$W(x_1, x_2) = E^*_{\text{out}}(x_1) E_{\text{out}}(x_2), \tag{2.23}$$

Since all realizations of $E_{\text{out}}(x)$ are identical, the ensemble average disappears, making the averaging process over multiple realizations equivalent to a single one. Using the amplitudes this can be represented as

$$W(x_1, x_2) = E_0(x_1) E_0(x_2) \exp\left\{-i\left[\varphi(x_2) - \varphi(x_1)\right]\right\}. \tag{2.24}$$

From Eq. (2.24), the phase information of the total field E_{out} can be retrieved, and the amplitude can be computed either from the diagonal terms of $W(x_1, x_2)$ or from the measured intensity. Once the complex field is reconstructed, the modal strengths can be determined from Eqs. (2.16) and (2.17).

Two numerical examples, shown in Figs. 2.8 and 2.9, are used to demonstrate the retrieval method. In Fig. 2.1, the refractive indices of the core (n_{f}), cladding (n_{c}), and substrate (n_{c}) are taken as 2.04, 1.00, and 1.46, respectively, at a wavelength of 520 nm. With the core thickness (h_{f}) set to 1.5 μm, the allowed modes (ϕ_p) and their profiles can be calculated using Eq. 2.13 written for the core, cladding and the substrate [47,48]. For the TE polarization, the (ϕ_p) can be expressed as

$$\phi_{y,p} = \begin{cases} A_p e^{(\rho_p x)}, & \text{for } x \leq 0, \\ A_p \cos(\kappa_p x) + B_p \sin(\kappa_p x), & \text{for } 0 \leq x \leq h_{\text{f}}, \\ \left[A_p \cos(\kappa_p h_{\text{f}}) + B_p \sin(\kappa_p h_{\text{f}})\right] e^{-\sigma_p (x - h_{\text{f}})}, & \text{for } x \geq h_{\text{f}}, \end{cases} \tag{2.25}$$

and the propagation constants for the modes are determined by solving the following transcendental equation

$$\tan(\kappa_p h) = \frac{\kappa_p(\sigma_p + \rho_p)}{\kappa_p^2 - \sigma_p \rho_p}, \tag{2.26}$$

In Eq. 2.26, ρ_p^2 is represented by $\rho_p^2 = \beta_p^2 - k_0^2 n_s^2$, κ_p^2 is defined as $\kappa_p^2 = k_0^2 n_g^2 - \beta_p^2$, and σ_p^2 is given by $\sigma_p^2 = \beta_p^2 - k_0^2 n_a^2$, The electric field component for the p-th mode, denoted by $\phi_{y,p}$ and polarized along the y-axis, has amplitude coefficients A_p and B_p. The coordinate x indicates the spatial dimension perpendicular to the waveguide layers, and h_{f} specifies the guiding layer's thickness. The decay rates in the substrate and cladding are given by the constants ρ_p, κ_p, and σ_p, with σ_p representing the propagation constant in the guiding layer. Moreover, k_0 represents the free-space wave number, and β_p is the propagation constant of the mode.

Eq. 2.26 gives the probation constants of the allowed modes while Eq. 2.25 gives the electric field profiles for the mode perpendicular to our plane of incidence. Mode properties such effective refractive index, mode field distribution and mode confinement factor (ptical power confined within the various regions of the waveguide) can be determined once the possible modes sustained by the waveguide are determined.

In figures 2.8 and 2.9, the modal strengths and the corresponding electric fields are analyzed using CSD and an overlap integral method to validate a retrieval algorithm. In figure 2.8, the modal strengths c_p follow a Gaussian distribution, meaning

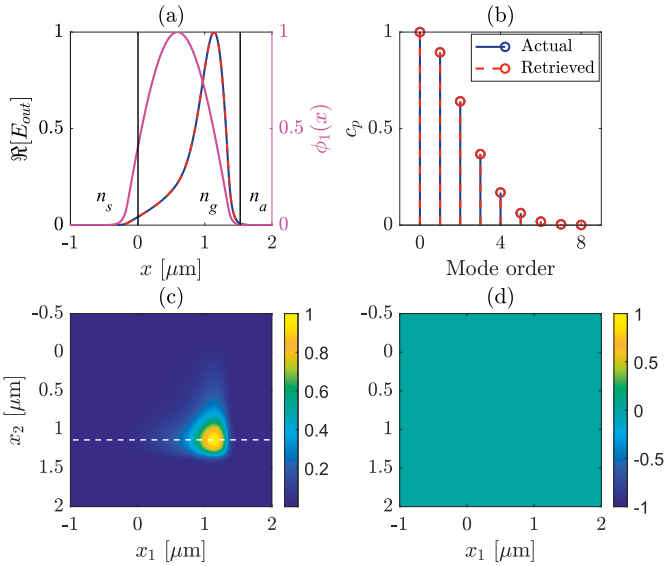

Figure 2.8: Demonstration of the retrieval algorithm with an asymmetric slab waveguide for TE polarisation. The fundamental mode has the highest weight in this case. In (a), the black lines indicate the waveguide boundaries, while the normalized real field for the fundamental mode (magenta) and the normalized total real field for both the actual (blue) and retrieved (red-dashed) fields are shown. (b) compares the actual mode strengths (blue) with the retrieved mode strengths (red-dashed). The real part of the CSD is displayed in (c), with a 1-D line profile used for mode retrieval. (d) shows the corresponding phase of the CSD.

the fundamental mode (TE_0) carries most of the energy, with higher-order modes contributing less. The total electric field is computed using Eq. (2.14), and the real part is shown in figure 2.8(a), while the modal weights (indicating how much energy each mode carries) are plotted in figure 2.8(b) as a solid blue line. The corresponding CSD is also calculated using Eq. (2.23), with its amplitude and phase shown in figure 2.8(c) and (d), respectively.

For this Gaussian distribution, the phase is flat, as seen in figure 2.8(d), meaning that the total electric field can be reconstructed from the intensity alone without needing phase information. A 1D line profile is selected from the CSD where the amplitude is highest (marked by a white dashed line in figure 2.8(c)), allowing the complex total electric field to be reconstructed. This retrieved field is compared to the actual one in figure 2.8(b) as a dashed red line, with both showing good agreement, validating the method.

In figure 2.9, the same process is applied but with random modal strengths, which can occur in real-world scenarios due to waveguide defects or misalignment of the in-coupling field. In this case, the phase plays a role in determining the mode strengths as the CSD becomes complex-valued. Since the modal phases α_p are real, the phase of both the CSD and total electric field exhibits a binary nature, as shown in Figure 2.9(d). Despite the randomness, the retrieved modal strengths and total field match the actual ones, confirming the reliability of the algorithm in handling both Gaussian and random distributions.

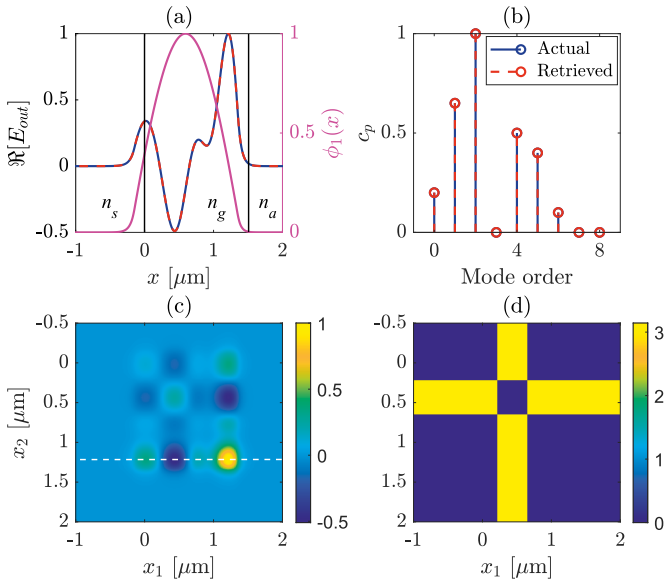

Figure 2.9: Demonstration of the retrieval algorithm with an asymmetric slab waveguide for TE polarisation. The second order mode has the highest weight in this case. In (a), the black lines indicate the waveguide boundaries, while the normalized real field for the fundamental mode (magenta) and the normalized total real field for both the actual (blue) and retrieved (red-dashed) fields are shown. (b) compares the actual mode strengths (blue) with the retrieved mode strengths (red-dashed). The real part of the CSD is displayed in (c), with a 1-D line profile used for mode retrieval. (d) shows the corresponding phase of the CSD.

The complex CSD can be measured experimentally using devices such as a wavefront interferometer (WFI) or a digital micro-mirror device (DMD) [49]. In both cases, a 1D line is selected (as marked by white-dashed line in Figs. 2.8(c) and 2.9(c)) of CSD where $|W(x_1, x_2)|$ is maximum. This particular choice is important from the experimental point of view, which will become clear in Sec. 4.2.2, where we show the detailed experimental procedure and results.

2.4 MACH-ZEHNDER INTERFEROMETERS

Mach-Zehnder interferometers (MZIs) are essential in sensing, telecommunications, and quantum computing [50–53]. An MZI consists of four straight waveguides and two Y-junctions as shown in 2.10 .

One of the waveguides serves as input and one also serves as an output of the MZI, The remaining two waveguides function as the sample and the reference arm of the MZI. One Y-junction acts as a splitter, and the other as a combiner. Light entering through the input waveguide is split into two beams by the Y-branch splitter. These beams travel through the reference and sample arms and are then recombined by the Y-combiner and comes out of the output waveguide. The combined signals from both arms create an interference of the output intensity, described by the following equation [54]

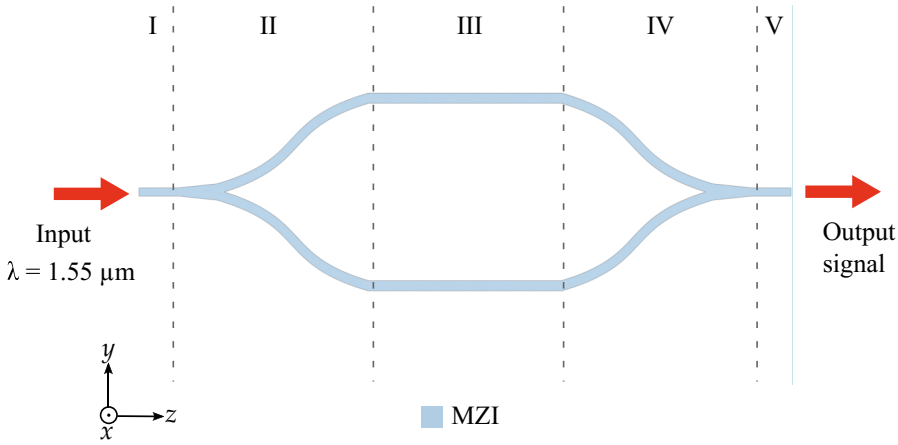

Figure 2.10: Balanced Mach-Zhender interferometer configuration (MZI). The MZI consists of different elements such as waveguides located in region I, III, and IV. Splitter in region II and combiner in region IV. The sample arm and the reference arm are located in region III.

$$I = I_S + I_R + 2\sqrt{I_S I_R}\cos(\Delta\phi). \tag{2.27}$$

The sensing and reference arms have intensities I_S and I_R, respectively, with the phase difference between the beams from the sample and reference arm given by $\Delta\phi$. The output intensity I of the MZI can be modulated by the variation of the effective index of the sample arm. This is done by an introduction of an analyte in our case ethanol-water solution which interacts with the light in the sample arm. As the concentration of the analyte is varied, light in the sample and reference arms interferes, generating a sinusoidal modulation of the output intensity that can be detected at the output of the device. During the characterization of the MZI, the MZI's output is measured in terms of power expressed as

$$P_{out} = P_{in}\cos^2(0.5\Delta\phi), \tag{2.28}$$

where P_{out} is the output power, while P_{in} corresponds to the input power. By varying the analyte's refractive index and measuring the output power, we can calculate the phase change $\Delta\phi$. This phase change is described by [55]

$$\Delta\phi = \frac{2\pi}{\lambda}L\Delta n_{\text{eff}}. \tag{2.29}$$

The interaction length is given by L, the wavelength of light in a vacuum by λ, and Δn_{eff} describes the difference in the effective index between the reference and sensing arms. From Eq. (2.29), it is evident that larger sensing areas increase sensitivity, even for slight refractive index changes, making the MZI highly effective for detecting small variations in the refractive index.

Sensor sensitivity is the rate at which the output intensity is modulated in response to changes in the refractive index of the surrounding medium (analyte). It indicates how the output of the MZI is affected by changes in the refractive index of the material in contact with the sensing arm. Specifically, sensitivity measures

how efficiently the sensor detects these changes by observing shifts in the phase difference between the reference and sensing arms.

In our proposed design, the ethanol-water solution (analyte) is deposited uniformly across the entire sensing region. This configuration ensures that the refractive index change affects the whole interaction region equally, leading to consistent and predictable modulation of the MZI's output signal. By covering the sensing area entirely with the analyte, we ensure "homogeneous sensitivity," meaning the sensor's response is uniform across the sensing region. This uniformity enhances measurement accuracy because it prevents localized variations in refractive index from skewing the overall output. This homogeneous sensitivity, S_H, is calculated as

$$S_H = \frac{\delta n_{\text{eff}}}{\delta n_{\text{c}}}, \tag{2.30}$$

where, S_H describes the change in the mode effective index, Δn_{eff}, as a function of the analyte's refractive index, Δn_{c}. This indicates how the output intensity, I, of the MZI is modulated by varying the refractive index of the analyte in the cladding. Experimentally, the homogeneous sensitivity is determined by the slope of the linear fit for Δn_{eff} versus Δn_{c}, which correlates with varying ethanol-water solution concentrations.

3 Fabrication techniques

Fabrication techniques encompass a variety of methods that are used to create material layers, devices, and structures with specific properties and functionalities. The development of photonics devices, for instance, relies heavily on these techniques to actualize the envisioned devices. Furthermore, to design effective photonic waveguides and devices, a fundamental understanding of the fabrication processes involved in their manufacture is essential. This chapter describes the fabrication and characterization techniques and methods relevant to the works presented in this thesis. It includes thin film deposition techniques and electron beam lithography for the main fabrication methods and prism coupler characterization for the investigation of the deposited thin films and the accurate measurement of the refractive index of the materials and the effective index of the modes allowing for a better design.

3.1 THIN FILM DEPOSITION METHODS

Thin film deposition methods are pivotal for the fabrication of waveguides, tailored to the specific materials and applications of the devices being produced. Various techniques are employed, including atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), and spin coating. In this work, due to the use of materials such as titanium dioxide (TiO_2) and nLoF polymer, the focus will be on describing ALD and spin coating as thin film deposition techniques.

3.1.1 Atomic layer deposition

Atomic layer deposition is a very precise method for adding extremely thin layers of material to a surface. This technique has been first introduced by Suntola and Antson [56]. It is a refined type of CVD that uses sequential introduction of precursor gases, unlike the simultaneous gas mixing seen in traditional CVD [57]. This step-by-step process leads to self-limiting surface reactions, allowing for control over film thickness and uniformity [58].

The ALD technique consists of cyclic steps: first, a precursor gas is introduced, which adsorbs onto the substrate, forming a monolayer. Any excess precursor is then purged using an inert carrier gas like nitrogen or argon. An oxidizing agent is then introduced to react with the adsorbed precursor, creating a thin film layer. Repeating this cycle builds up the film to the desired thickness [59–61].

In this thesis work, the TFS-200 ALD system from Beneq at the University of Eastern Finland (UEF) has been used to fabricate TiO_2 layers. This system precisely controls the introduction of gases, using intermediate flushing steps between each gas precursor pulse to ensure accurate deposition. Unlike conventional CVD, ALD's sequential process prevents premature reactions, thereby improving film uniformity. The TFS-200 system maintains optimal pressure and temperature conditions, with the reactor capable of heating up to 450°C, and uses nitrogen flow to purge the system between gas pulses.

ALD is highly regarded for its ability to produce uniform, conformal films, even on substrates with complex surfaces. This is crucial for applications requiring smooth, uniform thin layers, such as in semiconductor devices, micro-electromechanical systems (MEMS), and various nanotechnology applications. For example, TiO_2 can be deposited using thermal ALD with titanium tetrachloride ($TiCl_4$) as the precursor and water as the oxidant, achieving a growth rate of 0.0718 nm per cycle. This results in amorphous when deposited at 120 $^{\circ}C$ temperature and smooth layers which is essential in photonics application to minimise optical losses [62].

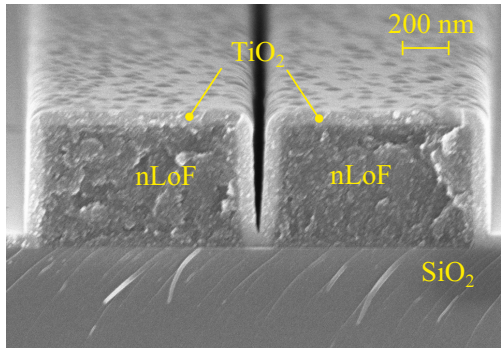

Figure 3.1: SEM image showing TiO_2 film deposited on top of nLoF using ALD technique.

3.1.2 Spin coating

Spin coating is a widely used method for applying photoresist thin films like nLoF polymer onto substrates. This technique involves depositing a small amount of liquid photoresist onto the center of a substrate, typically a silicon wafer with a silicon dioxide (SiO_2) layer, and then rapidly spinning the substrate. Centrifugal force spreads the photoresist solution across the substrate's surface, forming a thin layer [63–65].

AZ nLof 2070 from MicroResist is a negative photoresist commonly employed in microfabrication processes. Upon exposure to UV light through a photomask, the areas of the photoresist unexposed to light become soluble in the developer solution. The exposed regions becomes insoluble in the AR 300-47 (Allresist GmbH) developer solution forming a patterned mask on the substrate. The parameters involved in the spin coating such as spin time, spin speed, and photoresist concentration, significantly influence the thickness and uniformity of the deposited film. Optimization of these parameters is crucial for achieving the desired film thickness and quality.

In practice, spin coating offers several advantages, including simplicity, scalability, and versatility. It is a cost-effective and rapid method for depositing thin layer films over large areas, making it suitable for various microfabrication applications, including semiconductor device fabrication and photolithography processes [66].

3.2 THIN FILM CHARACTERIZATION WITH A PRISM COUPLER

Prism coupling is a well-established, technique used to characterize the refractive index and thickness of thin films. This method is particularly useful for evaluating the optical properties of materials deposited on substrates like fused silica and oxidized silicon wafers [67,68]. The Metricon Model 2010/M was used for measuring the refractive index of thin films used in this works. It is equipped with illumination sources at wavelengths of 0.532 μm, 0.636 μm, and 1.547 μm wavelength [69].

The process involves pressing the sample (guiding film) against a high-refractive index prism with an air gap of a few nanometers. A laser beam passes through one side of the prism and is aimed at the film. By varying the incidence angle, light is coupled at specific angles in the film, which corresponds to the guided modes. Measuring the coupling angles allow for the estimation of the modal effective index. This coupling appears as dips in the reflected signal detected by the system. These measurements enable the determination of both the substrate's refractive index, the film's effective and refractive indices, and the thickness of the film.

At least Two modes are required at a single wavelength to characterise a single homogeneous film with an unknown refractive index and thickness. Both TE and TM modes are used to retrieve the refractive index and the thickness. With a known film thickness, a single mode is sufficient to determine the refractive index at a wavelength.

Prism coupler effective index measurements confirms the quality of the fabricated films before proceeding with further patterning and device fabrication. This also ensures that the correct effective and refractive index values are used for the simulation ensuring accuracy in the numerical simulations [35]. Prism coupling demonstrated to be highly effective in characterizing our waveguide structures. The measurements performed on a TiO_2 and nLoF showed excellent agreement with theory.

3.3 ELECTRON BEAM LITHOGRAPHY

Electron beam lithography (EBL) is a technique used to create extremely fine patterns ranging in the micrometers and nanometers scale [70–72]. This process leverages the interaction of a focused electron beam with a resist material, which is sensitive to accelerated electrons.

Exposure to light alters the molecular structure of the resist, either increasing its solubility or reducing it, depending on whether a positive or negative resist is used. Positive tone resists become more soluble when exposed to the electron beam, enabling the exposed regions to be dissolved away during development. Negative tone resists, conversely, become less soluble, as the exposure causes the resist to polymerize and become cross-linked, making the exposed areas more resistant to dissolution [73,74].

The EBL technique allows for the fabrication of any desired patterns with high density, resolution, and reliability. Despite being a slow and expensive method, EBL is invaluable for creating masks for conventional optical lithography. It can be used to create templates for processes like roll-to-roll fabrication and nanoimprinting, which are suitable for high-volume production. [75].

We have used Vistec EBPG5000+HR EBL device at UEF to pattern nLoF in this work. During fabrication, the energy per unit area of the electron beam dosage is very important and controlled. The beam's current and acceleration voltage are kept

constant. The best dosage depends on the type of photoresist used and the density and size of the patterns to be made.

nLoF electron beam resist is well-known for its use in EBL due to its availability in different molecular weights. It was spin-coated and pre-baked on silicon wafers, then exposed to the electron beam after which it is developed [76,77].

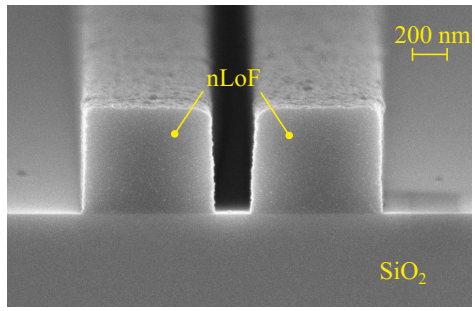

Figure 3.2: SEM image showing nLoF resist after electron beam exposure and development. The nLoF strips represents the exposed region which remain insoluble in the developer solution.

4 Waveguide characterizations and design

In this chapter, we present the characterization setup used in this work in Section 4.1. Through experiments, we demonstrate that the strength of modes in a waveguide can be estimated by measuring the phase and amplitude at the exit plane of the waveguide output, as detailed in Section 4.2. Finally, in Section 4.3, we discuss the specific challenges and solutions associated with designing and fabricating polymer-titania hybrid waveguides, a promising material combination for enhancing sensor performance. Insights gained from the detailed analysis of propagation losses and mode dynamics within these waveguides are crucial for refining their design and improving functionality in practical sensing scenarios.

4.1 CHARACTERIZATION SETUP FOR OPTICAL GUIDING

The characterization of the fabricated structures was performed using the end-fire coupling method, as illustrated in Figure 4.1. This technique involves injecting light directly into the waveguide and measuring the power transmitted at the output with a power meter. Two types of laser sources were utilized for these experiments: a fiber-coupled laser source with a wavelength of 520 nm and a tunable laser source operating in the range of 1510 nm to 1640 nm.

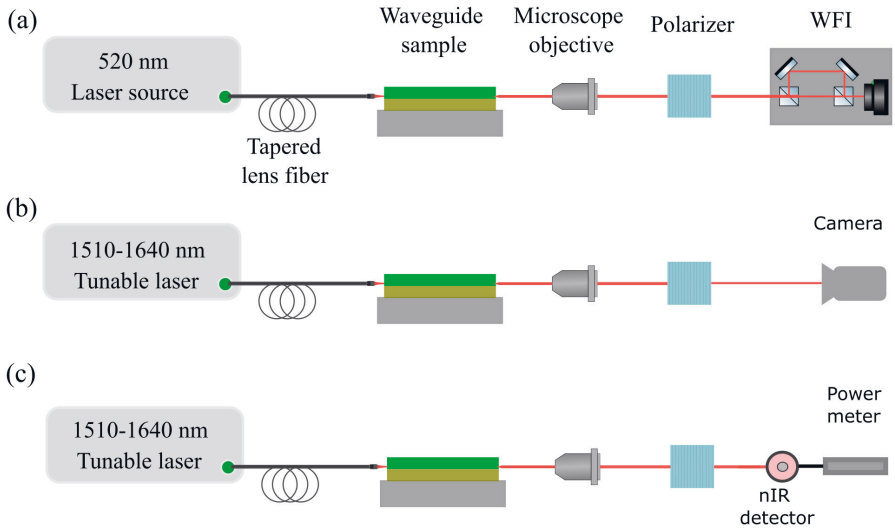

Figure 4.1: Characterisation setup for (a) mode retrieval (b) mode imaging and visualisation and (c) output power measurement.

Three configurations were employed in the experiments. In all cases, light was coupled into the waveguide using a tapered lens fiber (Oz Optics), which focused light onto the end-facet of the sample. This fiber had a spot size of about 2 ± 0.5 μm. The label Waveguide sample in figure 4.1 refers to both the fabricated chip and

the XYZ stage holding the chip. At the waveguide's output, a microscope objective was used to image the output mode(s). A Glan-Thompson polarizer GTH10M from Thorlabs was used to set the desired polarization, ensuring that only correctly polarized light reached the detector, thereby enhancing the accuracy of the measurements.

Figure 4.1(a) shows the setup used for determining mode strength. Light from the polarizer is directed onto a WFI, where scanning is performed. The mode images presented in this thesis for the work on near infrared hybrid waveguides were measured using the configuration shown in figure 4.1(b), where the output light from the polarizer is captured using a near-infrared (nIR) camera. In figure 4.1(c), the output of the polarizer is directed to an nIR detector, which is connected to a power meter. The power meter measures the output power from the sample imaged using the microscope objective.

These setup configurations allowed for precise characterization of the fabricated chips by measuring the light transmittance and ensuring reliable data collection.

4.2 MODE RETRIEVAL AND MODE ANALYSIS

Some devices may require multimode operation, e.g., in the tapering part of the input or output of a device or in the enlargement which may be needed to increase the surface area of a sensor as we will see later. Each mode is carrying a share of the total power injected in the waveguide. Understanding how light is balanced between each mode is important to optimize the design of waveguides, especially in the case of interferometers, for which the output signal depends on the interaction of the modes. In other words, analyzing modes allows designers to determine optimal waveguide parameters to support specific modes, controlling properties such as confinement and dispersion. Mode analysis also verifies waveguide performance by comparing simulated and experimental modes, identifying any fabrication errors. Additionally, it helps mitigate unwanted effects such as crosstalk and dispersion by understanding mode coupling and interference. This analysis also facilitates innovation by exploring novel waveguide structures, leading to the development of advanced photonic devices for various applications.

4.2.1 Concept of mode retrieval

We present an experimental demonstration to determine mode strengths in an asymmetric channel waveguide using self-referencing interferometric measurements at the waveguide's exit plane. A customized wavefront folding interferometer (WFI) [43, 44] is used to measure the CSD, from which the total electric field and mode strengths in an asymmetric strip channel waveguide are derived. The retrieval algorithm presented in Section 2.3 also provides phase information for each mode, derived directly from the measured electric field distribution.

The method is experimentally demonstrated with a silicon nitride strip waveguide (Fig. 2.2) designed to support several quasi-TE modes and the fundamental quasi-TM mode TM_{00}. The waveguide dimensions are h_{sw} = 330 nm, w_{sw} = 3 μm, and a vacuum wavelength of 520 nm for the coupled light. Light is injected through butt-coupling, and a linear polarizer is used to separate the TE and TM contributions. Only the intensity from the TE modes is considered. These modes are spatially coherent, mutually orthogonal, and correlated. Light enters a wavefront folding interferometer, which measures the CSD by analyzing the interference pattern. By

decoding the WFI output and correlating it with simulated field distributions, the mode weights are retrieved. Figure 4.2 illustrates the concept for retrieving mode strengths at the output of a channel waveguide.

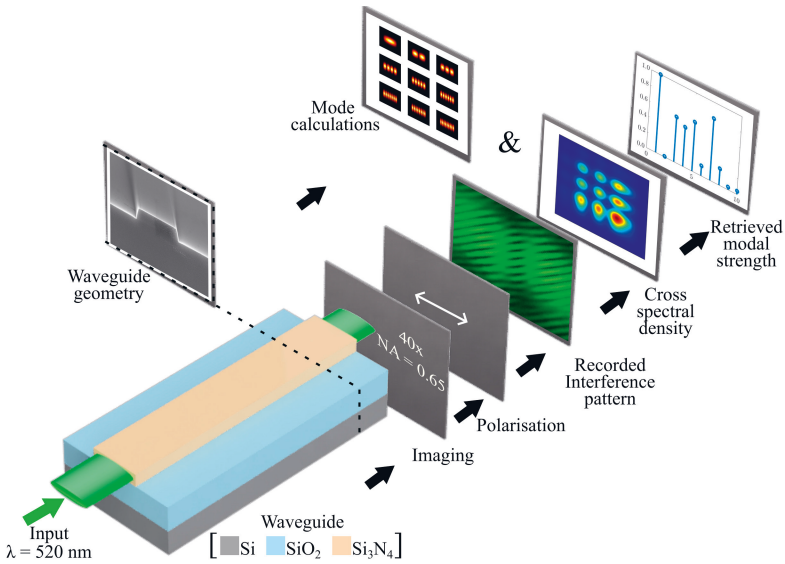

Figure 4.2: Experimental concept for retrieving mode strength in a channel waveguide for TE polarisation. Black arrows indicate the sequential steps in the mode retrieval process. A 520 nm illumination wavelength is coupled into a channel waveguide, and the output intensity is imaged using a microscope objective. The output is directed through a polarizer into a wavefront folding interferometer, where interference patterns are recorded. The cross-spectral density (CSD) is obtained from these interference patterns. The retrieved CSD, along with the calculated modes based on the waveguide dimensions, is then used to determine the mode strengths of the sustained modes inside the waveguide.(PAPER III)

4.2.2 Experimental mode visualization and retrieval

The experimental setup for mode measurement is shown in figure 4.1(a). The WFI captures an interference pattern to measure the complex CSD, which is mapped to absolute coordinates. By choosing a 1D cut that maximizes the signal-to-noise ratio, the modal strength distribution and modal dispersion can be retrieved. The retrieved field represents the output field with at most a uniform π phase shift ambiguity.

The number of observed modes is limited by the magnification of the microscope objective. Modes with propagation angles exceeding the objective's convergence angle (40.7°) are truncated. Despite this limitation, the method remains valid and is sufficiently demonstrated as the contribution of the higher-order modes is negligible.

Figures 4.3, 4.4, and 4.5 illustrate three cases that validate our retrieval algorithm, with each figure corresponding to a different coupling condition. In all cases, we investigate the impact of slightly tilting the input lensed fiber or adjusting its position relative to the waveguide to deviate from the optimal coupling condition.

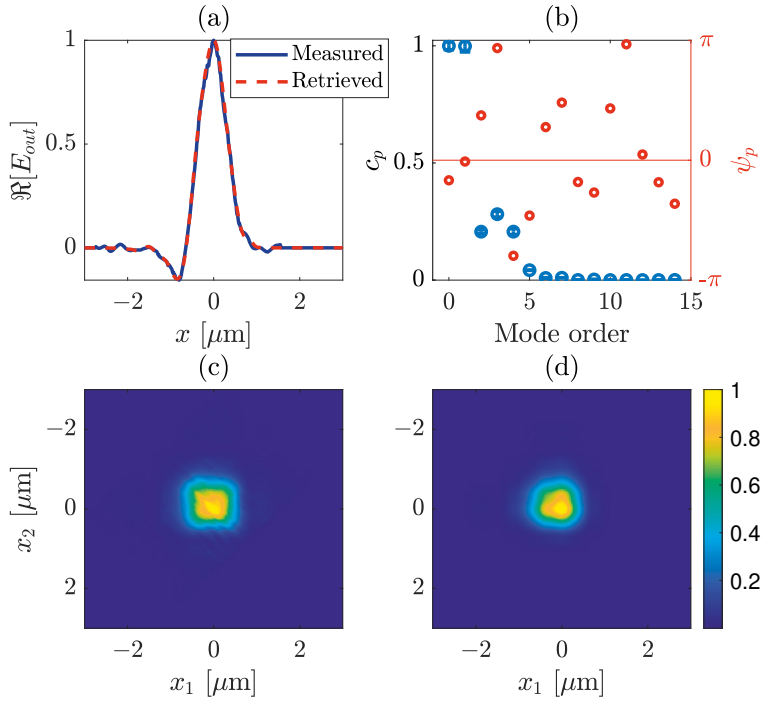

Figure 4.3: Mode retrieval results for a silicon nitride strip waveguide. In (a), the normalized real measured field is compared with the normalized real retrieved field. In (b), the retrieved mode strengths from the measured field and the corresponding phase of the modes are shown. (c) displays the measured cross-spectral density (CSD), and (d) shows the CSD reconstructed from the retrieved mode strengths.

The optimal coupling occurs when the fiber is perfectly aligned, resulting in the fundamental mode having the highest coupling efficiency and therefore the highest weight. However, when the fiber is tilted or slightly misaligned, as shown in these figures, the coupling efficiency shifts, leading to a different mode becoming dominant depending on the injection angle and position. Only the first 15 TE modes are presented.

In figure 4.3(a), both the measured (blue) and retrieved (red) electric field profiles (real part) are shown. The agreement between the measured and retrieved field profiles indicates the algorithm's robustness. Figures 4.3(c) and 4.3(d) show the measured and retrieved CSD, respectively.

A consistent display arrangement is maintained in figures 4.4 and 4.5 for the same waveguide with different injection conditions. This involves tilting the injection angle of the input lensed fiber. Variations in the modal strength of individual modes during measurements can be observed. In figure 4.3(b), the fundamental mode has the highest strength. In figure 4.4, the fundamental mode dominates, but several higher-order modes have significant weights. In figure 4.5(b), the fourth-order mode is dominant.

The agreement between the retrieved and measured electric field profiles confirms the method's reliability. Slight discrepancies between the retrieved and mea-

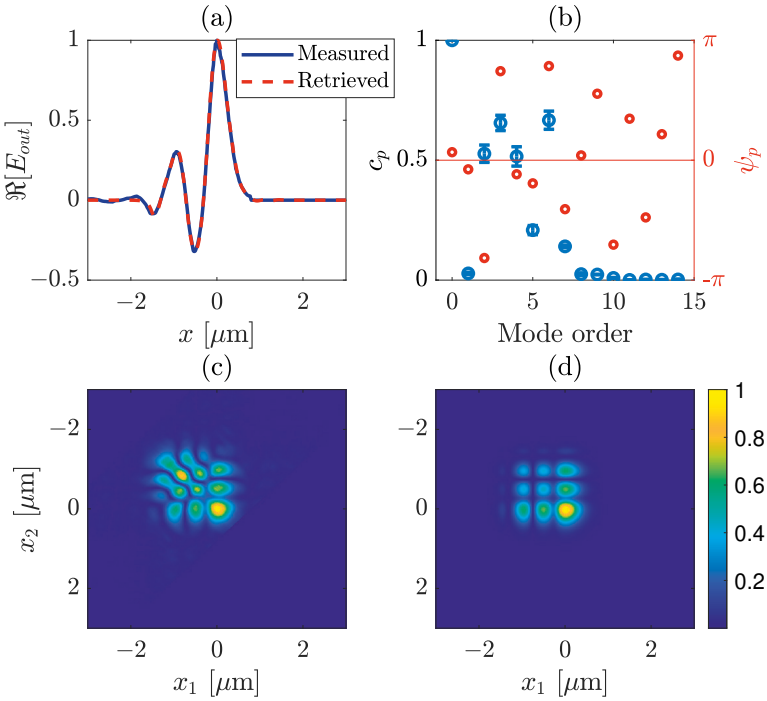

Figure 4.4: Mode retrieval results for a silicon nitride strip waveguide. In (a), the normalized real measured field is compared with the normalized real retrieved field. In (b), the retrieved mode strengths from the measured field and the corresponding phase of the modes are shown. (c) displays the measured cross-spectral density (CSD), and (d) shows the CSD reconstructed from the retrieved mode strengths.

sured CSDs may arise from the finite bandwidth of the laser and slight errors in the object plane alignment. This effect is illustrated in figure 4.6, where the CSD at the exit plane ($z = 0$) and at $z = 2\,\mu$m shows a slight stretching in the propagated CSD, explaining the discrepancies.

Error bars in the modal strengths represent one standard deviation from the mean, calculated from 20 independent realizations of numerically calculated electric field E_{out}. These error bars demonstrate the robustness of the method. Finally, from the retrieved phase distribution associated with each modal strength, the dispersion between different modes can be estimated, influenced by both in-coupling conditions and different mode propagation constants.

4.3 POLYMER-TITANIA HYBRID WAVEGUIDE

We present here the key results in the development of a novel polymer channel waveguide integrated with a titania core aimed at compact photonics circuitry for TM operation. We analyze the mode evolution and field confinements and bends design, confirming their suitability for evanescent field sensors. The hybrid structure demonstrates enhanced confinement and reduced propagation losses. Moreover it offers additional versatility in its use by increasing the degree of freedom in terms

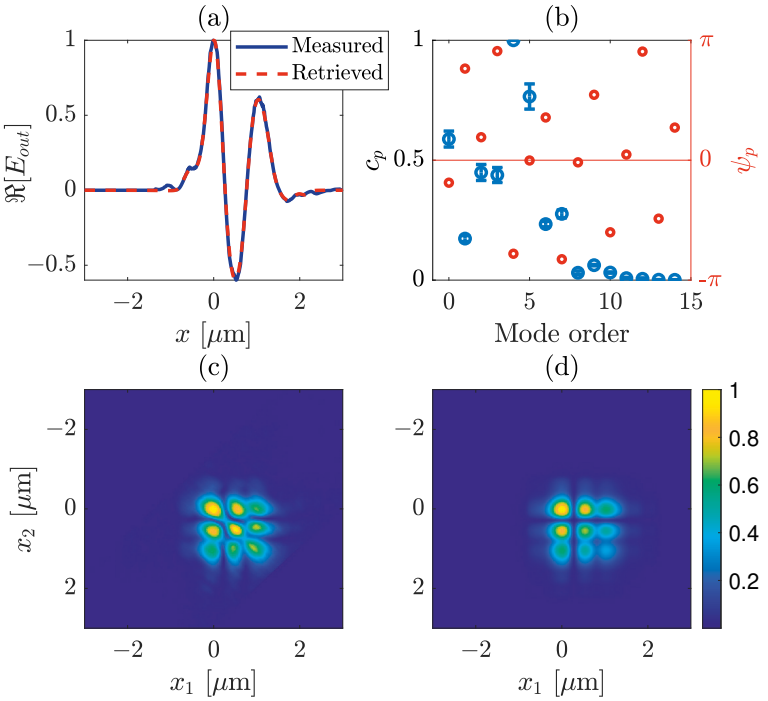

Figure 4.5: Mode retrieval results for a silicon nitride strip waveguide. In (a), the normalized real measured field is compared with the normalized real retrieved field. In (b), the retrieved mode strengths from the measured field and the corresponding phase of the modes are shown. (c) displays the measured cross-spectral density (CSD), and (d) shows the CSD reconstructed from the retrieved mode strengths.

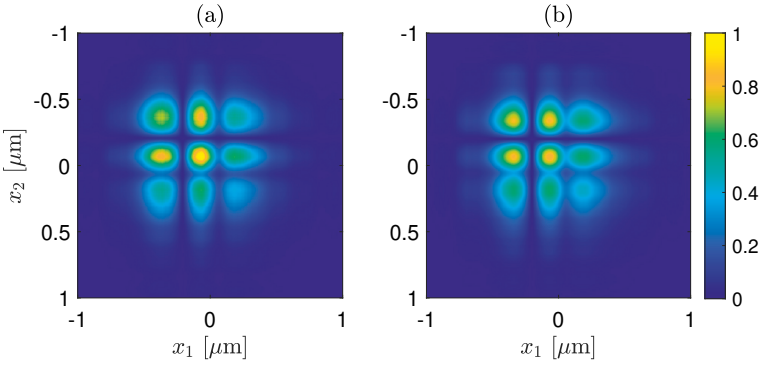

Figure 4.6: Illustration of the effect of propagation on the CSD. (a) At $z = 0$, and (b) at $z = 2\,\mu$m.

of geometry. The demonstration is done at the telecom wavelengths, although the waveguide can be scaled up or down to the entire VIS-NIR spectral range thanks to the transparency of the materials involved.

4.3.1 Design and simulation approaches

The waveguide cross-section is shown in figure 2.4. To ensure numerical convergence, a 20 nm mesh step size was used in all x, y, and z directions. Perfectly matched layer (PML) boundary conditions were applied to absorb any scattered fields [78,79]. The source wavelength was set to 1.55 μm. Optimizing waveguide parameters, especially to reduce propagation losses, involves achieving the best mode confinement (Γ) within the waveguide core. This refers to the percentage of the electric field magnitude maintained in the core [80]. For optimal confinement, the waveguide's thickness and width must be optimized.

Figure 4.7 shows the effective index evolution of possible sustained modes as a function of TiO_2 core width $2t$. Initial width (w) and height (h) were set to 0.8 μm and 0.9 μm, respectively, to ensure single-mode behavior in the vertical axis.

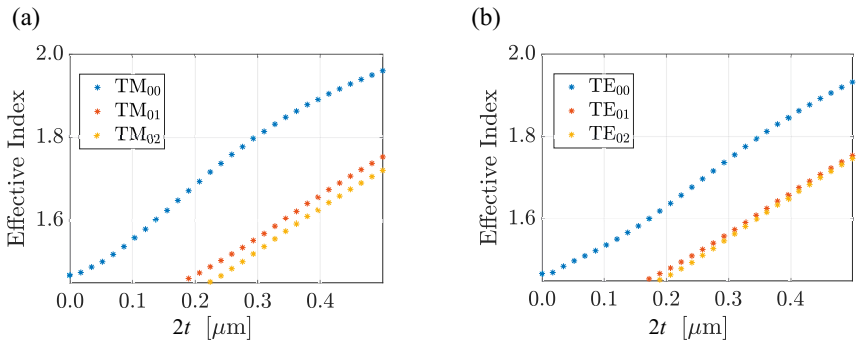

Figure 4.7: Effective index as a function of TiO_2 width $2t$ for (a) quasi-TM polarization and (b) quasi-TE polarization.

Figures 4.7(a) and 4.7(b) show the effective index dependence on TiO_2 thickness ($2t$) for the fundamental, first-order, and second-order modes for quasi-TM and quasi-TE polarization respectively. The effective index for both polarisation increases with $2t$. For TM polarization, the first-order mode appears at a width $2t =$ 195 nm. For TE polarization, the waveguide becomes multimode at $2t = 180$ nm. For a strictly single-mode waveguide, $2t$ values should be below these thresholds. However, due to fabrication restrictions with respect to the high aspect ratio which is requires, a slightly higher value ($2t$ = 200 nm) than the threshold has been chosen. It makes the waveguide slightly multimodal.

We then examined how the polymer geometry affects the effective index for both TM and TE fundamental modes by varying the nLoF width ($2w$) and height (h). The results of these variations are illustrated in figure 4.8.

From figure 4.9(a), it is evident that variations in nLoF width $2w$ have minimal impact on the effective index compared to variations in nLoF height. The mode field is primarily confined within the TiO_2 core. Figure 4.9(b) shows that effective guiding heights range from 0.7 μm to 1.5 μm. It is important to note that the confinement factor, which indicates the proportion of the optical power confined in the cladding and core, must also be considered. Therefore, deciding on the value of h based solely on effective index values is insufficient, especially for waveguides designed for evanescent field sensing applications. As h increases, the mode field becomes more confined in the core, leading to an increase in the effective index, but simulta-

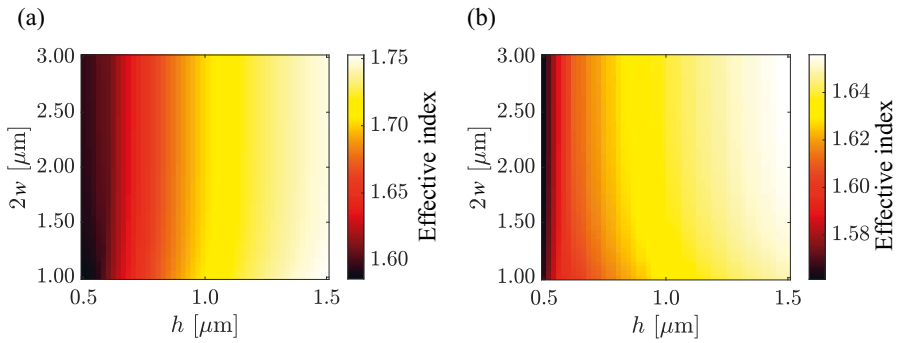

Figure 4.8: Effective index as a function of nLoF width w and height h for (a) quasi-TM polarization and (b) quasi-TE polarization.

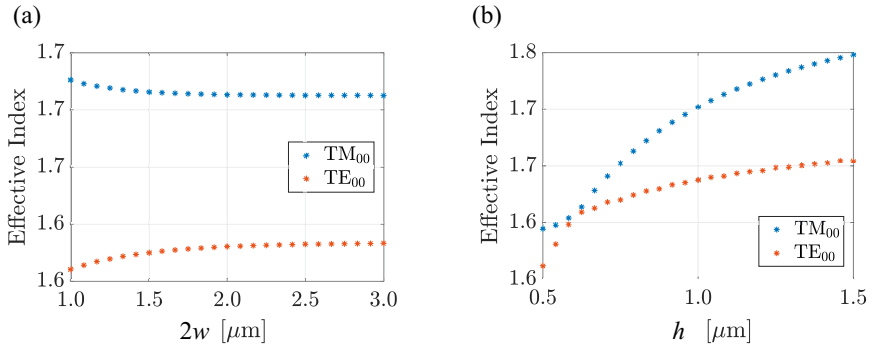

Figure 4.9: Effective index of fundamental quasi-TM and quasi-TE modes as a function of (a) nLoF width w and (b) nLoF height h.

neously, the confinement factor in the cladding, crucial for sensing, decreases, and vice versa.

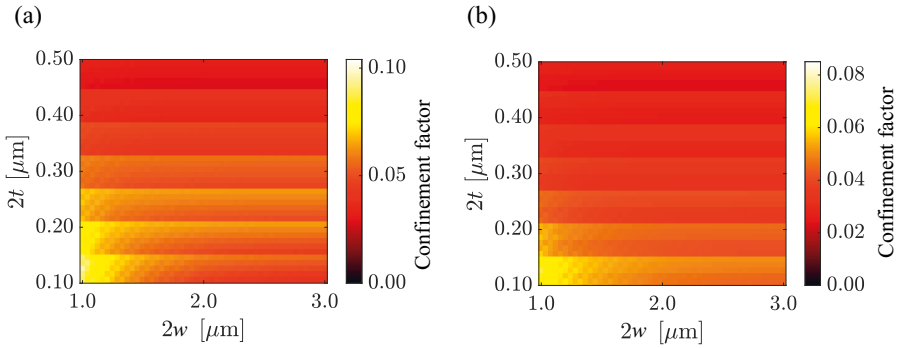

Figure 4.10: Confinement factor in the air cladding as a function of TiO_2 width $2t$ and nLoF width $2w$ for (a) quasi-TM and (b) quasi-TE polarization.

Figure 4.10(a) shows that the confinement factor in the cladding decreases with increasing $2t$ and $2w$ for the quasi-TM mode, dropping below 8% for $2t > 200$ nm

and below 7% for $2w > 1$ μm. Similarly, for the quasi-TE mode, the confinement factor decreases with increasing $2w$ (figure 4.10(b)). Therefore, the optimal nLoF height for sensing is below 1 μm, specifically in the range of 0.7 μm to 1 μm.

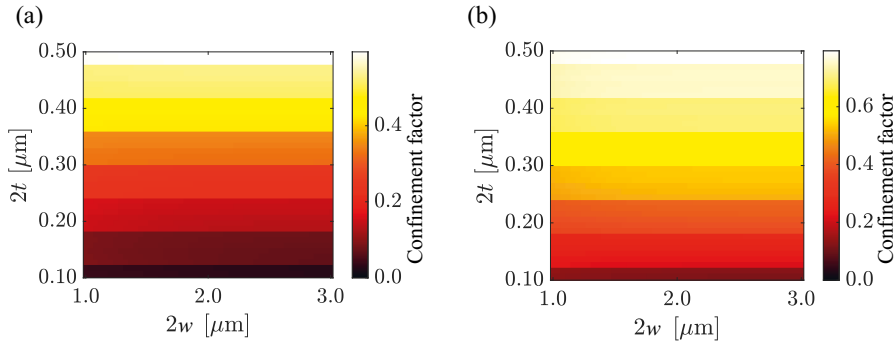

Figure 4.11: Confinement factor in the TiO_2 core as a function of TiO_2 width $2t$ and nLoF width w for (a) quasi-TM and (b) quasi-TE polarization.

Figures 4.11(a) and 4.11(b) illustrate the confinement factor in the TiO_2 core. In both figures, The mode field changes uniformly with the TiO_2 width up to a certain point, after which it adapts to the structure, resulting in the observed zigzag pattern in the confinement factor along $2t$ axis.

Based on the effective index and confinement factor calculations, we selected a $2t$ to be 200 nm, an $2w$ of 1800 nm, and h of 900 nm. This means t = 100 nm and w = 900 nm. Figures 4.12(a) and 4.12(b) show the electric field distribution for the fundamental quasi-TM and quasi-TE modes, respectively.

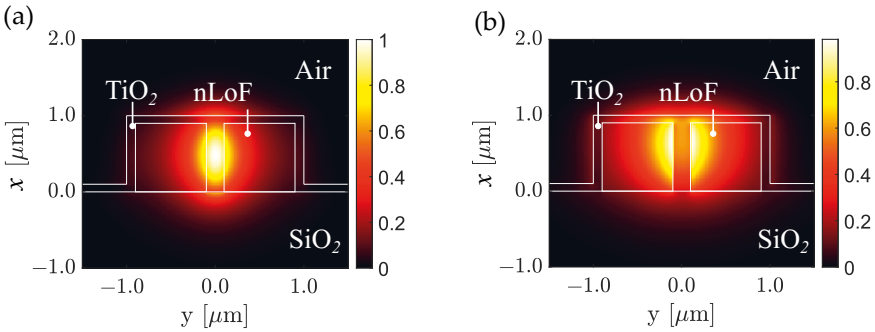

Figure 4.12: Mode distribution of the fundamental (a) quasi-TM mode and (b) quasi-TE mode. TiO_2 width $2t$ = 200 nm, w = 800 nm, and h = 900 nm.

The quasi-TM mode is highly localized inside the TiO_2 core region, while The quasi-TE mode is mainly confined in the polymer region. The quasi-TM mode is predominant and has the highest effective index. The confinement factor in the cladding in both polarisation enables sensing capabilities with optimised parameters.

The dispersion in the waveguide is comparable to other waveguides used in silicon photonics in the C-band (figure 4.13(a)) [81]. The high birefringence (close to 0.1 refractive index units) could enable some polarization management in the

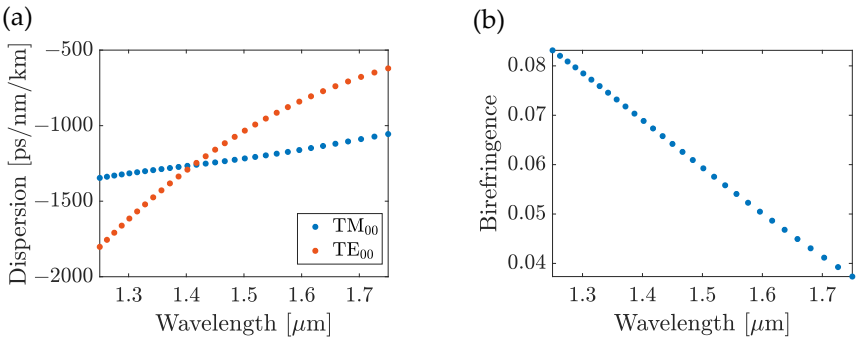

Figure 4.13: Mode distribution of the fundamental (a) TM mode and (b) TE mode. TiO_2 width $2t$ = 200 nm, w = 800 nm, and h = 900 nm.

waveguide (figure 4.13(b)). The quasi-TM fundamental mode shows low dispersion variation within the wavelength range studied, which makes the concept suitable for broadband devices [82, 83].

To address the challenge of increased footprint with longer sensing regions, some designs use bends and spirals instead of linear extensions. High index contrast waveguides allow for tight bends with low radiative losses. Our hybrid waveguide can move the TiO_2 trench horizontally to enable smaller bend radii and reduced bend loss. Figure 4.14(a) and 4.14(b) show the TM_{00} and TE_{00} mode profiles along the y-direction as a function of trench position y_T. The mode distribution remains well-confined despite being off-centered, which shows indirectly a low influence on the width of the polymer.

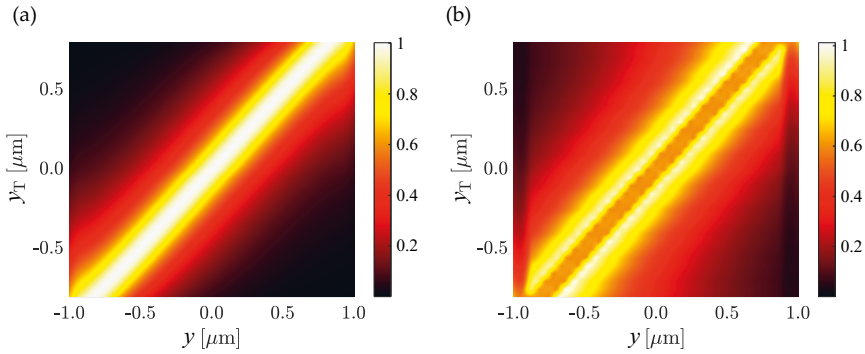

Figure 4.14: (a) TM_{00}-mode profile ($|E_x|$) and (b) TE_{00}-mode profile ($|E_y|$) along the y-direction as a function of the trench position y_T. TiO_2 width $2t$ = 200 nm, w = 800 nm, and h = 900 nm.

Bend loss is critical for photonic circuitry, affecting device footprint and integration. Simulations for different trench positions (0 nm, 400 nm, 600 nm) and bend radii R between 1 μm and 20 μm show that shifting the trench help reducing bend loss. Figure 4.15(a) and 4.15(b) illustrate these variations, revealing negligible losses when the trench is shifted only in the bend portion.

Figure 4.16 shows the TM fundamental mode propagation for y_T=0 nm in the straight section and y_T = 400 nm in the bend section for (a) single bend and (b)

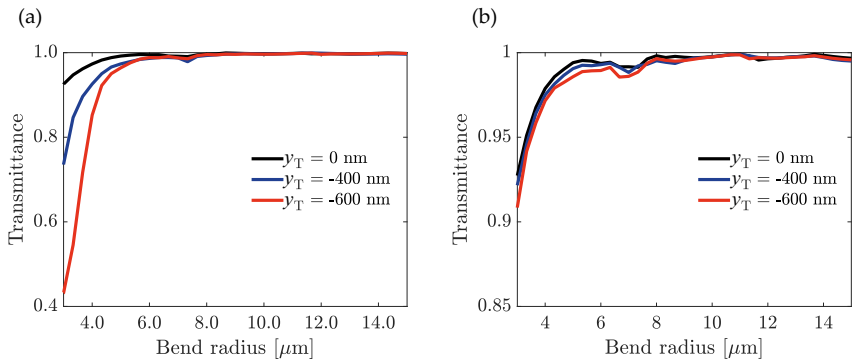

Figure 4.15: The transmission of the TM_{00} mode after a U-turn bend is shown for three horizontal displacements: y_T=0 nm (black), y_T=-400 nm (blue), and y_T=-600 nm (red). The results are presented for two cases: with y_T varying along the entire U-turn bend waveguide, and (b) with y_T changing only in the bend region, while y_T is kept at 0 for the remaining sections of the waveguide

multiple bends. The mode remains well-confined and stable, with minor differences in straight and bend sections.

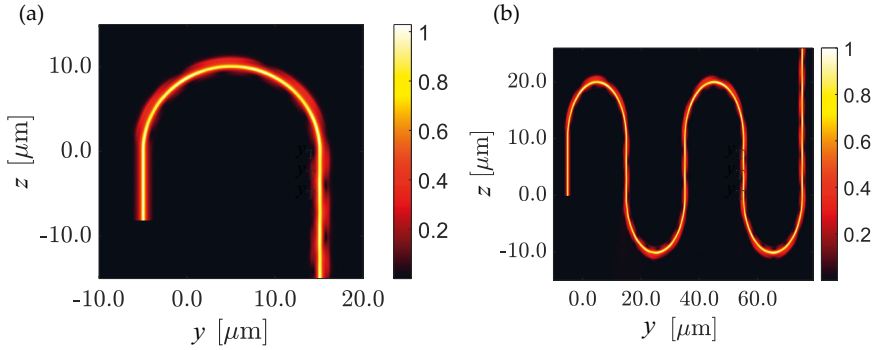

Figure 4.16: TM_{00} mode propagation for y_T=0 nm in the straight section and y_T = 400 nm in the bend section for (a) single bend and (b) multiple bends.

These insights are highly valuable for designing waveguides in photonic circuits, particularly if bends are unavoidable, such as in compact optical sensors or signal routing across components. Understanding the impact of bending on mode shape and losses directly affects photonic device performance.

4.3.2 Fabrication process flow and device

The fabrication starts by spin coating nLoF resist on an oxidized silicon wafer spinning at 1100 rpm. Using a hot plate at 100 °C, the thin layer is soft baked for 1 min. Using EBL, the predefined waveguide structure is transferred on to the resist layer with 40 $\mu C/cm^2$ exposure dose and 100 kV acceleration voltage. The sample is hard-baked for 1 min at 110 °C after exposure. The pattern is then developed for 90 s in AR 300-47, then rinsed for 30 s in de-ionized (DI) water. The areas exposed

remain after development since nLoF is a negative tone resist. We then further deposit 100 nm layer of TiO_2 on the sample using ALD. This is done at a temperature of 120 °C and at a rate of deposition of 0.0718 nm per cycle. Figure 4.17 illustrates a simplified process flow for fabricating the hybrid waveguides.

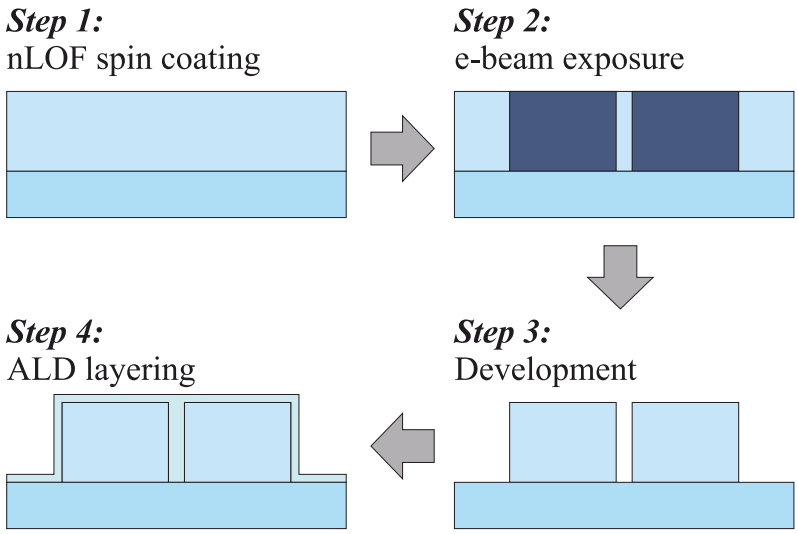

Figure 4.17: Process flow for waveguide fabrication. Step 1: deposition of resist on an oxidized silicon wafer; Step 2: electron-beam patterning of the nLoF resist; Step 3: development of the pattern and Step 4: atomic layer deposition of titania [35].

Bends with different radius of curvature were fabricated. To enable efficient light coupling, we implemented a tapering part at both the input and output ends, employing a 3 μm wide taper with a length of 500 μm at the both ends. Figure 4.18(a) presents a SEM image of the waveguide cross-section before the ALD coating. Despite some trench footing, likely due to limitations in resist performance during fabrication, the geometry of the nLoF polymer channel is properly maintained as designed. Figure 4.18(b) shows the cross-section after TiO_2 deposition, which has been cleaved within the waveguide taper section. The TiO_2 film conformally coats the structure, with a slight thickness reduction above the trench due to edge rounding during the ALD process [84]. To address this, an additional 10 nm ALD coating was applied to fill the gap. Contamination, likely from residual particles, chemical residues, or impurities during the fabrication and deposition processes, is noticeable at the top of the waveguide both before and after TiO_2 deposition. Figure 4.18(c) depicts the bent waveguides used for loss measurements, highlighting the precise positioning of the trench within the waveguide.

4.3.3 Analysis of propagation losses

Our main objective in this study was to demonstrate a waveguide where the quasi-TM fundamental mode predominates. Consequently, all characterizations were conducted with TM polarization. The schematic diagram in Fig. 4.1c illustrates the setup used to measure propagation loss. Some waveguides could not be measured due to damages incurred during fabrication, cleaving, or handling. Figure 4.19(a)

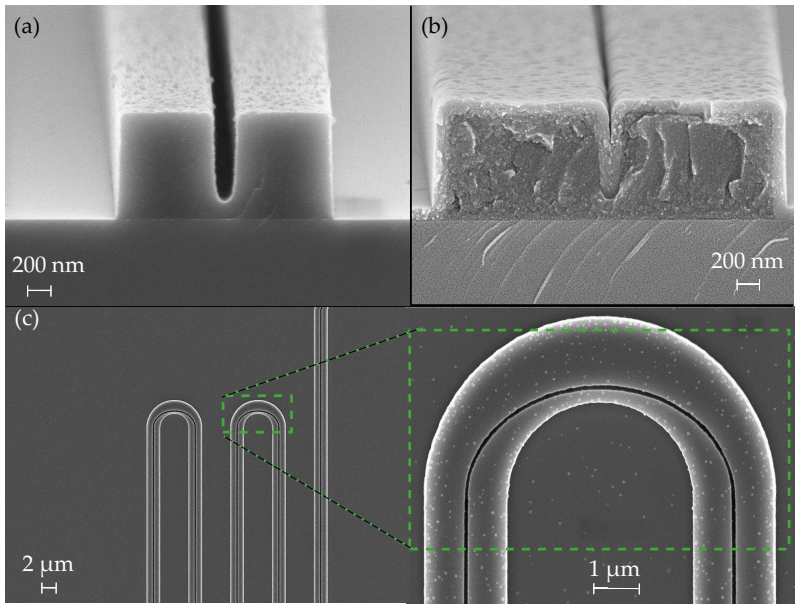

Figure 4.18: Scanning electron microscope (SEM) images of fabricated waveguides. (a) Cross-section before ALD coating. (b) Cross-section after ALD coating. (c) Top view of a multi-bend waveguide with a zoom-in on the bend section before TiO_2 deposition.

shows the captured output image for the fundamental quasi-TM mode of the waveguide operating at 1550 nm using schematic diagram in figure 4.1(b). The large size of the mode is mainly due to the use of the tapers at the output.

The propagation losses of the waveguide were measured using the cutback method [85]. This is done by assuming that there is consistent insertion loss across all the fabricated samples. The normalized output power were measured for different length of the waveguide. The measurement was repeated 5 times and the uncertainty in the measurement was estimated as shown in figure 4.19(b). The slope of the measured data gives the propagation losses of the waveguide. The measured propagation loss was 1.75 $\pm$ 0.32 dB/cm which is very close to the state of the art for titania nanowaveguide [86–88].

Additionally, we evaluated the transmission through waveguides with different bend radii. The transmission remained stable (with an error margin similar to that of the propagation loss measurements) down to a bend radius of 7 μm, as expected theoretically, before experiencing a sharp drop to non-measurable values [35].

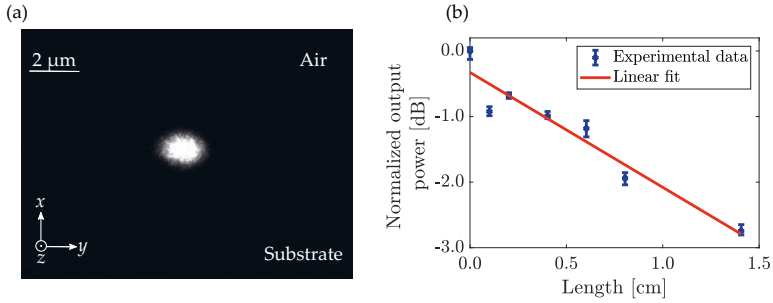

Figure 4.19: (a) Image of the fundamental quasi-TM mode at the output of a waveguide with a 5 μm bend radius. (b) Plot showing normalized output power versus waveguide length (blue circles) with a linear fit (red line) based on the cut-back method. The propagation losses are calculated to be 1.75 $\pm$ 0.32 dB/cm.

5 Strip-loaded Mach-Zehnder interferometer

In this chapter, we explore the key results in a strip-loaded Mach-Zehnder interferometer (strip-loaded MZI) used for refractive index sensing, as detailed in the third paper. It includes the design and simulation processes, fabrication steps, and practical application for measuring ethanol concentration in a water-ethanol mixture. We begin with the waveguide optimization, the functional design of the Y-branch splitter, and specific design considerations for mainly the sensing region. The chapter then details the fabrication process and key results from both fabrication and experimentation.

5.1 CONCEPTUAL DESIGN AND THEORETICAL BACKGROUND

Figure 5.1(a) illustrates the cross-section of the strip-loaded MZI input, while Figure 5.1(b) presents the 3D configuration of the entire strip-loaded MZI.

Designing MZI, particularly with strip-loaded waveguides, poses unique challenges. Although strip-loaded waveguides are highly effective at confining light and minimizing loss, their design inherently limits the interaction between the guided light and the external analyte. This makes evanescent sensing difficult because only a small portion of the light extends beyond the waveguide core as shown in figure 5.1(a). The presence of the low-index strip further reduces this light-analyte interaction, which is critical for sensing applications where a strong interaction is necessary to detect changes in the refractive index.

The strip-loaded MZI design overcomes this limitation by introducing a deliberate opening in the sensing arm, which enhances the interaction between light and the surrounding analyte. This structural modification allows a larger portion of the guided light to interact with the liquid analyte, improving the measurement of refractive index changes. In this design, the analyte itself serves as the strip material in the sensing region, significantly increasing light-matter interaction within a controlled section of the sensor arm. Meanwhile, the rest of the device remains shielded from environmental fluctuations, preserving its overall stability.

The key advantages of this design include enhanced sensitivity due to increased evanescent tail exposure, alongside a larger available surface area for functionalization. These features make it particularly well-suited for highly sensitive biochemical sensing applications. The strip that facilitates mode propagation in the TiO_2 slab is constructed from nLoF polymer, with a refractive index of 1.601 at a wavelength of 1550 nm. The entire circuit, including the reference and sample arms, as well as the input and output waveguides, is fabricated using nLoF electron beam resist. Since almost no light escapes above the strip, no protective layer is required, allowing the analyte to cover the entire sample without interfering with the measurement.

5.2 SIMULATIONS

To ensure precise results, the simulation boundaries were set with PML to absorb scattered fields and prevent reflections. A 20 nm mesh size in all directions balanced

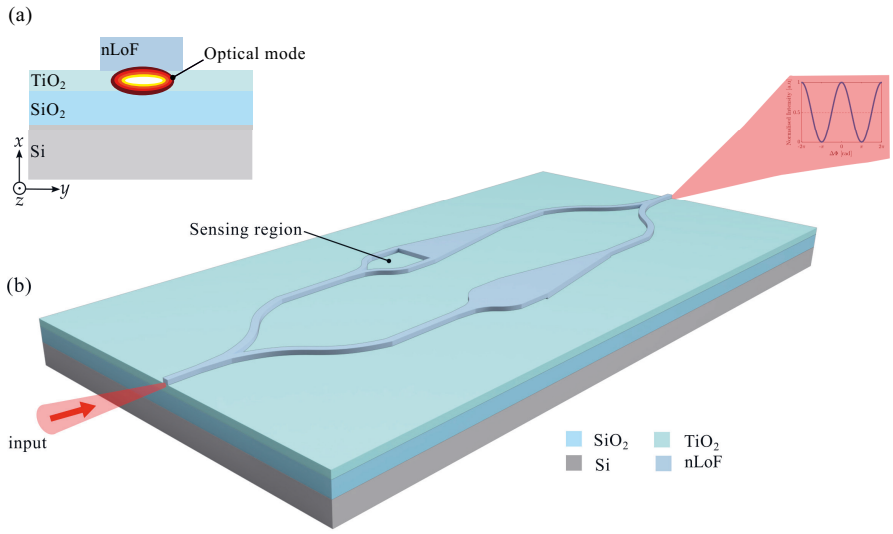

Figure 5.1: The configuration of the strip-loaded Mach-Zehnder Interferometer. (a) A cross-sectional view of the input waveguide, describing the material layers. (b) A 3D representation of the strip-loaded Mach-Zehnder Interferometer, with the sensing region clearly labeled. The field in the reference arm remains unaffected as it propagates through the high-index TiO_2.

numerical accuracy and computational efficiency, with an illumination wavelength of 1550 nm. To fully realize the MZI, an efficient design of all its elements is required, as described in Section 2.4. For clarity, we present the design in SEGMENTS and focus on the main considerations made in each SEGMENT as shown in figure 5.2.

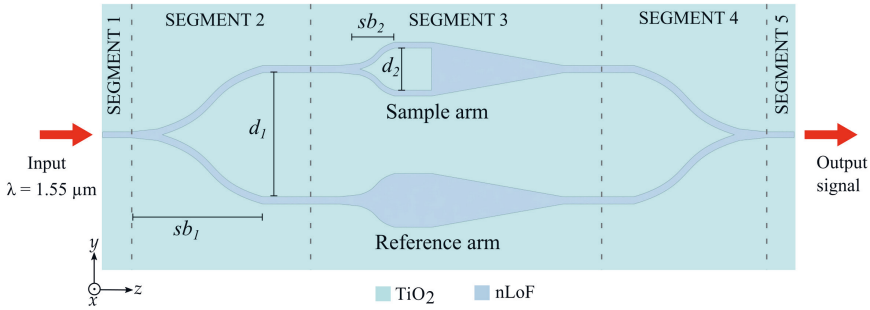

Figure 5.2: Top view of the strip-loaded MZI showing different sections of the device. SEGMENT 1 is the waveguide design, SEGMENT 2 involves splitter design, SEGMENT 3 involves sensing arm and reference arm design and SEGMENT 4 is the combiner while SEGMENT 5 is also a waveguide

SEGMENT 1 (Section 5.2.1) involves the design and optimisation of an appropriate waveguide geometry needed for both guiding and sensing. SEGMENT 2 (Section 5.2.2) involves the design of the Y-branch splitter, SEGMENT 3 (Section 5.2.3) involves the the design of the sensing region and the SEGMENT 4 and 5 involves a

combiner and a waveguide which is already presented in SEGMENT 1 and 2.

5.2.1 Waveguide optimization

In a Mach-Zehnder interferometer designs, the dimensions and materials of the sensor are chosen to ensure single-mode operation. Single-mode conditions are necessary for the MZI to eliminate ambiguity in mode interactions with the electric field. The design process begins by selecting a waveguide structure that supports only the fundamental mode, either TM or TE. After determining the analyte and waveguide configuration, the next step is optimizing the waveguide serving as the primary guide in the interferometer. This involves determining the appropriate thickness and desired polarization. In addition to achieving monomodal behavior, the waveguide must have low losses and effective mode confinement. Therefore, there is always a trade-off between efficient light confinement in the core and interaction with the analyte in the cladding for evanescent sensing, as established in Chapter 4.

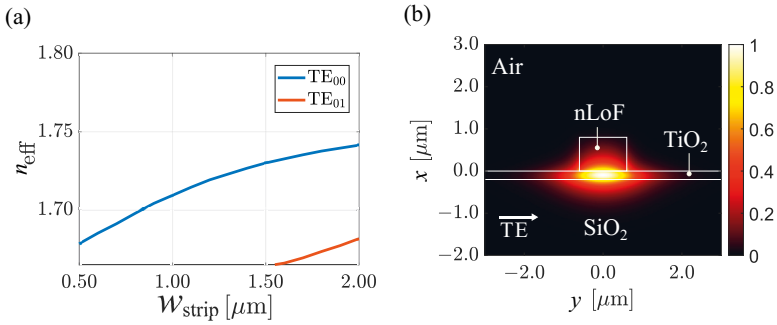

Figure 5.3: Optimization of the strip-loaded waveguide. (a) The relationship between the effective index n_{eff} and the width of the loading strip w_{strip} for the TE_{00} and TE_{01} modes. The nLoF polymer thickness h_{strip} and TiO_2 thickness t_f are fixed at 800 nm and 200 nm, respectively. (b) Mode field distribution of the fundamental TE_{00} mode at a loading strip width $w_{\text{strip}} = 1200$ nm.

We studied the effective index variation and confinement factors for the fundamental TE_{00} mode and the first-order TE_{01} mode in order to optimise the geometry for a single mode condition. Figure 5.3(a) shows that the effective index of both modes increases with the strip width (w_{strip}) when the strip height (h_{strip}) is kept at 800 nm and the TiO_2 thickness (t_f) at 200 nm. Choosing the appropriate waveguide thickness for evanescent sensing involves considering both the effective index and the electric field confinement in the cladding and analyte. When the strip width is less than 1500 nm, the waveguide supports only the fundamental TE_{00} mode. Therefore, we set the strip width w_{strip} to 1.2 μm to ensure proper mode confinement while also accommodating for fabrication tolerances. Figure 5.3(b) shows the electric field magnitude for the fundamental TE mode.

5.2.2 The Y-branch splitter

The Y-branch splitter primary function is to divide the incoming light power into two equal halves [89, 90]: one leading to the sensing arm and the other to the refer-

ence arm.. Uneven splitting can affect performance by altering the sensing length.

In designing the splitter, a Bézier curve is used [91]. This curve provides flexibility and is defined by the S-bend length (sb_1) and the distance between the two arms (d_1) as shown in figure 5.2. As the S-bend length increases, the control points are adjusted according to the S-bend equation, decreasing the angle between the two arms. The Bézier curve equation is given as

$$C(u) = \sum_{k=0}^{m} N_{k,m}(u) P_k, \tag{5.1}$$

where the function $C(u)$ describes the Bézier curve and u is a parameter that varies from 0 to 1. The control points of the Bézier curve are represented as P_k, and m is the degree of the Bernstein polynomial described as $N_{k,m}(u)$, and is given as

$$N_{k,m}(u) = \binom{m}{k} u^k (1-u)^{m-k}, \quad k = 0, 1, 2, \ldots, m, \tag{5.2}$$

where $\binom{m}{k}$ is the binomial coefficient, u is the parameter along the curve, and k is the index of the control point. The symmetric half of the splitter is defined by 4 control points, making it a cubic Bézier curve. This can be expanded as

$$R(u) = P_0 N_{3,0}(u) + P_1 N_{3,1}(u) + P_2 N_{3,2}(u) + P_3 N_{3,3}(u), \tag{5.3}$$

or equivalently,

$$R(u) = P_0(1-u)^3 + P_1 3u(1-u)^2 + P_2 3u^2(1-u) + P_3 u^3, \tag{5.4}$$

where $R(u)$ is the position on the curve at parameter u, and P_0, P_1, P_2, P_3 are the control points for the cubic Bézier curve. The terms $N_{3,k}(u)$ represent the Bernstein polynomials for a cubic Bézier curve.

The gap between the arms of the splitter must also be considered to avoid cross-talk. While wider gaps can reduce cross-talk, they also increase the device's footprint and may reduce the MZI's overall response. Figure 5.4 illustrates the simulation results of the splitter. Figure 5.4(a), shows that the transmittance generally increases with the S-bend length. However, for $sb_1 > 150$ μm, transmittance stabilizes and does not significantly change with further increase in S-bend length, given a fixed d_1 of 50 μm between the arms.

Figure 5.4(b) displays the magnitude of the electric field profile for the Y-branch splitter, showing precise splitting of the input light.

5.2.3 Sensing region

The phase change induced in an MZI depends significantly on the length of the sensing region, often referred to as the interaction length. In our MZI design, we ensure both the reference arm and the sensing arm are of equal length.

In our strip-loaded waveguide, the mode is buried beneath the strip, which prevents the evanescent field from extending above the strip and interacting with the analyte. While this buried configuration offers protection against environmental influences, it also limits the waveguide's ability to sense changes in the surrounding medium. To address this limitation, we applied a reverse technique from the Y-branch splitter development, allowing the evanescent field propagating in the TiO_2 layer to be exposed to the analyte for effective sensing.

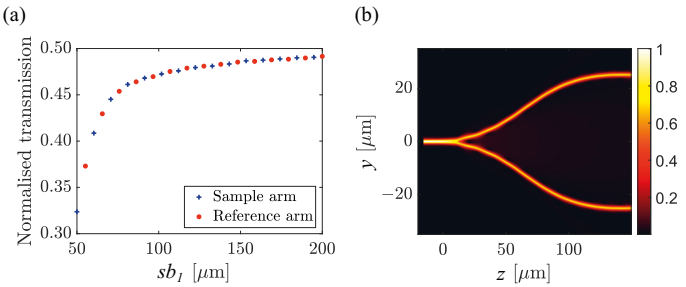

Figure 5.4: Splitter design results. (a) Normalized transmittance as a function of the S-bend length (sb_1) at a fixed gap (d_1) of 50 μm. (b) Electric field profile for the splitter at sb_1 of 150 μm and a gap d_1 of 50 μm.

To achieve the desired design, we adjusted the S-bend length sb_2 and the distance between the two arms d_2. Figure 5.5 shows a simulation of the sensor region design. We observed earlier in the Section 5.2.2 that, for a Y-branch splitter, transmittance increases with a longer S-bend length and vice versa. For this reason, in the sensing region design, we reduced the sb_2 until no light split under the arm, forcing the beam to continue straight in the TiO_2 material in the open area.

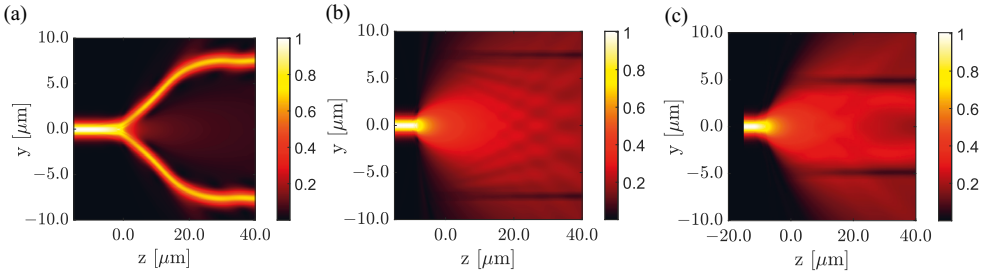

Figure 5.5: Electric field distribution in the sensing region viewed from above: (a) S-bend length (sb_2) of 40 μm and a gap d_2 of 15 μm. (b) sb_2 of 5 μm and d_2 of 15 μm. (c) sb_2 of 5 μm and d_2 of 10 μm.

Figure 5.5(a) shows an S-bend length (sb_2) of 40 μm and d_2 of 15 μm. Figure 5.5(b) shows an sb_2 of 5 μm and d_2 of 15 μm, showing diverging field behavior in the TiO_2 region. Figure 5.5(c) shows an sb_2 of 5 μm and d of 10 μm, showing a finer field in between d_2. Reducing d_2 ensures the mode field does not diverge significantly before reaching the taper section. With the evanescent field exposed, we can calculate the confinement factor in the sensing region.

Figure 5.6(a) shows confinement factor calculations for different cladding materials (nLoF, air, water, and ethanol) as a function of TiO_2 thickness (t_f). Generally, the confinement factor decreases with increasing t_f because the mode becomes more confined in the TiO_2 and less in the cladding. At a thickness of 200 nm, the confinement factor in the cladding for both water and ethanol is about 23% with ethanol slightly higher than water. Figure 5.6(b) shows confinement factor calculations using a mixture of ethanol and water refractive indices, showing a decrease in confinement factor values with increasing ethanol concentration.

Light from the sensing region is guided back under the strip using a taper. The

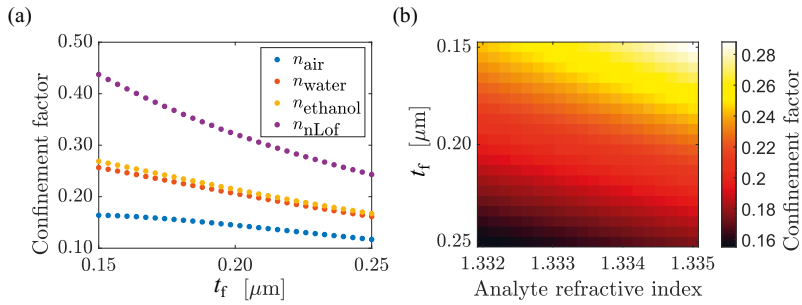

Figure 5.6: Confinement factor calculations in the sensing region: (a) Calculations of the confinement factor in the cladding of the sensing region with different cladding materials (air, nLoF, ethanol, and water) as a function of TiO_2 thickness t_f. (b) Confinement factor calculations in the cladding using the refractive index range of the ethanol-water mixture as the analyte, also as a function of t_f.

transmittance of the taper is proportional to its length, longer tapers are thus preferred. As light propagates through a taper, its mode gradually adjusts to the changing waveguide geometry. In a long taper, this transition happens more gradually, allowing the light to remain in its fundamental mode with minimal scattering or mode mismatch [92]. To account for the low index contrast of the strip-loaded waveguide, the taper length was set to 2 mm. To enhance light and matter interaction, the length of the sensing region was set to 5 mm. The strip in the reference arm remains nLoF and shares the same geometry as the sensing arm, ensuring an equal path length for the split beam. The combiner in SEGMENT 4 has the same sb_1 of 150 μm and gap d_1 of 50 μm as the splitter. The output waveguide in SEGMENT 5 is also the same design as in SEGMENT 1. The overall length of the chip is 1.33 cm, excluding the input and output tapers.

5.3 DETAILS OF THE FABRICATED DEVICE

Three-inch oxidized silicon wafers with a 3 μm thermal oxide layer (Si-Mat, Kaufering, Germany) serve as substrates for this process. Initially, to remove surface impurities, an O_2 plasma cleaning is performed using a PE-50 plasma cleaner as shown in figure 5.7. By using an ALD, a 200 nm film of TiO2 is deposited. This leads to TiO_2 amorphous layer at 120 °C. The wafer undergoes another O_2 plasma treatment after ALD deposition. Next, we spin-coat onto the wafer approximately 800 μl of AZ 2070 nLoF and spun for 60 s at 1500 rpm. The wafer is then soft-baked for 60s on a hot plate at 110 °C. A profilometer (Dektak 150) measures the final resist coating thickness as 799 nm. The subsequent step involves EBL with doses of 40 $\mu C/cm^2$ and 60 $\mu C/cm^2$. After exposure, the wafer is post-baked for 60 s at 110 °C and developed for 90 s in AR 300-47 (Allresist GmbH). The development is stopped and rinsed with deionized water for 30 seconds and it is blown dry using nitrogen. Finally, the wafer is manually cleaved into individual chips for further characterization.

Figure 5.8 shows an SEM images of the key components of the fabricated MZI. Zoom-ins are also shown to appreciate the finest details of the structure. Figure 5.8(a) shows the Y-branch splitter with a long sb_1. Figure 5.8(b) depicts the entry point of the sensing region with a short sb_2. Figure 5.8(c) illustrates the boundary

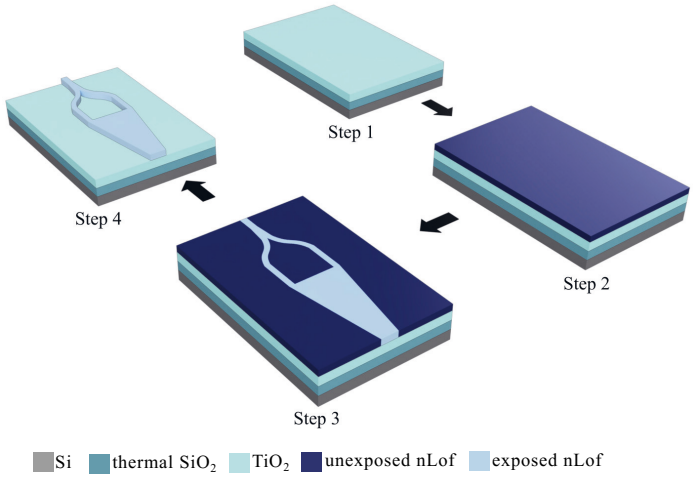

Figure 5.7: Step 1: The fabrication process begins with coating the substrate with a 200 nm layer of TiO_2 using ALD. Step 2: a layer of nLoF resist is applied by spin coating. Step 3: The pattern is written onto the nLoF resist using EBL. Step 4: the nLoF resist is developed, the wafer is cleaned, and the chips are cleaved from the wafer.

between the sensing region and the taper section, highlighting that one arm of the device has an opening while the other does not.

Figure 5.8(a) presents the SEM image of the splitter section of the MZI, zooming in on the Y-junction. Figure 5.8(b) displays the start of the splitter section, while figure 5.8(c) shows the end of the sensing region and the beginning of the splitter for both the reference and sensing arm.

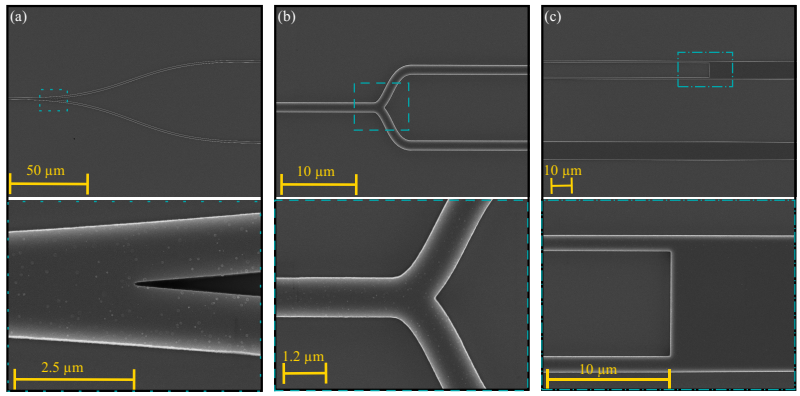

Figure 5.8: Top-view scanning electron microscope (SEM) images of various sections of the MZI: (a) A Y-branch splitter with a close-up view of the arm separation. (b) The start of the sensing region, including the sample arm section. (c) The end of the sensing region and the beginning of the taper section.

5.4 ETHANOL CONCENTRATION MEASUREMENT

5.4.1 Analyte sample preparation

Ethanol-water solutions were prepared with mass concentrations of c_1 = 0.00178 wt%, c_2 = 0.0167 wt%, and c_3 = 0.178 wt% [93]. Ethanol-water mixtures were precisely measured with a Sartorius Entris analytical balance (ENTRIS224i-1S/0.1mg) to ensure that each solution had a total mass of 100 g. Each mixture was thoroughly homogenized. The refractive indices at $\lambda = 1550$ nm for pure water and ethanol were measured as 1.3154 and 1.3520, respectively. As the concentration of ethanol in the solution increases, the refractive index of the ethanol-water mixture also rises. The resulting change in refractive index can be determined using the formula $\Delta n_c = 0.0006 \times C_{eth}$, where Δn_c represents the variation in refractive index and C_{eth} is the mass concentration of ethanol [94, 95]. Once Δn_c is calculated, the corresponding refractive index of the solution can be estimated.

5.4.2 Experimental characterization

The MZI was characterised using the setup shown in figure 4.1(a) and (b). Key components such as the splitter were also tested to ensure effective splitting of the beam. A splitter with the same specification as the splitter in the MZI were fabricated to test the splitting and to capture the output mode images. In figure 5.9(a), the output image of the fundamental TE mode from the splitter is captured using a near-infrared camera, showing how the splitter performs. Figure 5.9(b) presents the output of the entire MZI device, also imaged with a near-infrared camera with air as the cladding material.

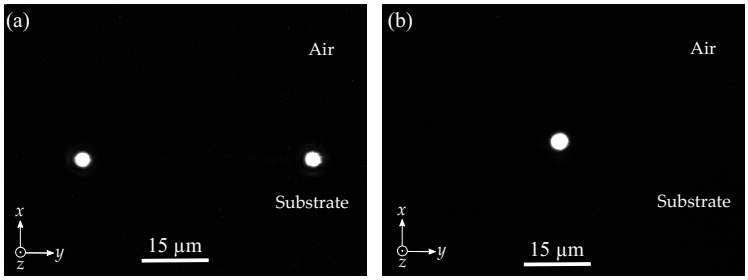

Figure 5.9: Characterization results: (a) The output of the TE fundamental mode from the Y-branch splitter, which has an sb_1 of 150 μm and d_1 of 50 μm separation between the arms. (b) The output of the TE fundamental mode from the strip-loaded Mach-Zehnder interferometer, with air as the cladding material in the sensing region.

The experiment process began by establishing a baseline. The MZI chip was carefully mounted onto a translational stage, secured with custom-built sample holders to ensure stability throughout the experiment. At this stage, the chip was exposed to air with no solution applied. Power monitoring was initiated using Thorlab powermeter software to record baseline measurements of the chip under air-cladding conditions. These measurements served as a reference for comparison with subsequent experimental steps.

Next, de-ionized water was carefully deposited onto the MZI chip using a separate pipette designated solely for the control solution. The water was allowed to sit on the chip's surface for waiting time of few minutes minutes, after which it was removed using the same pipette by suction. This process was repeated three times to ensure measurement accuracy and consistency. The control phase established the sensor's behavior when exposed to de-ionized water.

After the control measurements were completed, the first of the three prepared solutions was applied to the same MZI chip using a separate pipette assigned to that solution. The solution was left on the chip for approximately two minutes and then removed. This process was repeated for the second and third solutions, with each solution handled using its respective pipette. All measurements were taken under identical conditions to maintain consistency and enable accurate comparisons between the solutions and the control.

Throughout the experiment, a TEM light source with TE polarization fractions (17.73 mW/49.96 μW) was injected into the input of the Mach-Zehnder Interferometer sensor. Tunable lasers and polarization controllers were utilized to ensure precise control over the light. The sensor's output was captured using a near-infrared camera, and the output power was recorded for subsequent analysis. Upon completing the experiment with all solutions, the data were saved in an Excel file. To ensure repeatability, the experiment was conducted three times using three different fabricated chips. Figure 5.10 illustrates the real-time power measurements obtained for the two MZI sensors.

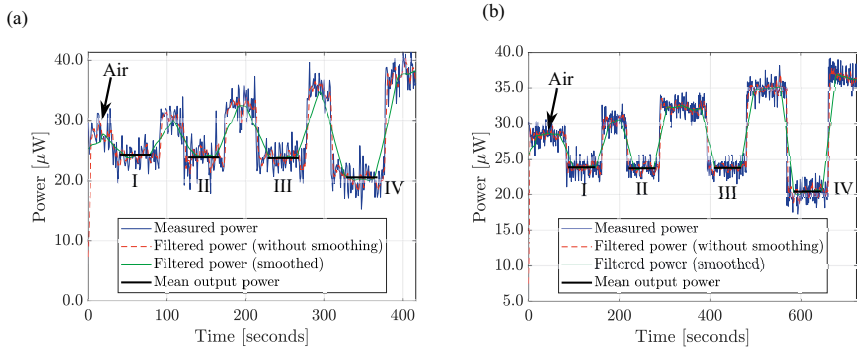

Figure 5.10: Characterization results: Real-time output power measurement for two MZIs with identical experimental conditions, using the same ethanol-water mixtures and different waiting times. Red dashed lines (4-point moving average) and green lines (30-point moving average) represent filtered results, while the black dashed line indicates the mean output power due to changes in the analytes. The waiting time between deposition and removal of analyte is shorter in (a) than in (b). I, II, III, and IV represent deionized water, c_1, c_2, and c_3 respectively.

The recorded power data were analyzed to identify distinct high and low peaks corresponding to changes in the analyte concentration and air. A filtering process, using a 4-point moving average, was applied as a low-level smoothing filter to minimize high-frequency noise and small fluctuations while maintaining the integrity of the signal. Subsequently, a 30-point moving average was used for further smoothing. The findpeaks function was employed to detect high and low peaks in the smoothed data, with parameters set for minimum peak prominence and distance to

ensure only significant peaks were detected. The mean peak power values for the analyte were calculated to quantify the variations, as depicted in figure 5.10. The sensor's response to changes in analyte refractive index was consistent across all measurements. Notably, the mean power values were similar across the different MZI sensors, indicating that the experimental conditions remained stable and the analyte was not contaminated during the measurements.

As shown in Figures 5.10(a) and (b), the MZI exhibited a response time of approximately one second to changes in the refractive index of the ethanol-water mixture. However, this response time may be shorter, as it is limited by the powermeter's integration time, which is also around one second. The results presented in Figures 5.10(a) and (b) demonstrate that the MZI responded promptly to variations in the refractive index of the analyte.

In all measurements, It could be observed that the output power did not return to the baseline after the ethanol-water mixture was removed from the sensing region. This residual power shift is likely caused by a thin layer of the analyte remaining on the sensor surface, which alters the effective refractive index in the sensing area, preventing the output power from returning to its original baseline. Additionally, it was observed that this shift increased as the concentration of ethanol in the mixture increased.

To ensure accurate measurements, the ethanol-water mixtures were introduced in increasing concentrations. This procedure ensured that the output power corresponded to the current refractive index change and was not influenced by any residual analyte from previous measurements.

Figure 5.11(a) shows the experimental mean output power as a function of the analyte's refractive index, with a sine curve fitted using Eq. (2.28). Simultaneously, the corresponding phase change values were determined using the same equation. The change in effective index was calculated with Eq. (2.29), and figure 5.11(b) shows the effective index change as a function of the analyte refractive index change, with a linear fit applied. The slope of the fit represents the sensor's sensitivity, determined as 0.128, with a correlation coefficient (ρ) of 0.9987, indicating a strong fit. The detection range of the sensor is calculated to be 1.1×10^{-3} RIU.

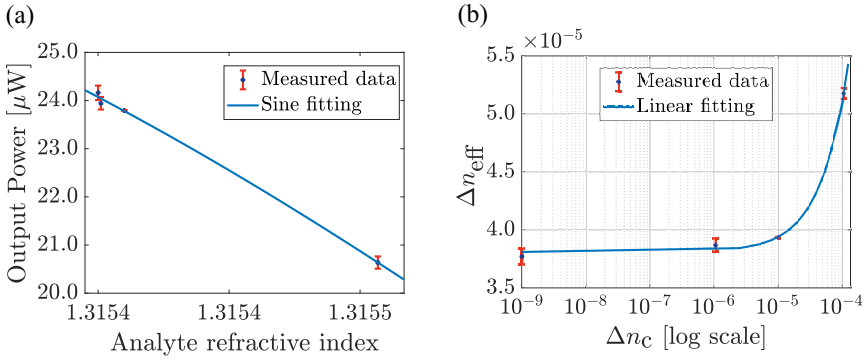

Figure 5.11: Characterization results: (a) The measured output power as a function of the analyte's refractive index, with sine curve fitting. (b) The calculated change in the effective index as a function of the analyte's refractive index change, with a linear fit. The slope indicates a homogeneous sensitivity of 0.128.

These findings demonstrate that the MZI's sensitivity is predominantly influ-

enced by the refractive index change of the analyte. The validation confirms that external factors have a minimal impact on the sensor's sensitivity, allowing the device to accurately measure changes in the refractive index of the ethanol-water mixture. In comparison, the state-of-the-art hybrid polymer refractive index sensor reports a sensitivity of 0.051 [93], while our sensor achieves a sensitivity of 0.128, approximately 2.5 times higher, indicating a significant improvement in performance. This high level of sensitivity and reliability indicates the MZI's potential as a robust tool for precise sensing applications.

6 Conclusions and outlook

This thesis has presented a comprehensive study on the design, fabrication, and characterization of advanced waveguide structures, specifically focusing on polymer-titania hybrid waveguides and strip-loaded MZIs for refractive index sensing. The key findings and contributions of this research are summarized as follows: Through extensive simulations and theoretical analysis, we optimized the design parameters for polymer-titania hybrid waveguides. The integration of titanium dioxide with polymer materials significantly improved the optical confinement and reduced propagation losses, making these waveguides highly efficient for photonics applications.

Advanced fabrication techniques, including ALD and EBL, were employed to achieve precise control over the waveguide dimensions. These techniques were pivotal in realizing the high-performance waveguides and MZIs discussed in this thesis. Detailed experimental setups were developed to characterize the optical guiding properties of the fabricated waveguides. The end-fire coupling method and mode retrieval analysis provided insights into the propagation losses and mode dynamics, confirming the theoretical predictions and demonstrating the effectiveness of the design. The strip-loaded MZI was successfully applied for ethanol concentration measurement, showcasing its practical utility. The high sensitivity and reliability of the MZI make it a robust tool for precise sensing applications, particularly in biochemical sensing where detecting small changes in refractive index is crucial.

The research presented in this thesis opens several avenues for future exploration and development. Further research into alternative high-refractive-index materials that can be integrated with polymers could yield waveguides with even lower losses and better confinement properties. Materials such as silicon nitride and aluminum oxide are promising candidates for such investigations. Continued advancements in nanofabrication technologies, such as nanoimprint lithography and roll-to-roll processing, could enable mass production of these waveguides, making them more accessible for commercial applications. Expanding the application scope of strip-loaded MZIs to include various biochemical and environmental sensors could significantly impact fields such as medical diagnostics and environmental monitoring.

Further improvements on the sensor are currently in progress. It involves mainly fiber optic connections to the input and output waveguides to stabilize the measurements. A control in temperature combined with microfluidic circuit could be also envisioned in some cases, e.g., volatile analytes, but the design remains relatively robust since only the open sensing region is sensitive to the refractive index change, i.e., the environment of the sensor.

The advancements in waveguide technology and sensor applications discussed in this thesis represent significant contributions to the field of photonics. By contributing to bridging the gap between theoretical designs and practical implementations, this research paves the way for the development of more efficient and versatile optical devices. The insights gained from this work will undoubtedly inspire further innovations and applications in integrated optics, driving forward the capabilities of photonics technologies.

BIBLIOGRAPHY

[1] H. Mukundan et al., "Planar optical waveguide-based biosensor for the quantitative detection of tumor markers," *Sensors and Actuators B: Chemical* **138**, 453–460 (2009).

[2] J. P. Hulme et al., "Simple leaky-waveguide devices for the detection of bacterial spores," *Sensors and Actuators B: Chemical* **160**, 1508–1513 (2011).

[3] R. Horvath et al., "Optical waveguide sensor for on-line monitoring of bacteria," *Optics Letters* **28**, 1233–1235 (2003).

[4] K.-H. Chen, C.-C. Hsu, and D.-C. Su, "Interferometric optical sensor for measuring glucose concentration," *Applied optics* **42**, 5774–5776 (2003).

[5] D. Marcuse, *Theory of Dielectric Optical Waveguides* (Elsevier, 2013).

[6] X. C. Tong Ph. D and X. C. Tong, "Fundamentals and design guides for optical waveguides," *Advanced Materials for Integrated Optical Waveguides* 1–51 (2014).

[7] J. Schrage and T. Bierhoff, "Embedded optical waveguides for on-board interconnections," in *Frontiers in Optics* (Optica Publishing Group, Tucson, Arizona, USA, 2003), p. WDD4.

[8] J. Sun, Y. Hao, L. Zhang, et al., "Brief review of Lithium Niobate crystal and its applications," *Journal of Synthetic Crystals* **49**, 947–964 (2020).

[9] B. Gao, M. Ren, D. Zheng, et al., "Long-lived Lithium Niobate: history and progress," *Journal of Synthetic Crystals* **50**, 1183–1199 (2021).

[10] D. Runde et al., "Mode-selective coupler for wavelength multiplexing using LiNbO3: Ti optical waveguides," *Central European Journal of Physics* **6**, 588–592 (2008).

[11] J.-E. Broquin and S. Honkanen, "Integrated photonics on glass: A review of the ion-exchange technology achievements," *Applied Sciences* **11**, 4472 (2021).

[12] J.-H. Ryou, R. Kanjolia, and R. D. Dupuis, "CVD of III-V Compound Semiconductors," in *Chemical Vapour Deposition: Precursors, Processes and Applications* (Royal Society of Chemistry, 2008), pp. 272–319.

[13] G. E. Stillman, V. M. Robbins, and N. Tabatabaie, "III-V compound semiconductor devices: Optical detectors," *IEEE Transactions on Electron Devices* **31**, 1643–1655 (1984).

[14] T. Miya, "Silica-based planar lightwave circuits: passive and thermally active devices," *IEEE Journal of Selected Topics in Quantum Electronics* **6**, 38–45 (2000).

[15] H. Yamada et al., "Si photonic wire waveguide devices," *IEEE Journal of Selected Topics in Quantum Electronics* **12**, 1371–1379 (2006).

[16] R. Kashahara et al., "New structures of silica-based planar light wave circuits for low power thermooptic switch and its application to 8 × 8 optical matrix switch," *Journal of Lightwave Technology* **20**, 993–1000 (2002).

[17] K. Worhoff et al., "Design, tolerance analysis and fabrication of silicon oxynitride based planar optical waveguides for communication devices," *Journal of Lightwave Technology* **17**, 1401–1407 (1999).

[18] A. V. Mule et al., "Photopolymer-based diffractive and MMI waveguide couplers," *IEEE Photonics Technology Letters* **16**, 2490–2492 (2004).

[19] M. H. Ibrahim et al., "A novel 1 × 2 multimode interference optical wavelength filter based on photodefinable benzocyclobuene polymer," *Journal of Microwave and Optical Technology Letters* **49**, 1024–1028 (2007).

[20] H. P. Chan et al., "A wide angle X-junction polymeric thermo optic digital switch with low crosstalk," *IEEE Photonics Technology Letters* **15**, 1210–1212 (2003).

[21] S. N. Khonina, G. S. Voronkov, E. P. Grakhova, N. L. Kazanskiy, R. V. Kutluyarov, and M. A. Butt, "Polymer waveguide-based optical sensors—interest in bio, gas, temperature, and mechanical sensing applications," *Coatings* **13**, 549 (2023).

[22] K. Okamoto, *Fundamentals of Optical Waveguides* (Elsevier Science & Technology, 2006), Maxwell's Equations.

[23] Z. Zhu and T. G. Brown, "Full-vectorial finite-difference analysis of microstructured optical fibers," *Optics express* **10**, 853–864 (2002).

[24] F. Teixeira, C. Sarris, Y. Zhang, D.-Y. Na, J.-P. Berenger, Y. Su, M. Okoniewski, W. Chew, V. Backman, and J. J. Simpson, "Finite-difference time-domain methods," *Nature Reviews Methods Primers* **3**, 75 (2023).

[25] D. M. Sullivan, *Electromagnetic simulation using the FDTD method* (John Wiley & Sons, 2013).

[26] U. S. Inan and R. A. Marshall, *Numerical electromagnetics: the FDTD method* (Cambridge University Press, 2011).

[27] F. Kong, K. Li, and X. Liu, "Accurate analysis of planar optical waveguide devices using higher-order FDTD scheme," *Optics Express* **14**, 11796–11803 (2006).

[28] A. Taflove, A. Oskooi, and S. G. Johnson, *Advances in FDTD computational electrodynamics: photonics and nanotechnology* (Artech house, 2013).

[29] A. W. Snyder, J. D. Love, et al., *Optical waveguide theory*, Vol. 175, (Chapman and hall London, 1983).

[30] M. Hammer and O. V. Ivanova, "Effective index approximations of photonic crystal slabs: a 2-to-1-D assessment," *Optical and quantum electronics* **41**, 267–283 (2009).

[31] A. F. Oskooi and S. G. Johnson, *Numerical Methods in Photonics* (Cambridge University Press, 2013).

[32] V. Ramaswamy, "Strip-loaded film waveguide," *The Bell System Technical Journal* **53**, 697–704 (1974).

[33] Y. Wang, Z. Chen, and H. Hu, "Analysis of waveguides on lithium niobate thin films," *Crystals* **8**, 191 (2018).

[34] W. Qiu, M.-P. Bernal, A. Ndao, C. Guyot, N. M. Hameed, N. Courjal, H. Maillotte, and F. Baida, "Analysis of ultra-compact waveguide modes in thin film lithium niobate," *Applied Physics B* **118**, 261–267 (2015).

[35] I. Doughan, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Low loss vertical tio2/polymer hybrid nano-waveguides," *Nanomaterials* **13**, 469 (2023).

[36] H. Furuta, H. Noda, and A. Ihaya, "Novel optical waveguide for integrated optics," *Applied optics* **13**, 322–326 (1974).

[37] L. Ahmadi, V. Kontturi, J. Laukkanen, J. Saarinen, S. Honkanen, M. Kuittinen, and M. Roussey, "Strip-loaded waveguide on titanium dioxide thin films by nanoimprint replication," *Optics Letters* **42**, 527–530 (2017).

[38] J. Turunen, E. Tervonen, and A. T. Friberg, "Coherence theoretic algorithm to determine the transverse-mode structure of lasers," *Opt. Lett.* **14**, 627–629 (1989).

[39] E. Tervonen, J. Turunen, and A. T. Friberg, "Transverse laser-mode structure determination from spatial coherence measurements: Experimental results," *Appl. Phys. B* **49**, 409–414 (1989).

[40] W.-P. Huang and J. Mu, "Complex coupled-mode theory for optical waveguides," *Optics express* **17**, 19134–19152 (2009).

[41] W.-P. Huang, "Coupled-mode theory for optical waveguides: an overview," *JOSA A* **11**, 963–983 (1994).

[42] D. Malacara, *Optical Shop Testing* (Wiley, 2007).

[43] M. Koivurova, H. Partanen, J. Lahyani, N. Cariou, and J. Turunen, "Scanning wavefront folding interferometers," *Opt. Express* **27**, 7738–7750 (2019).

[44] A. Halder, H. Partanen, A. Leinonen, M. Koivurova, T. K. Hakala, T. Setälä, J. Turunen, and A. T. Friberg, "Mirror-based scanning wavefront-folding interferometer for coherence measurements," *Opt. Lett.* **45**, 4260–4263 (2020).

[45] A. T. Friberg, T. Setälä, et al., "Complete coherence of random, nonstationary electromagnetic fields," *Optics Letters* **46**, 1756–1759 (2021).

[46] A. Halder, "Intensity-induced phase in linear optics," *JOSA A* **40**, 1583–1586 (2023).

[47] H. Partanen, J. Tervo, and J. Turunen, "Coupling of spatially partially coherent beams into planar waveguides," *Optics Express* **23**, 7879–7893 (2015).

[48] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics* (Wiley, 1991).

[49] H. Partanen, J. Turunen, and J. Tervo, "Coherence measurement with digital micromirror device," *Optics letters* **39**, 1034–1037 (2014).

[50] A. Psarouli, A. Salapatas, A. Botsialas, P. Petrou, I. Raptis, E. Makarona, G. Jobst, K. Tukkiniemi, M. Sopanen, R. Stoffer, et al., "Monolithically integrated broad-band Mach-Zehnder interferometers for highly sensitive label-free detection of biomolecules through dual polarization optics," *Scientific reports* **5**, 17600 (2015).

[51] M. Janik, M. Koba, A. Celebańska, W. J. Bock, and M. Śmietana, "Live E. coli bacteria label-free sensing using a microcavity in-line Mach-Zehnder interferometer," *Scientific reports* **8**, 17176 (2018).

[52] D. Pandey, L. Bellentani, M. Villani, G. Albareda, P. Bordone, A. Bertoni, and X. Oriols, "A proposal for evading the measurement uncertainty in classical and quantum computing: Application to a resonant tunneling diode and a mach-zehnder interferometer," *Applied Sciences* **9**, 2300 (2019).

[53] C. Taraphdar, T. Chattopadhyay, and J. N. Roy, "Mach–Zehnder interferometer-based all-optical reversible logic gate," *Optics & Laser Technology* **42**, 249–259 (2010).

[54] H. Y. Choi, M. J. Kim, and B. H. Lee, "All-fiber Mach-Zehnder type interferometers formed in photonic crystal fiber," *Optics Express* **15**, 5711–5720 (2007).

[55] R. K. Musa, S. Al deen Adnan, and A. M. Hammadi, "Evanescent field Mach-Zehnder interferometer sensor for concentration measurement," *Journal of Madenat Alelem University College* **10**, 44–55 (2018).

[56] T. Suntola and J. Hyvärinen, "Atomic layer epitaxy," *Ann. Rev. Mater. Sci.* **15**, 95–177 (1985).

[57] R. W. Johnson, A. Hultqvist, and S. F. Bent, "A brief review of atomic layer deposition: from fundamentals to applications," *Materials Today* **17**, 236–246 (2014).

[58] N. Pinna and M. Knez, *Atomic Layer Deposition of Nanostructured Materials* (Wiley-VCH, 2012).

[59] H. B. Profijt, S. E. Potts, M. C. M. van de Sanden, and W. M. M. Kessels, "Plasma-assisted atomic layer deposition: Basics, opportunities, and challenges," *Journal of Vacuum Science & Technology A* **29**, 050801 (2011).

[60] J. Lu, J. W. Elam, and P. C. Stair, "Atomic layer deposition—Sequential self-limiting surface reactions for advanced catalyst "bottom-up" synthesis," *Surface Science Reports* **71**, 410–472 (2016).

[61] R. Zazpe, M. Knaut, H. Sopha, L. Hromadko, M. Albert, J. Prikryl, V. Gartnerova, J. W. Bartha, and J. M. Macak, "Atomic layer deposition for coating of high aspect ratio TiO2 nanotube layers," *Langmuir* **32**, 10551–10558 (2016).

[62] S. Pélisset, *Strip-loaded slot waveguides: design, fabrication, and characterization*, PhD thesis (Itä-Suomen yliopisto, 2021).

[63] D. Birnie III, "Spin coating technique," *Sol-gel technologies for glass producers and users* 49–55 (2004).

[64] N. Sahu, B. Parija, and S. Panigrahi, "Fundamental understanding and modeling of spin coating process: A review," *Indian Journal of Physics* **83**, 493–502 (2009).

[65] J. Moreira, A. C. Vale, and N. M. Alves, "Spin-coated freestanding films for biomedical applications," *Journal of materials chemistry B* **9**, 3778–3799 (2021).

[66] V. T. Lukong, K. Ukoba, and T.-C. Jen, "Review of self-cleaning TiO2 thin films deposited with spin coating," *The International Journal of Advanced Manufacturing Technology* **122**, 3525–3546 (2022).

[67] S. Agan, F. Ay, A. Kocabas, and A. Aydinli, "Stress effects in prism coupling measurements of thin polymer films," *Applied Physics A* **80**, 341–345 (2005).

[68] J. Cardin and D. Leduc, "Determination of refractive index, thickness, and the optical losses of thin films from prism-film coupling measurements," *Appl. Opt.* **47**, 894–900 (2008).

[69] Metricon Corporation, "Metricon Model 2010/M," (2010), Available at: `http://www.metricon.com/model2010m.htm`.

[70] Y. Chen, "Nanofabrication by electron beam lithography and its applications: A review," *Microelectronic Engineering* **135**, 57–72 (2015).

[71] E. Sharma, R. Rathi, J. Misharwal, B. Sinhmar, S. Kumari, J. Dalal, and A. Kumar, "Evolution in lithography techniques: microlithography to nanolithography," *Nanomaterials* **12**, 2754 (2022).

[72] M. A. Mohammad, M. Muhammad, S. K. Dew, and M. Stepanova, "Fundamentals of electron beam exposure and development," in *Nanofabrication: techniques and principles* (Springer, 2011), pp. 11–41.

[73] Z. Cui, *Nanofabrication: Principles, Capabilities and Limits* (Springer US, 2008).

[74] A.-C. Tseng, K. Chen, C.-D. Chen, and K. J. Ma, "Electron beam lithography in nanoscale fabrication: recent development," *IEEE Transactions on Advanced Packaging* **26**, 141–149 (2003).

[75] S. Okazaki, "High resolution optical lithography or high throughput electron beam lithography: The technical struggle from the micro to the nanofabrication evolution," *Microelectronic Engineering* **133**, 23–35 (2015).

[76] N. Gadegaard and D. McCloy, "Direct stamp fabrication for NIL and hot embossing using HSQ," *Microelectronic Engineering* **84**, 2785–2789 (2007).

[77] H.-C. Liou and J. Pretzer, "Effect of curing temperature on the mechanical properties of hydrogen silsesquioxane thin films," *Thin Solid Films* **335**, 186–191 (1998).

[78] D. Komatitsch and J. Tromp, "A perfectly matched layer absorbing boundary condition for the second-order seismic wave equation," *Geophysical Journal International* **154**, 146–153 (2003).

[79] W. Shin and S. Fan, "Choice of the perfectly matched layer boundary condition for frequency-domain Maxwell's equations solvers," *Journal of Computational Physics* **231**, 3406–3431 (2012).

[80] M. Aellen and D. J. Norris, "Understanding optical gain: which confinement factor is correct?," *ACS Photonics* **9**, 3498–3505 (2022).

[81] R. R. Singh, "Dispersion tailoring of silicon nanowire optical rectangular waveguide (SNORW)," *SN Applied Sciences* **2**, 502 (2020).

[82] C. Yuan, J. Dai, H. Jia, J. Ding, L. Zhang, X. Fu, and L. Yang, "Design of a C-band polarization rotator-splitter based on a mode-evolution structure and an asymmetric directional coupler," *Journal of Semiconductors* **39**, 124008 (2018).

[83] M. N. Erim, N. Erim, and H. Kurt, "Optical surface modes of photonic crystals for dual-polarization waveguide," *Photonics and Nanostructures-Fundamentals and Applications* **11**, 123–131 (2013).

[84] M. Häyrinen, M. Roussey, A. Bera, M. Kuittinen, and S. Honkanen, "Atomic layer re-deposition for nanoscale devices," *Encyclopedia of Plasma Technology* **1**, 143–151 (2016).

[85] Y. A. Vlasov and S. J. McNab, "Losses in single-mode silicon-on-insulator strip waveguides and bends," *Optics express* **12**, 1622–1631 (2004).

[86] M. Häyrinen, M. Roussey, V. Gandhi, P. Stenberg, A. Säynätjoki, L. Karvonen, M. Kuittinen, and S. Honkanen, "Low-loss titanium dioxide strip waveguides fabricated by atomic layer deposition," *Journal of Lightwave Technology* **32**, 208–212 (2013).

[87] I. Hegeman, M. Dijkstra, F. Segerink, W. Lee, and S. Garcia-Blanco, "Development of low-loss TiO 2 waveguides," *Optics express* **28**, 5982–5990 (2020).

[88] C. C. Evans, C. Liu, and J. Suntivich, "Low-loss titanium dioxide waveguides and resonators using a dielectric lift-off fabrication process," *Optics express* **23**, 11160–11169 (2015).

[89] C. Chaudhari, D. S. Patil, and D. Gautam, "A new technique for the reduction of the power loss in the Y-branch optical power splitter," *Optics communications* **193**, 121–125 (2001).

[90] L. Han, B. P.-P. Kuo, N. Alic, and S. Radic, "Ultra-broadband multimode 3dB optical power splitter using an adiabatic coupler and a Y-branch," *Optics Express* **26**, 14800–14809 (2018).

[91] M. Duncan, "Bézier Curves I," in *Applied Geometry for Computer Graphics and CAD* (Springer, 2005), pp. 135–160.

[92] A. Melikyan and P. Dong, "Adiabatic mode converters for silicon photonics: Power and polarization broadband manipulators," *APL Photonics* **4** (2019).

[93] L. Ahmadi, M. Hiltunen, P. Stenberg, J. Hiltunen, S. Aikio, M. Roussey, J. Saarinen, and S. Honkanen, "Hybrid layered polymer slot waveguide Young interferometer," *Optics Express* **24**, 10275–10285 (2016).

[94] S. Kedenburg, M. Vieweg, T. Gissibl, and H. Giessen, "Linear refractive index and absorption measurements of nonlinear optical liquids in the visible and near-infrared spectral region," *Optical Materials Express* **2**, 1588–1611 (2012).

[95] D. R. Lide, *CRC Handbook of Chemistry and Physics*, 85 ed. (CRC Press, 2004).

A Appendix

Other publications and presentations at national and international conferences:

- **I. Doughan**, A. L. Asilevi, A. Halder, T.-L. Guo, E. Mogni, M. Celebrano, M. Fi-nazzi, G. Pellegrini, P. Biagioni, E. Descrovi,et al., “Temporal and spectral signatures of the interaction between ultrashort laser pulses and Bloch surface waves,” *APL Photonics* **9** (2024).
- **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing", PREIN Workshop on On-Chip Photonics for Sensing, Vantaa, Finland, 2024; Oral presentation.
- **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing", Optics and Photonics Days, Helsinki, Finland, 2024; Poster presentation.
- **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing", SPIE Photonics West, San Francisco, United States of America, 2024; Poster presentation.
- **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Hybrid polymer-titania waveguides for highly integrated circuits", European Optical Society Annual Meeting, Porto, Portugal, 2022; Poster presentation.
- **I. Doughan**, K. Oyemakinwa, O. Ovaskainen, and M. Roussey, "Hybrid polymer-titania waveguides for highly integrated circuits", Optics and Photonics Days, Tampere, Finland, 2022; Poster presentation.

Paper **I**

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"Low loss vertical tio2/polymer hybrid nano-waveguides"
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Low Loss Vertical TiO_2/Polymer Hybrid Nano-Waveguides

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Abstract: This article proposes a novel demonstration of a low-loss polymer channel hybridized with a titania core leading to a nano-waveguide elongated in the normal direction to the substrate. It is aimed at using the quasi-transverse magnetic (TM) mode as the predominant mode in compact photonic circuitry. A detailed design analysis shows how a thin layer of a higher-refractive index material in a trench within the core of the waveguide can increase the confinement and reduce the propagation losses. This thin layer, produced by atomic layer deposition, covers the entire polymer structure in a conformal manner, ensuring both a reduction of the surface roughness and a stronger field confinement. The trench can be made at any place within the polymer channel and therefore its position can be tuned to obtain asymmetric modal distribution. The waveguide is demonstrated at telecom wavelengths, although the material's properties enable operation over a large part of the electromagnetic spectrum. We measured propagation losses as low as 1.75 ± 0.32 dB/cm in a 200 nm $\times$ 900 nm section of the waveguide core. All processes being mass-production compatible, this study opens a path towards easier integrated-component manufacture.

Keywords: optical waveguides; integrated optics; micro fabrication; polymer components

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1. Introduction

A channel waveguide on a chip is composed of a core, a substrate, and a cladding. Except for rare exceptions, e.g., slot waveguides [1,2] and anti-resonant reflective waveguides [3,4], the core is usually constituted of one or several materials of a higher refractive index than the substrate and superstrate, ensuring the required light confinement. The refractive index difference between the three parts of the waveguide is a measure of the confinement strength of the propagating modes inside the core. Low refractive index contrast waveguides, such as glass waveguides made by ion exchange, usually have a large and high cross section to ensure good confinement of the optical modes within the core of the waveguide [5,6]. This is a synonym of low loss when the material constituting the waveguide is homogeneous and exempt from scatterers possibly perturbating the field propagation. However, in such a case, tight bends are prohibited to avoid extensive losses. On the other hand, nano-waveguides enabled by high refractive index contrast materials, such as silicon or germanium, allow very tiny and compact photonic circuitry [7–9]. Propagation loss is higher, but the reduced footprint of the components compensates for it. Polymers offer a trade-off between these two extremes since they have a refractive index ranging from 1.2 to 2 [10–12]. Moreover, direct electron beam lithography (EBL) is possible for some of them [13,14], and nanoimprinting is an interesting option for the fabrication of channel waveguides at large scale [15,16]. However, because of a refractive index close to that of the substrate, polymers may not be ideal for small-footprint devices.

The cross-sectional geometry of a waveguide is the other key parameter to tuning the effective index of a guided mode. In particular, the shape, size, and symmetry of the core will directly affect the birefringence, dispersion, confinement, and localization of a guided mode. Large core-based waveguides, usually the low refractive index contrast waveguides or the large 3 µm thick silicon waveguides, present modes with low dependence on the

fine core modifications. In contrast, nano-waveguides are guiding modes reacting strongly to any tiny modifications such as a sidewall corrugation or a tapering.

We present, here, an alternative geometry allowing the use of polymer waveguides even at a small-footprint scale. The cross section of the waveguide is presented in Figure 1. The geometry, i.e., the cross section of the waveguide, consists of a polymer waveguide of width $2w + 2t$ made of AZ-2070 (nLOF) electron beam resist of refractive index $n_P = 1.601$ at $\lambda = 1550$ nm [17–19]. A trench of width $2t$ is opened in the channel at a position y_T varying from $-a$ to a, with $a = w + t$ and $y_T = 0$ corresponding to the center of the waveguide. To ensure confinement of the field in the central core of the waveguide, the trench is filled up with a high-refractive index material. We opted for titania, TiO_2 with $n_{TiO_2} = 2.24$ at $\lambda = 1550$ nm, deposited by atomic layer deposition (ALD) in a conformal manner all over the polymer channel [20–22]. This fabrication method leads to a coating of thickness t on the sides and lateral unguiding regions of the waveguide, but also to a filling of the high aspect ratio trench, as sketched in Figure 1. The entire structure is fabricated on a thermally oxidized silicon wafer. The thermal oxide layer of 3 µm thickness ensures isolation of the optical mode with the silicon wafer. Therefore, a vertical high-refractive index nano-waveguide can be created.

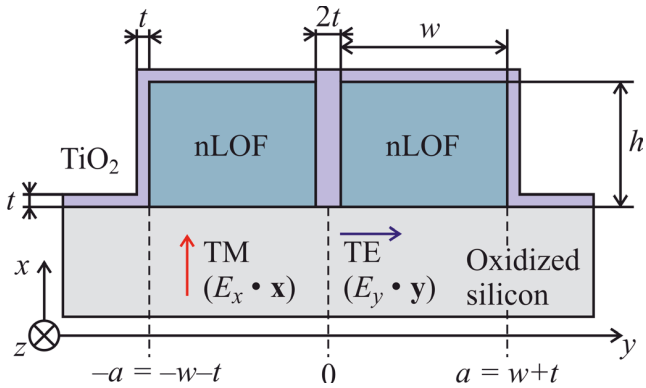

Figure 1. Cross-section geometry of the vertical nano-waveguide. The substrate is an oxidized silicon wafer. The polymer is AZ-nLof. The coating is TiO_2. The coating thickness is t, the height and width of each part of the polymer region are h and w, respectively. The trench thickness is $2t$ and its position with respect to the waveguide is at $y_T = 0$ when centered.

In this article, we focus on the demonstration of the waveguide with a titania core. However, it is worth mentioning that as long as the refractive indices match the condition $n_{POLYMER} < n_{TRENCH}$, any changes in the materials are prevented. One can imagine a multitude of applications for which a hybrid waveguide is necessary but it is not easy at wafer scale. For instance, one can include a gain medium within the trench in order to create an on-chip amplifier [23]. Another application could be the use of quantum dots or dyes inside the trench to provide the up or down conversion of a signal in order to reach other wavelength ranges [24]. Emitters at very low concentration within the trench can provide very small amounts of photons and with a proper design of the polymer section, a single photon source could be achieved [25] for quantum photonics purposes.

2. Materials and Methods

2.1. Designing Tools

The finite-difference time-domain method (FDTD) is a tool for modeling and simulating nanoscale optical devices. It is well capable of handling dielectric structures that are inhomogeneous and complicated while producing field solutions that are well defined [26,27]. The waveguides studied in this work have been investigated with the commercial software,

Ansys Lumerical suite. The configuration was set up as a structure group controlled with script as shown in Figure 1. The boundary conditions of the simulation region were set as perfectly matched layers (PML), which act as absorbers for any scattered field. The mesh step size used in the simulations was 20 nm in x, y, and z directions. This mesh size allows for numerical convergence of the results. The central illumination wavelength was 1550 nm. Due to high computational power and memory requirements with the 3D-FDTD, leading to time-consuming simulations, the waveguide and bend variation simulations were mostly carried out in 2D-FDTD using effective index approximation. The results were analyzed with MATLAB R2018b, MathWorks (Natick, MA, USA). The mode calculations were performed using the Lumerical in-built mode solver based on the finite difference method. In all calculations, the dispersion of the materials, measured by ellipsometry, was taken into account.

2.2. Fabrication Techniques

A layer of diluted negative resist (MicroChemicals AZ nLof 2070, Ulm, Germany) was spin-coated (Laurell WS-650MZ-8NPPB, Lansdale, PA, USA) at 1100 rpm on an oxidized silicon wafer. The polymer layer was soft baked at 110 °C on a hot plate for 1 min. The predefined waveguide structure was then transferred onto the resist layer by electron beam lithography (EBL, Vistec EBPG5000+HR, Dortmund, Germany). The acceleration voltage was set at 100 kV, and an exposure dose of 40 μC/cm^2 was used. After exposure, the sample was baked hard at 110 °C for 1 min. The pattern was then developed in AR 300-47, AllResist (Strausberg, Germany), for 90 s and then stopped in de-ionized (DI) water for 30 s before rinsing with DI-water. nLof is a negative EBL resist, which means that exposed areas remain after development. The sample was further coated by atomic layer deposition (ALD) for a 100 nm titanium dioxide film. In the ALD process, the precursors were titanium tetrachloride ($TiCl_4$) and water (H_2O). The deposition rate was 0.0718 nm/cycle. The process temperature was set to 120 °C to ensure the deposition of an amorphous material.

Figure 2 describes the 4-step process flow for the manufacturing of the waveguides. The relatively large features of the structure would allow the use of nanoimprint technology instead of EBL for a faster, lower cost, and larger scale fabrication.

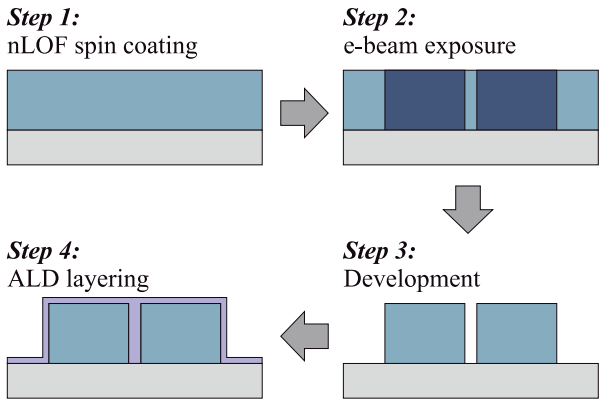

Figure 2. Process flow of the waveguide fabrication: resist deposition on an oxidized silicon wafer, electron-beam patterning of the nLof resist, development of the pattern, ALD coating of titania.

Several patterns were fabricated. This included bends of different radii of curvature (r) as well as long waveguides. These patterns were categorised into three chips as shown in Figure 3. Each waveguide had an input and output taper part of 3 μm width and 500 μm length. In chip 1, the bend radius was kept at 100 μm and the length L was varied from 500 μm to 5 mm. Chip 2 presents waveguides with varying bend radii from 2 μm to

20 µm whilst the length L was kept constant at 1 mm. In chip 3, the position of the trench filled by TiO_2 was varied linearly within the bend for three different bend radii.

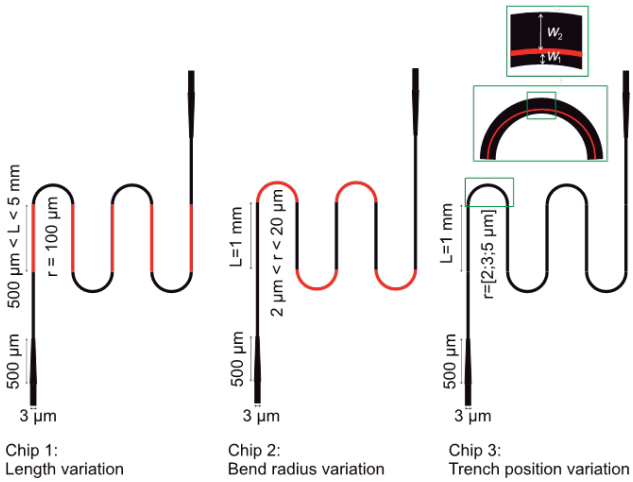

Figure 3. Layout of the chip. Chip 1: length variation from 500 µm to 5 mm. Chip 2: bend radius variation from 2 µm to 20 µm. Chip 3: TiO_2 filled trench position variation W = 200 nm, 300 nm, 400 nm, and 500 nm for three bend radii of 2 µm, 3 µm and, 5 µm.

2.3. Characterization Methods

The final structure was characterized by the end-fire coupling method [28] described in the set-up shown in Figure 4. Light from a laser source was coupled through a tapered lens fiber into the waveguide and the transmittance power was measured using a power meter. The laser source used was a Tunics plus tunable laser source with wavelength accuracy of 1 pm. Its operation wavelength range was 1510 nm to 1640 nm. A polarization controller was used to set the desired polarization of light. The microscope objective (Olympus Plan N, 20×, NA = 0.4) was used to image the output of the waveguide on a near-infrared camera (Point Grey Chameleon CMLN-13S2M). The focused spot size of the lensed fiber used was 2 ± 0.5 µm. A Glan-Thompson GTH10M polarizer ensured the desired polarized light passed through to the NIR detector and power meter (Thorlabs S122c, PM100D) to measure the transmitted power. Alternatively, to the detector, the output mode could be imaged allowing a check at any time of the experiment that a well-coupling to the waveguide is performed without any modification of the measurement system. We used the cut-back method [29,30] to estimate the propagation losses by comparing transmissions through waveguides of varying lengths and fitting the length dependence under similar coupling conditions and surface roughness.

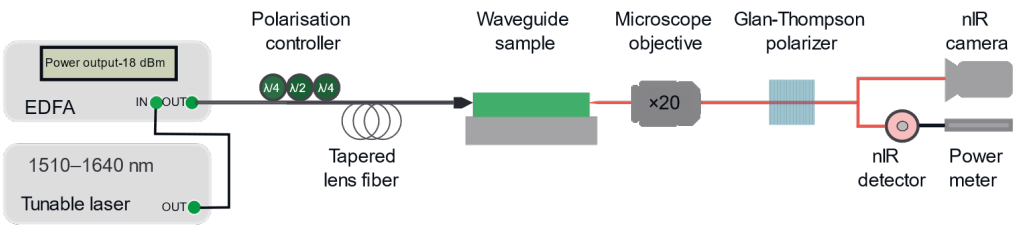

Figure 4. Schematics of the fiber characterization setup used to estimate the propagation losses in the waveguides using cut-back method. The detector of the power meter can be switched to a NIR camera for mode imaging.

Different height lengths of waveguides were created for the propagation loss measurements. The transmission through each waveguide was measured five times, changing the injection side and the order of the measurement, i.e., from the longest to the shortest waveguide and vice versa. Results were averaged to obtain a single value used for the estimation of the propagation losses.

3. Results and Discussion

3.1. Optimization of the Waveguide Parameters

One figure of merit for many waveguide structures, especially as it relates to reducing propagation losses, is the mode confinement (Γ) within the core of the waveguide. To obtain the highest confinement, the parameters of the waveguide were optimized.

3.1.1. Modal Effective Index

The optimization of the waveguide parameters was performed at the operation wavelength $\lambda = 1550$ nm, i.e., the central telecom wavelength of the C-band, 1525 nm $< \lambda <$ 1575 nm. The evolution with w, h, and t of the characteristics of the waveguides, such as the effective index, the confinement factor, and the birefringence between TE_{00}- and TM_{00}-modes were studied. The width and height of each polymer section were set to $w = 800$ nm and $h = 900$ nm to ensure single modal behavior of the waveguide, when considering titania inside the trench. It should be noted that several solutions exist for the dimensions of the polymer sections to maintain a single mode waveguide. We opted for these dimensions for fabrication constraints and for a better mode confinement in the trench, for which the height plays the most important role.

In Figure 5a,b, we show the dependence on the effective index of the two fundamental modes as a function of the TiO_2 thickness $2t$ inside the trench. When $2t$ increased, both effective indices increased. For $2t > 200$ nm, the waveguide became multimodal. Figure 5c,d shows the field intensity distribution for both fundamental modes. One can remark that the quasi-TM fundamental mode had a higher effective index and a better confinement in the TiO_2-filled trench than the quasi-TE mode. While the quasi-TE mode was mainly confined in the polymer region on the interfaces TiO_2/polymer ($\Gamma^{TE}_{trench} \simeq 14\%$), similar to standard nanowires, the fundamental quasi-TM mode was highly located inside the high refractive index region ($\Gamma^{TM}_{trench} \simeq 39\%$), similar to a 90°-rotated (vertical) standard ridge nano-waveguide. Moreover, the TM nature of the mode allowed a significant confinement factor ($\Gamma^{TM}_{above} \simeq 5\%$) above the waveguide allowing for sensing with optimized parameters, e.g., a reduction of the height of the waveguide.

3.1.2. Dispersion and Birefringence of the Waveguide

The dispersion, Figure 5e, in the waveguide was comparable to that observed in other waveguides, such as those used in silicon photonics, in the C-band [31]. Most importantly, the dispersion variation for the fundamental quasi-TM mode was relatively low over the studied wavelength range, which opens the path for broadband operating devices. The birefringence observed in Figure 5f was high (close to 0.1 refractive index unit). Such a difference between the modal effective indices of the modes would enable a polarization management in the waveguide for future applications, such as polarization rotation [32], or polarization maintaining waveguide [33], by acting on the shape, size, and position of the trench.

3.1.3. Effect of the Position of the Trench

The TiO_2 trench surrounded by polymer enabled a reduced bend loss and, therefore, a smaller bend radius. This comes from the fact that the trench yields a higher localization of the mode in the center of the waveguide, especially for the fundamental quasi-TM mode. The trench can be easily placed at any position within the waveguide leading to a tuning of the mode distribution central position. Figure 6 shows the TM_{00}-mode profile ($|E_x|$) and the TE_{00}-mode profile ($|E_y|$) along the y-direction as a function of the position y_T of the trench.

One can see that the mode distribution can be off centered with little or no modification of the overall modal distribution that remains well confined within the waveguide.

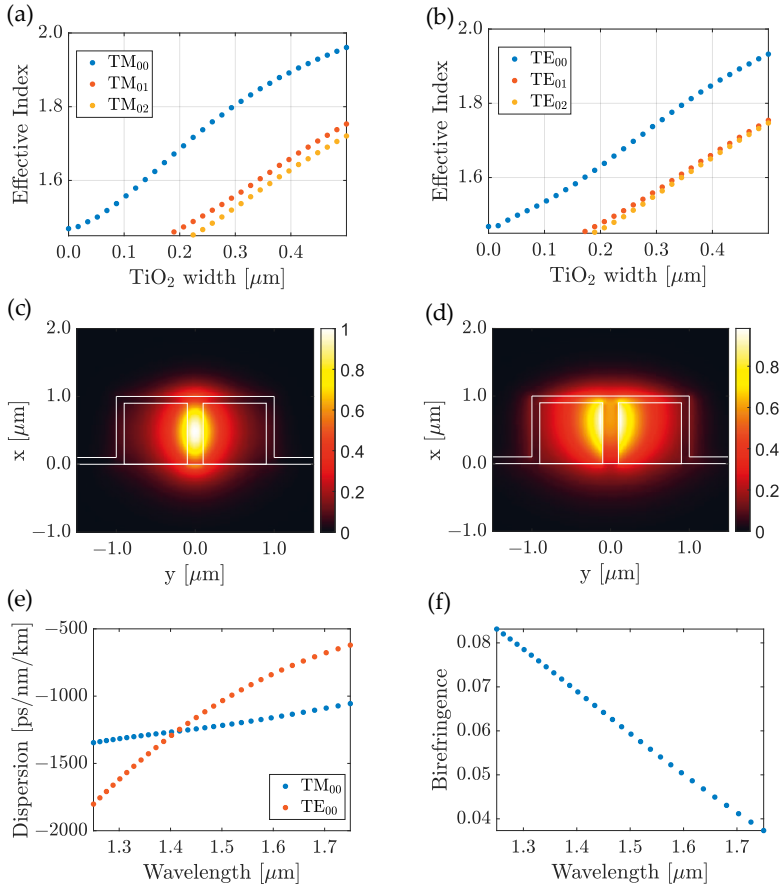

Figure 5. Waveguide structure design. (**a**,**b**) Effective index as a function of the TiO_2 thickness $2t$ for quasi-TM and quasi-TE polarizations, respectively. (**c**,**d**) TM_{00} and TE_{00} mode distributions, respectively. (**e**) Dispersion of the TM_{00} (dotted red curve) and TE_{00} (dotted blue curve) modes. (**f**) Birefringence.

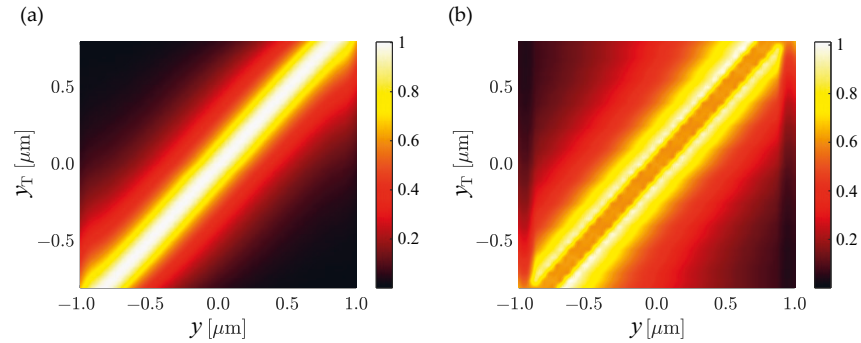

Figure 6. (**a**) TM_{00}-mode profile ($|E_x|$) and (**b**) TE_{00}-mode profile ($|E_y|$) along the y-direction as a function of the position y_T of the trench.

An important aspect to consider while designing a waveguide is the bend loss since bends are nearly unavoidable in most photonics circuitry. The tighter the bend, the smaller the footprint, and the higher the integration of the devices.

We performed simulations for three distinct positions of the trench, namely: $y_T = 0$ nm, -400 nm, and -600 nm, keeping w = 800 nm, h = 900 nm, and t = 100 nm constant, and observed the transmission after a U-turn (180°) of the waveguide for different bend radii 1 µm < r < 20 µm. The results are shown in Figure 7. One can clearly see the expected trend for the transmission with an asymptotic limit for large values of r. In Figure 7a, the trench was shifted even on straight portions of the waveguide, while in Figure 7b, the trench was shifted only inside the bend portion. In the latter case, the titania part became elliptical to remain inside the polymer. One can remark that losses increased as soon as the trench was no longer centered; however, by operating a shift only in the bend part, losses were nearly negligible compared to the centered case. For the minimal radius $r = 1$ µm, the loss for the bend was about 8% and rapidly decreased when r increased until reaching an asymptotic value below 1% for $r > 7$ µm. Such a feature is a key towards waveguide coupling regions and other classical waveguide management that could be used in further applications. No major differences could be observed between the different core positions within the bend. This allows us to expect relaxed fabrication constraints and alignment, especially when considering nano-imprinting as a manufacturing method.

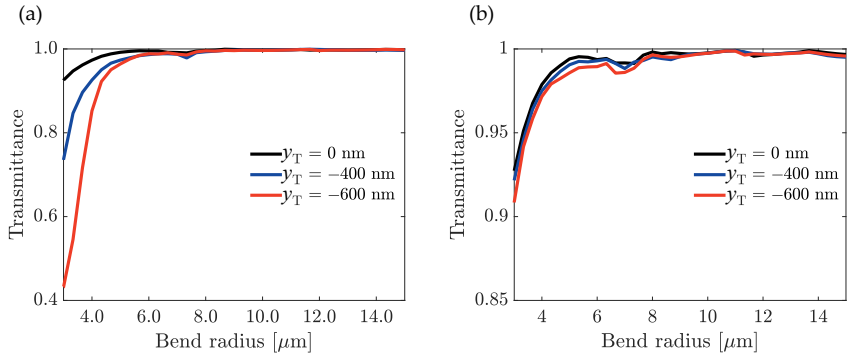

Figure 7. TM_{00} mode transmission after a U-turn bend for $y_T = 0$ nm (black), $y_T = -400$ nm (blue), and $y_T = -600$ nm (red) for: (**a**) when y_T is varied in the entire U-turn bend waveguide and (**b**) when $y_T = 0$ is varied only at the bend section.

3.2. Fabricated Sample

The target parameters chosen for the fabrication of the structure were: w = 800 nm, h = 900 nm, and t = 100 nm. A less than 5% error in these dimensions was measured from the SEM pictures. Figure 8 presents several SEM pictures acquired on different samples.

Figure 8a is a scanning electron microscopy (SEM) picture of the cross section of the waveguide prior to atomic layer deposition (ALD) coating. The dimensions of the polymer-based channel were well respected despite the trench having a footing. Such an issue arises from limitations in the resist performance and fabrication. Figure 8b is an SEM picture of the cross section of the waveguide after TiO_2 deposition (cleaved within the tapered section of the waveguide, which explains a larger waveguide than in Figure 8a). As expected, the material conformally coated the structure and a slight drop in the thickness was observable above the trench due to the unavoidable rounding of the edge of the waveguide [34] which was followed during the ALD process. The layer was thickened afterwards to fill this gap by an additional ALD coating of 10 nm. The roughness of the waveguide seems acceptable although one can see some contamination at the top of the waveguide both before and after the TiO_2 deposition. Figure 8c is an SEM picture of the bent waveguides used for the loss measurements. The zoomed-in section on the bend highlights the precision with which the trench can be positioned within the waveguide. Among the tests we carried out prior to

the fabrication of these waveguides, we realized that the exposure of the bottom part of the trench in the polymer channel is not obvious for all aspect ratios of the trench. We had to make a trade-off between the designed device and the fabrication possibilities, the most important of which was to ensure a single modal behavior of the waveguide; we chose to leave a partly unexposed resist inside the trench in order to keep the titania width as small as possible. Additional simulations have been performed including this default and no major change in the trends or even in the effective indices of the fundamental quasi-TM mode was observed. Finally, one should also note that such a bias is to be expected if the waveguide patterning is made by the nano-imprinting technique.

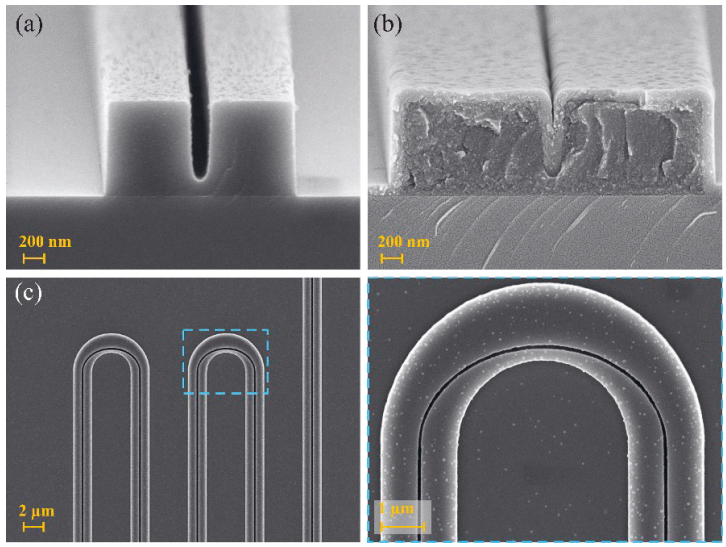

Figure 8. Scanning electron microscope pictures of examples of fabricated waveguide. (**a**) Cross section before ALD coating along the actual waveguide section. (**b**) Cross section after ALD coating along the tapered section of the waveguide. (**c**) Top view of a multi-bend waveguide with a zoomed-in section on the bend before TiO_2 deposition.

3.3. Experimental Determination of the Propagation Losses

The characterization setup used for measuring the propagation loss is shown schematically in Figure 4. In this article, we are mainly interested in demonstrating a waveguide for which the quasi-TM fundamental mode is predominant over the others. Therefore, the polarization was set to TM for all characterizations. Figure 9a shows an image of the quasi-TM fundamental mode at the output of one of the waveguides at λ = 1550 nm. We recall that tapers are used for the in- and out-coupling, which explains the size of the observed mode. To measure the propagation losses, several waveguides of different lengths with bend radius of 100 µm were measured while assuming that insertion losses are similar for all waveguides. Due to some damage (because of some fabrication, cleaving, handling issues), some waveguides were not measurable. Repeating the transmission measurements five times through the waveguides allowed us to give error bars on our measurements and give a better estimate of the propagation losses using the cut-back method from the slope of the curve presented in Figure 9b. The estimated propagation loss was measured to be 1.75 $\pm$ 0.32 dB/cm, which is close to the state-of-the-art for a titania nano-waveguide [35–37]. We would like to remark that due to the very thin additional re-coating of 10 nm of titania performed to fill the remaining gap, the waveguide became slightly multimodal (as Figure 5a predicted). However, this did not influence the loss measurements because we observed that the higher order TM mode was filtered out after the first bend.

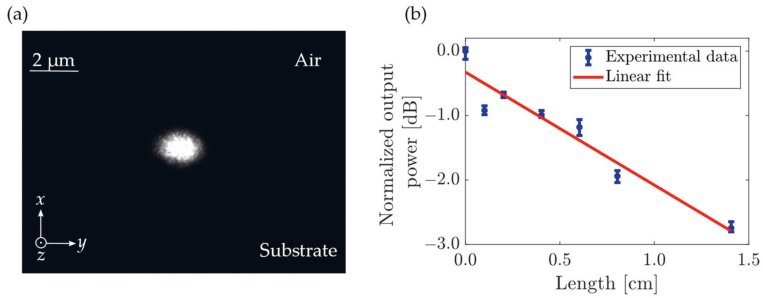

Figure 9. (**a**) Photograph of the output quasi-TM fundamental mode at the output of waveguide (chip 2) with bend radius of 5 μm. (**b**) Normalized power measured at the output of the waveguides (chip 1) as a function of the length of the waveguide (blue circles) and a linear fit (red line) using the cut-back method. Propagation losses are estimated at 1.75 ± 0.32 dB/cm.

In addition to the propagation losses, we measured the transmission through waveguides with varying bend radius. We observed constant transmission (with regard to an error similar to the one observed during the propagation loss measurements) down to 7 μm radius (as theoretically expected), followed by a sudden drop leading to non-measurable values.

4. Conclusions

In this paper, we have presented a detailed analysis of the geometric characteristics of a low loss TiO_2/polymer waveguide. This structure is highly flexible in terms of operation over a large wavelength range. Furthermore, because such a waveguide is able to operate with very tight bends (radius < 7 μm), it offers a tradeoff between the quality of transmitted light, the device footprint, and relaxed fabrication and therefore can suit different applications. The trench feature enabled us to shift the position of the mode away from the center of the waveguide while still maintaining a high confinement without any modification to the geometry of the waveguide. Due to the confinement of the mode within the high index region, small footprint devices are achievable enabling mass production of compact photonic circuitry. The ALD fabrication process ensures a conformal deposition of a low-loss material inside the trench. These features allowed us to achieve waveguides with a low value for propagation loss (1.75 ± 0.32 dB/cm). The geometry of the waveguide itself allows high versatility in terms of the hybridization of materials. We have shown in this paper that despite a thin layer of high refractive index all over the sample, the mode remained confined inside the core of the waveguide. It yields an opening for many possible material combinations to be deposited with or without the conformal method. Finally, we point out that the use of polymer makes the core of the waveguide more robust and resistant to any contamination, mechanical stress and, therefore, such a concept could be used, for instance, in flexible photonics for wearable sensing devices.

Author Contributions: Design, I.D. and K.O.; fabrication, O.O.; characterization, O.O., I.D. and M.R.; data analysis, I.D. and M.R.; writing, I.D., M.R. and K.O.; conceptualization, and supervision, M.R.; all authors contributed to the finalization of the manuscript. All authors have read and agreed to the published version of the manuscript.

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References

1. Almeida, V.R.; Xu, Q.; Barrios, C.A.; Lipson, M. Guiding and Confining Light in Void Nanostructure. *Opt. Lett.* **2004**, *29*, 1209. [CrossRef] [PubMed]
2. Kargar, A.; Chao, C.-Y. Design and Optimization of Waveguide Sensitivity in Slot Microring Sensors. *J. Opt. Soc. Am. A* **2011**, *28*, 596. [CrossRef] [PubMed]
3. Duguay, M.A.; Kokubun, Y.; Koch, T.L.; Pfeiffer, L. Antiresonant Reflecting Optical Waveguides in SiO_2-Si Multilayer Structures. *Appl. Phys. Lett.* **1986**, *49*, 13–15. [CrossRef]
4. Lai, C.-H.; You, B.; Lu, J.-Y.; Liu, T.-A.; Peng, J.-L.; Sun, C.-K.; Chang, H. Modal Characteristics of Antiresonant Reflecting Pipe Waveguides for Terahertz Waveguiding. *Opt. Express* **2010**, *18*, 309. [CrossRef] [PubMed]
5. Asquini, R.; Buzzin, A.; Caputo, D.; de Cesare, G. Integrated Evanescent Waveguide Detector for Optical Sensing. *IEEE Trans. Compon. Packag. Manuf. Technol.* **2018**, *8*, 1180–1186. [CrossRef]
6. Geoffray, F.; Bastard, L.; Broquin, J.-E. High Confinement Ion-Exchanged Waveguides for Nonlinear Applications. *Ceram. Int.* **2015**, *41*, 8034–8039. [CrossRef]
7. Vlasov, Y.A.; McNab, S.J. Losses in Single-Mode Silicon-on-Insulator Strip Waveguides and Bends. *Opt. Express* **2004**, *12*, 1622. [CrossRef]
8. Li, W.; Anantha, P.; Bao, S.; Lee, K.H.; Guo, X.; Hu, T.; Zhang, L.; Wang, H.; Soref, R.; Tan, C.S. Germanium-on-Silicon Nitride Waveguides for Mid-Infrared Integrated Photonics. *Appl. Phys. Lett.* **2016**, *109*, 241101. [CrossRef]
9. Kaplan, A.E.; Vitali, V.; Demontis, V.; Rossella, F.; Fontana, A.; Cornia, S.; Petropoulos, P.; Bellani, V.; Lacava, C.; Cristiani, I. Polarization Control in Integrated Silicon Waveguides Using Semiconductor Nanowires. *Nanomaterials* **2022**, *12*, 2438. [CrossRef]
10. Jabeen, F.; Chen, M.; Rasulev, B.; Ossowski, M.; Boudjouk, P. Refractive Indices of Diverse Data Set of Polymers: A Computational QSPR Based Study. *Comput. Mater. Sci.* **2017**, *137*, 215–224. [CrossRef]
11. Katritzky, A.R.; Sild, S.; Karelson, M. Correlation and Prediction of the Refractive Indices of Polymers by QSPR. *J. Chem. Inf. Comput. Sci.* **1998**, *38*, 1171–1176. [CrossRef]
12. Gao, J.; Xu, J.; Chen, B.; Zhang, Q. A Quantitative Structure-Property Relationship Study for Refractive Indices of Conjugated Polymers. *J. Mol. Model.* **2007**, *13*, 573–578. [CrossRef] [PubMed]
13. Geka, G.; Papageorgiou, G.; Chatzichristidi, M.; Karydas, A.G.; Psycharis, V.; Makarona, E. CuO/PMMA Polymer Nanocomposites as Novel Resist Materials for E-Beam Lithography. *Nanomaterials* **2021**, *11*, 762. [CrossRef] [PubMed]
14. Chen, K.-S.; Lin, I.-K.; Ko, F.-H. Fabrication of 3D Polymer Microstructures Using Electron Beam Lithography and Nanoimprinting Technologies. *J. Micromech. Microeng.* **2005**, *15*, 1894–1903. [CrossRef]
15. Sotomayor Torres, C.M.; Zankovych, S.; Seekamp, J.; Kam, A.P.; Clavijo Cedeño, C.; Hoffmann, T.; Ahopelto, J.; Reuther, F.; Pfeiffer, K.; Bleidiessel, G.; et al. Nanoimprint Lithography: An Alternative Nanofabrication Approach. *Mater. Sci. Eng. C* **2003**, *23*, 23–31. [CrossRef]
16. Alkaisi, M.M.; Jayatissa, W.; Konijn, M. Multilevel Nanoimprint Lithography. *Curr. Appl. Phys.* **2004**, *4*, 111–114. [CrossRef]
17. Pélisset, S.; Laukkanen, J.; Kuittinen, M.; Honkanen, S.; Roussey, M. Modal Properties of a Strip-Loaded Horizontal Slot Waveguide. *J. Eur. Opt. Soc.-Rapid Publ.* **2017**, *13*, 37. [CrossRef]
18. Pélisset, S.; Roussey, M. Strip-Loaded Nano-Photonics on Horizontal Slot Waveguide. *J. Eur. Opt. Soc.-Rapid Publ.* **2019**, *15*, 6. [CrossRef]
19. Roussey, M.; Ahmadi, L.; Pélisset, S.; Häyrinen, M.; Bera, A.; Kontturi, V.; Laukkanen, J.; Vartiainen, I.; Honkanen, S.; Kuittinen, M. Strip-Loaded Horizontal Slot Waveguide. *Opt. Lett.* **2017**, *42*, 211. [CrossRef]
20. Meng, X.; Byun, Y.-C.; Kim, H.; Lee, J.; Lucero, A.; Cheng, L.; Kim, J. Atomic Layer Deposition of Silicon Nitride Thin Films: A Review of Recent Progress, Challenges, and Outlooks. *Materials* **2016**, *9*, 1007. [CrossRef]
21. Kariniemi, M.; Niinistö, J.; Vehkamäki, M.; Kemell, M.; Ritala, M.; Leskelä, M.; Putkonen, M. Conformality of Remote Plasma-Enhanced Atomic Layer Deposition Processes: An Experimental Study. *J. Vac. Sci. Technol. A Vac. Surf. Film.* **2012**, *30*, 01A115. [CrossRef]
22. Pfeiffer, K.; Schulz, U.; Tünnermann, A.; Szeghalmi, A. Antireflection Coatings for Strongly Curved Glass Lenses by Atomic Layer Deposition. *Coatings* **2017**, *7*, 118. [CrossRef]
23. Mu, J.; Dijkstra, M.; Korterik, J.; Offerhaus, H.; García-Blanco, S.M. High-gain waveguide amplifiers in Si_3N_4 technology via double-layer monolithic integration. *Photon. Res.* **2020**, *8*, 1634–1641. [CrossRef]
24. Huang, X.; Cutinha, N.; Alcázar de Velasco, A.; Chandler, P.J.; Townsend, P.D. Upconversion in erbium doped YAG ion-implanted waveguides. *Nucl. Instrum. Methods Phys. Res. B* **1998**, *142*, 50–60. [CrossRef]
25. Papon, C.; Wang, Y.; Uppu, R.; Scholz, S.; Wieck, A.D.; Ludwig, A.; Lodahl, P.; Midolo, L. Independent operation of two waveguide-integrated single-photon sources. *arXiv* **2022**, arXiv:2210.09826.
26. Weiland, T. Time domain electromagnetic field computation with finite difference methods. *Int. J. Numer. Model. Electron. Netw. Devices Fields* **1996**, *9*, 295–319. [CrossRef]
27. Kong, F.; Li, K.; Liu, X. Accurate Analysis of Planar Optical Waveguide Devices Using Higher-Order FDTD Scheme. *Opt. Express* **2006**, *14*, 11796. [CrossRef]
28. Ramponi, R.; Osellame, R.; Marangoni, M. Two Straightforward Methods for the Measurement of Optical Losses in Planar Waveguides. *Rev. Sci. Instrum.* **2002**, *73*, 1117–1120. [CrossRef]

29. Daldosso, N.; Melchiorri, M.; Riboli, F.; Girardini, M.; Pucker, G.; Crivellari, M.; Bellutti, P.; Lui, A.; Pavesi, L. Comparison among Various Si/sub 3/N/sub 4 Waveguide Geometries Grown within a CMOS Fabrication Pilot Line. *J. Light. Technol.* **2004**, *22*, 1734–1740. [CrossRef]
30. Peczek, A.; Mai, C.; Winzer, G.; Zimmermann, L. Comparison of Cut-Back Method and Optical Backscatter Reflectometry for Wafer Level Waveguide Characterization. In Proceedings of the 2020 IEEE 33rd International Conference on Microelectronic Test Structures (ICMTS), Edinburgh, UK, 4–18 May 2020; pp. 1–3.
31. Singh, R.R. Dispersion Tailoring of Silicon Nanowire Optical Rectangular Waveguide (SNORW). *SN Appl. Sci.* **2020**, *2*, 502. [CrossRef]
32. Yuan, C.; Dai, J.; Jia, H.; Ding, J.; Zhang, L.; Fu, X.; Yang, L. Design of a C-Band Polarization Rotator-Splitter Based on a Mode-Evolution Structure and an Asymmetric Directional Coupler. *J. Semicond.* **2018**, *39*, 124008. [CrossRef]
33. Erim, M.N.; Erim, N.; Kurt, H. Optical Surface Modes of Photonic Crystals for Dual-Polarization Waveguide. *Photonics Nanostruct.-Fundam. Appl.* **2013**, *11*, 123–131. [CrossRef]
34. Ebers, L.; Hammer, M.; Förstner, J. Oblique Incidence of Semi-Guided Planar Waves on Slab Waveguide Steps: Effects of Rounded Edges. *Opt. Express* **2018**, *26*, 18621. [CrossRef] [PubMed]
35. Hayrinen, M.; Roussey, M.; Gandhi, V.; Stenberg, P.; Saynatjoki, A.; Karvonen, L.; Kuittinen, M.; Honkanen, S. Low-Loss Titanium Dioxide Strip Waveguides Fabricated by Atomic Layer Deposition. *J. Light. Technol.* **2014**, *32*, 208–212. [CrossRef]
36. Hegeman, I.; Dijkstra, M.; Segerink, F.B.; Lee, W.; Garcia-Blanco, S.M. Development of Low-Loss TiO_2 Waveguides. *Opt. Express* **2020**, *28*, 5982. [CrossRef]
37. Evans, C.C.; Liu, C.; Suntivich, J. Low-Loss Titanium Dioxide Waveguides and Resonators Using a Dielectric Lift-Off Fabrication Process. *Opt. Express* **2015**, *23*, 11160. [CrossRef]

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Paper **II**

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"Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing"
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Strip-loaded Mach–Zehnder interferometer for absolute refractive index sensing

Isaac Doughan✉, Kehinde Oyemakinwa, Olli Ovaskainen & Matthieu Roussey

This paper presents the design, fabrication, and characterization of a Mach–Zehnder interferometer (MZI) on a strip-loaded platform specifically developed for the measurement of refractive index variations in liquids. A novel approach of large opening in the sensing region enhances the sensitivity by using the analyte as the loading strip. This allows surpassing the performances of the conventional perforations and trenched-based MZIs and prevent protecting the reference arm. The implementation of this design resulted in a high confinement factor of approximately 23% in the cladding, enabling an effective interaction between the evanescent field and analyte. Ethanol-water solutions with varying concentrations were used as analyte for the characterization of the device. The strip-loaded waveguide was a 200 nm TiO_2 thin film on a silicon dioxide wafer and a 1.2 µm wide for 800 nm-thick nLOF resist served as loading strip. The sensing area featured a 10 µm wide open-ing in the loading material with a sensing length of 5 mm. The homogeneous sensitivity was experimentally determined to be 0.128, demonstrating the effectiveness of the proposed design, which enabled a refractive index change monitoring of 10^{-6}.

Mach–Zehnder interferometers (MZIs) offer numerous applications in several fields including for instance sensing[1,2], telecommunications[3,4], and quantum computing[5,6]. However, the design and integration of Mach–Zehnder interferometer sensors in integrated photonics platforms remain challenging, particularly with strip-loaded platforms[7]. A strip-loaded waveguide consists of a thin film of high refractive index confining vertically light and a strip patterned atop this slab waveguide allowing the channeling (2D confinement) of light[8]. This configuration creates a waveguide structure that guides and confines light along the strip region due to a change in the effective index of the slab waveguide in the region under the strip[9]. Strip-loaded waveguides offer advantages such as effective modal field confinement[10,11] and low loss[12,13]. It means the mode is buried and nearly independent from the surrounding media. This represents the challenge to overcome to make such a waveguide efficient for evanescent sensing applications. One of the challenges arises from the insufficient presence of light in the cladding for interactions with the analyte. Regular channel MZIs address this challenge by utilizing the analyte as the cladding[14,15] for strong effective index or phase modulation. It is not applicable to strip-loaded waveguides, where light is only weakly guided in the low-index strip atop the guiding layer, which serves to induce a local effective index change on the slab mode. The strip thickness improves light confinement within the guiding core but simultaneously reduces the light field in the cladding, which is crucial for evanescent field sensing. A direct transfer of concept from one platform to another is therefore not possible and the overall light path has to be rethought compared to classical channel photonic circuits in order to increase the performance of the device. In the realm of strip-loaded Mach–Zehnder interferometers, the prevailing method employs perforations and trenches, such as variously sized circles or diagonally placed ellipses[16]. However, these designs often lead to interferometers with reduced sensitivity, particularly for biochemical sensing, due to the difficulties in surface functionalization caused by the small size of the openings, which restricts liquid infiltration. This paper introduces an enhanced strip-loaded Mach–Zehnder interferometer design, distinguished by a substantial opening in its sensing arm, to measure refractive index variations in liquids and potentially gases. This innovation substantially boosts the interferometer's sensitivity by employing the analyte as the strip material, enhancing light-matter interaction within a precisely controlled segment of the sensor's arm, while the rest of the device remains largely unaffected by environmental changes. This design offers two main advantages: firstly, opening the low-index channel exposes the evanescent tail of the mode propagating in the slab, leading to an enhanced light and analyte interaction and an increased sensitivity. Secondly, it provides a larger surface area for surface functionalization, making it suitable for developing ultra-sensitive biochemical sensors with specific

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functionalization of the surface, in the future. Figure 1a is the schematic of cross section of the proposed strip-loaded waveguide. The substrate is an oxidized silicon wafer with 3 μm thick silicon dioxide (SiO_2). The guiding layer is titania (TiO_2) thin film with n_{TiO_2} =2.24 at λ =1550 nm created using atomic layer deposition (ALD).

The strip, enabling the channeling of the guided mode propagating in the TiO_2 slab, is made of a polymer resist AZ-2070 (nLOF) with refractive index n_{nLOF} = 1.601 at λ = 1550 nm[17]. Figure 1b shows the 3D sketch of the strip-loaded MZI configuration proposed and studied in this paper. The entire circuit is made of electron beam resist. It includes both the reference and the sample arms as well as the input and output channel waveguides allowing to inject and collect light. It is worth notifying that since nearly no light escapes above the loading strip, there is no need for a protection layer above it, i.e., the analyte can cover the entire sample without disturbing the measurement.

Materials and methods

Designing tools

Simulations leading to the design parameters of the Mach–Zehnder interferometer were carried out using the Finite Difference Time Domain (FDTD) method[18] through the commercial software Ansys Optics 2023 R1.2 suite. The choice of FDTD is because of the importance of precise design analysis. Moreso, Mach–Zehnder interferometers (MZIs) employed in our design leverage the interaction of optical waves over defined path lengths. This underscores the significance of employing time-domain simulations, a domain for which FDTD is quite notable. The adoption of FDTD, therefore, aligns with our objective of critically analyzing the behavior of optical signals within the MZI structure.

To ensure accurate and efficient simulations, the boundary conditions of the simulation region were defined as a perfectly matched layer (PML)[19,20], which are effectively absorbing scattered fields and unwanted reflections from the boundaries. A mesh size of 20 nm in x-, y-, and z-directions was chosen to achieve numerical convergence of the results while limiting the required computer power. The central illumination wavelength used in the simulations was set at λ = 1550 nm. The wavelength is chosen because of the low absorption by water in the short wavelength infrared range. Given the high computational demands and memory requirements for 3D-FDTD simulations, sections of the sensor, such as the splitter and the sensing region, were primarily simulated by 2D-FDTD using the effective index approximation[21]. This method helped to reduce computational time without compromising the accuracy of the results. Note that the effective index approximation is valid since the effective index contrast of the chosen waveguide is low, c.a., 0.1 refractive index unit. The data analysis was facilitated by using MATLAB. Mach–Zehnder Interferometer. The Mach–Zehnder interferometer (MZI) consists, here, of two Y-junctions placed head to tail and linked together. One is acting as a 50/50 power splitter separating the input light evenly between the two arms of the interferometer. The second one recombines the two signals and feeds the output waveguide. Their combination forms the two arms of the interferometer both having the same external geometry. One arm serves as a reference and no opening of the loading strip is made. The other arm

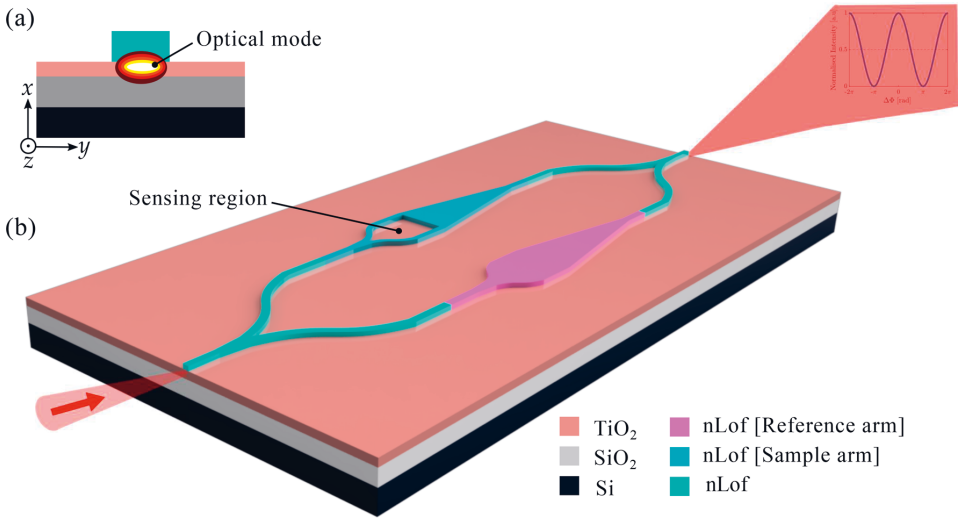

Figure 1. Strip-loaded Mach–Zehnder interferometer configuration. (**a**) Cross section geometry of the strip-loaded waveguide showing the region of mode propagation confinement. The substrate is an oxidized silicon wafer. The guiding layer is TiO_2 thin film with thickness of 200 nm. The strip atop is AZ-2070 - nLOF polymer. The dimensions of the polymer strip are 1200 nm wide and 800 nm high. (**b**). A 3D drawing of the strip-loaded Mach–Zehnder interferometer. Both the reference and sample arm are made of the same nLOF polymer. A large opening is performed in the sample arm to let the analyte in contact with the titania film. The reference arm is not influenced by the refractive index surrounding the MZI.

includes a large opening, detailed later in this paper, used to perform the sensing. The difference in refractive index between the reference and sample arms leads to a modulation of the output intensity, which depends on the optical path difference between the two arms and, therefore, only the refractive index of the analyte, which is the only varying parameter. The optical path must be extremely well defined in an interferometer with the Y-branch playing a crucial role as it sets the contrast between the reference and the sensing arm. Various types of optical splitters, including Y-branch splitters with varying complexities, have been employed to distribute optical signals across multiple outputs, serving diverse applications ranging from 1×2 to 1×128[22–27] configurations. A notable challenge with Y-branch splitters lies in achieving a balanced splitting ratio in the arms, as asymmetry can lead to an uneven distribution of power in the output waveguide. However, it has been demonstrated that these minor non-uniformities do not significantly impact the splitting characteristics of Y-branch splitters, particularly for configurations below 1×16[28]. In the context of this paper, a 1×2 splitter is sufficient. The standard predefined shapes for Y-branches are typically the S-bend arc and the S-bend sine. In the design of our 1×2 Y-branch splitter, we utilized the Bézier curve to define the geometric structure of our Y- branch, which is described by the equation[29] :

$$B(t) = \sum_{i=0}^{n} b_{i,n}(t) B_i \tag{1}$$

Here, B_i represents the control points of the curve and the Bernstein polynomial of degree n, denoted as $b_{i,n}(t)$, is expressed as follows:

$$b_{i,n}(t) = \left(\frac{n}{i}\right) t^i (1-t)^{n-1}, i = 0, 1, 2, \ldots, n \tag{2}$$

Compared to other predefined shapes, the Bézier curve offers more degrees of freedom by adjusting the control points and the degree of the Bézier curve in the Ansys Optics design environment. The design of the sensing arm involves opening the nLOF strip formed by two nLOF walls. It resembles a Y-branch with no guiding in the branches but only in the gap between them. It acts then as a taper enlarging the mode in the TiO_2 slab waveguide to form a larger sensing zone, while preventing losses due to leakage, since light remains trapped in the slab. When the gap between the two walls in the sensing region is far larger than the wavelength of the propagating mode, one can approximate the sensing region as a slab waveguide consisting of SiO_2 as the substrate, TiO_2 as the guiding layer and air as the cladding. The air cladding an open window for the analyte, i.e., the ethanol-water solution, application on the sensor. The reference arm has the same geometry as the sample arm except that no opening area is performed on the nLOF of strip, which acts as the reference material for the MZI sensor. The end of the sensing region consists of a tapper designed to collect and converge light from the sensing region making it guide under the nLOF strip again. The signals from the sensing arm and reference arm are coupled back through the combiner. The modes interfere with each other. The interference I of the two signals is given as[30,31]

$$I = I_S + I_R + 2\sqrt{(I_S I_R)} cos(\Delta\phi) \tag{3}$$

where I_S and I_R are the intensities of the sensing and the reference arm, respectively. $\Delta\phi$ is the phase difference between the reference and sensing signals, which is due to the difference of refractive index between the analyte and the reference material, i.e., nLOF. The refractive index of the analyte depends directly on the concentration of ethanol in water, which is our targeted measure. During the characterization process, the MZI's output is measured in terms of power. The output power of the MZI is given as

$$P_{out} = P_{in} cos^2(0.5\Delta\phi) \tag{4}$$

where P_{out} and P_{in} are the output and input power of the MZI, respectively. By varying the refractive index of the analyte and measuring the output power, the phase change $\Delta\phi$ can be calculated. The phase change between the reference arm and the sensing arm is described by[32]

$$\Delta\phi = \frac{2\pi}{\lambda} L \Delta n_{\text{eff}} \tag{5}$$

where L is the interaction length, λ is the wavelength of light in vacuum and Δn_{eff} is the difference in effective refractive index between the guiding modes. From Eq. 5, one can deduce that long sensing areas ensure high sensitivity even for very small refractive index variations in the sample arm. This makes MZI a very powerful tool to detect minute changes of the refractive index. Sensitivity refers to how much the sensor's output changes in response to a change in the refractive index of the surrounding medium (analyte). It is a measure of the responsiveness of the sensor to changes in the parameter being measured. In the proposed strip-loaded MZI, the analyte is deposited on the whole sensing region making the sensitivity of the device homogeneous. The homogeneous sensitivity S_H of the MZI is calculated as[33]

$$S_H = \frac{\delta n_{\text{eff}}}{\delta n_{\text{c}}} \tag{6}$$

which represents how the effective index of the propagating mode changes with the refractive index of the analyte in the cladding (n_c). The homogeneous sensitivity of the device obtained experimentally is simply the slope of the linear fitting for the calculated values of Δn_{eff} versus Δn_{c} relating to the varying concentrations of the ethanol-water solutions.

Fabrication techniques

3" oxidized silicon wafers (3 µm of thermal SiO_2, Si-Mat, Kaufering, Germany) are used as substrates. An O_2 plasma (PE-50 plasma cleaner, Plasma Etch) is performed to remove impurities at the surface of the wafer. The process flow is depicted in Fig. 2.

The required 200 nm thick TiO_2 film was formed by ALD (Beneq TFS200) using a thermal process with $TiCl_4$ and O2 as precursors (2857 cycles) at 120°C to ensure an amorphous material. After the ALD process, the sample was treated again with O2 plasma and spin-coated with the resist AZ 2070 nLOF diluted with AZ EBR (2:3). 800 µl of the dilution was pipetted and spin-coated at 1500 rpm for 60 s. The sample was soft baked on a hot plate at 110 °C for 60 s. A coating thickness of 799 nm was measured afterwards with a profilometer (Dektak 150). The sample was exposed with electron beam lithography (EBL) system (Raith EPBG5000+HR). Doses of 40 and 60 $\mu C/cm^2$ were used in the exposure. The sample was post baked at 110 °C for 60 s and developed in AR 300-47 (Allresist GmbH) for 90 s, stopped in DI water for 30 s and rinsed under DI water before being dried with N2 blow. Finally, the several chips on the wafer were manually cleaved to be ready for characterization.

Analyte sample preparation

Three Ethanol-water solutions were prepared with mass concentrations of $c_1 = 0.00178$ % (w/w), $c_2 = 0.0167$% (w/w), and $c_3 = 0.178$% (w/w)[34]. The required amounts of ethanol (solute) and deionized water (solvent) were measured accurately using a precise Sartorius Entris analytical balance (ENTRIS224i-1S/0.1mg). For each solution, a specified mass of ethanol was diluted with an appropriate amount of deionized water to achieve a total mass of 100 g. A thorough mixing ensured the homogeneity of the solutions. The refractive index of the solvent at $\lambda = 1550$ nm was taken as 1.3154 and the refractive index of ethanol as 1.3520 at the same wavelength[35]. The refractive index change of the ethanol-water solutions increases with the ethanol concentration and is calculated using the formula $\Delta n_c = 0.0006 \times C_{eth}$, where Δn_c represents the refractive index change and C_{eth} is the mass concentration of ethanol[36].

Characterization methods

The strip-loaded MZI was characterized using the setup configuration shown in Fig. 3[37]. A power meter is connected to a computer for the recording of the measured power. The end-fire coupling method was used to inject and collect light. Light from a tunable laser source (Tunics plus), set at $\lambda = 1550$ nm, is coupled into the input of the sensor using a tapered lens fiber (OZ Optics). The output power is measured and recorded using a power meter with near infrared detector (Thorlabs PM100D with S122c detector) and stored on a computer. The desired polarization (TE) of the input light was set with the help of a polarization controller and verified with a

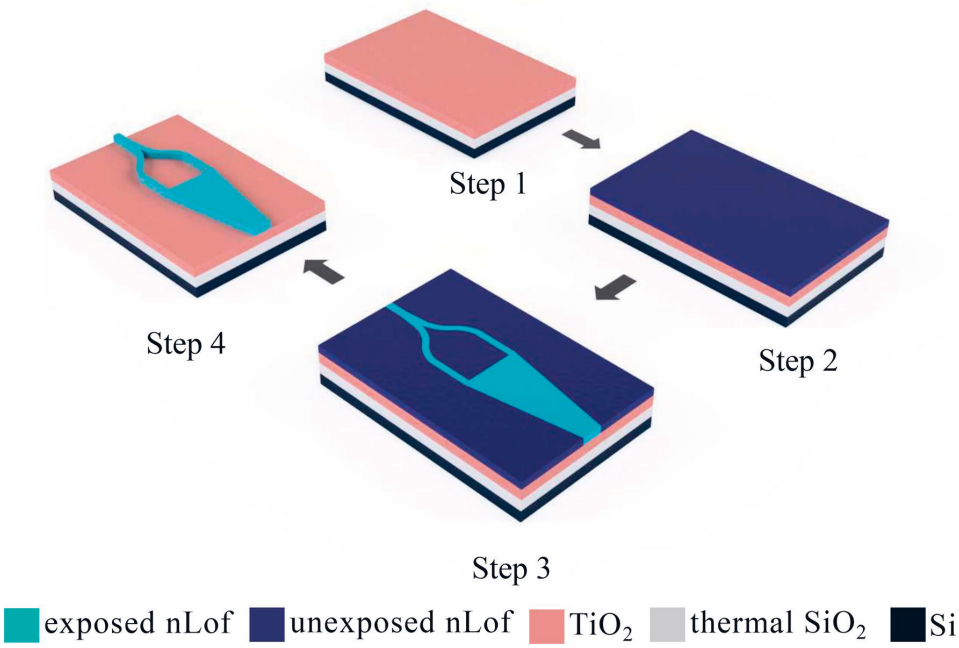

Figure 2. Fabrication process flow. Step1: ALD coating of the substrate with a 200 nm TiO_2. Step 2: spin coating of the nLOF resist. Step 3: electron beam lithography. Step 4: development of nLOF, cleaning of the wafer, and cleaving of the chips prior to characterization.

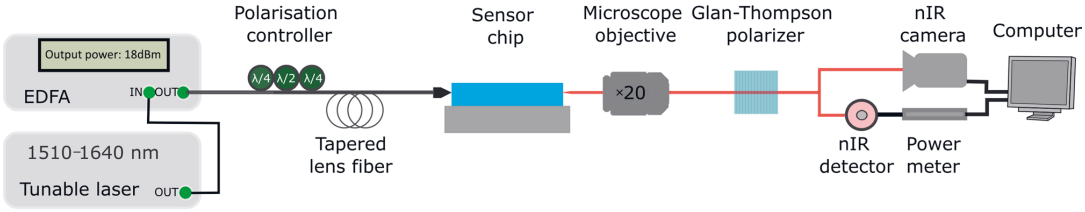

Figure 3. Schematics of the sensor characterization setup. The detector end can be switched from power meter to camera.

Glan-Thompson GTH10M polarizer from Thorlabs. A microscope objective (Olympus Plan N, N.A. = 0.4, 20×) was used to image the output mode on the detector of the power meter. Alternatively, to the detector, the mode can be observed using Point Grey Chameleon CMLN-13S2M camera, allowing a verification, at any time of the experiment, a well-coupling to the waveguide without any modification of the measurement system. The power values were recorded for various concentrations of the ethanol-water solution. Ideally, one would consider the use of a fluidic cell, but the nature of our sensor does not necessarily require it, since the reference arm is not dependent on the outer medium. Experimental procedure. To ensure precision and avoid contamination, we carefully transferred samples from each prepared solution into separate containers, using pipettes dedicated to each solution. The MZI chip was mounted onto a translational stage using custom-built 3D printed sample holders. To illuminate the sensor, a TE light source (TE/TM polarization fractions of 99.7%) was injected into the input of the MZI. The power measurement using the software was started. The output with air as cladding was recorded and used as the baseline for all measurements. With the help of a pipette (FINNPIPETTE), we deposited 20 µl of water on the sensing region and the output power was recorded. With the help of a balloon end pipette, water was sucked up of the chip and the power measurement was also observed till the power value stabilized to the baseline. The process was repeated for all the solutions. All the results were recorded in one file for further analysis under MATLAB.

Results and discussion

Optimization of the waveguide parameters. When selecting the waveguide parameters for refractive index sensing applications, an essential figure of merit is the confinement factor of light in the cladding. This factor determines how much the optical mode extends outside the slab waveguide, which dictates first the light/analyte interaction in the opening area and second, the no-interaction with the very external medium when the nLOF strip is present. Additionally, it is crucial to ensure that the waveguide supports only the fundamental quasi-TE mode for enhanced signal strength. Moreover, the variation of effective index between the regions with analyte and with nLOF resist is also linked to the confinement factor, which then affects the modulation of the output signal. Therefore, the geometry of the waveguide is optimized to meet these requirements.

Modal effective index

The waveguide configuration used in this work has initially been studied through nano-imprint replication technique. The results showed propagation losses for the fundamental mode in TiO_2 slab waveguide of about 1 dB/cm using prism coupler measurement, which makes it a suitable waveguide for this work. We have studied through FDTD simulations the effective index variation of the fundamental TE_{00} mode and the first order TE_{01} mode. The refractive indices considered for the materials have been measured using prism coupler (Metricon) and are at $\lambda = 1550$ nm: $n_{TiO_2} = 2.2452$, for titanium dioxide deposited by ALD. $n_{SiO_2} = 1.444$, for the silicon oxide layer on the silicon wafer substrate. $n_{nLOF} = 1.5991$, for the strip material. To ensure proper operation of the MZI, the evanescent tail of the guided mode should not interact with the medium above the loading-strip in the non-sensing regions. It means that the thickness of nLOF should be large enough. For this reason, the thickness of the nLOF strip was chosen to be 800 nm. Figure 4a shows the variation of the effective index of the fundamental quasi-TE mode as a function of the nLOF width for a fixed thickness of TiO_2 and nLOF of 200 nm and 800 nm, respectively. It is obvious from Fig. 4a that, below a strip width of 1500 nm, only the fundamental TE_{00} mode is sustained and guided in the waveguide.

It can also be observed that within the nLOF width of interest, effective index of both the fundamental and first order TE modes increases with an increase in the nLOF width. For this reason, we chose the width of the nLOF strip to be 1200 nm. This ensures good confinement of the mode with weak amount of light on the side of the strip and also considers the fabrication tolerances. Figure 4b shows the electric field profile of the fundamental mode solved using the Finite Difference Element (FDE) mode solver in Ansys Optics mode. The choice of appropriate geometry for a waveguide for evanescent sensing goes beyond considering the effective index only. One must also consider the confinement factor of the electric field in the cladding of the waveguide. The confinement factor in the cladding is important as it determines the amount of light that interacts with the analyte. This plays a key role in determining the sensitivity of the sensor. The relationship between the effective index and the confinement factor is inversely proportional. This means that as the effective index of the waveguide increases, the confinement factor (in the cladding) reduces in general. It implies that, the stronger the field in the cladding region for an efficient analyte and light interaction, the lower the confinement of light in the TiO_2 guiding layer, which is usually synonym of higher propagation losses. One must make a trade off between the

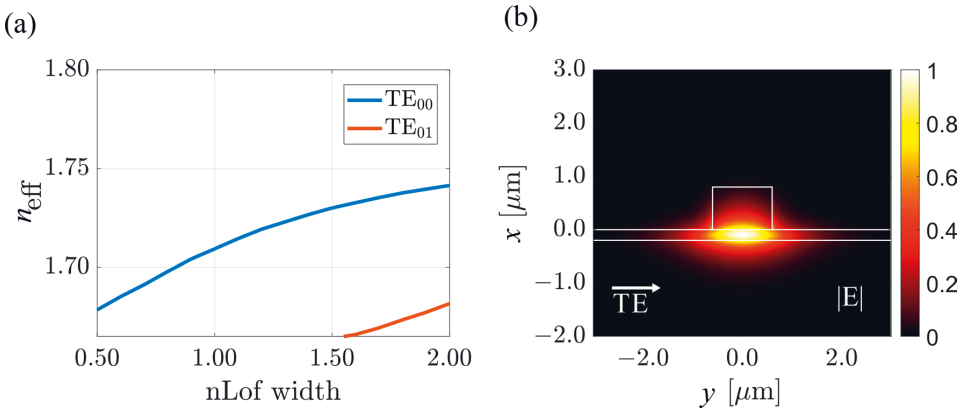

Figure 4. Waveguide optimization. (**a**) Effective index of the strip-loaded waveguide as a function of the nLOF strip width for the fundamental TE_{00} and the first order TE_{01} mode. The thicknesses of both TiO_2 and nLOF polymer are kept constant at 200 nm and 800 nm, respectively. (**b**) TE_{00} mode distributions with nLOF width of 1200 nm.

effective index and the confinement factor to obtain waveguide structure satisfying both a high sensitivity and low propagation losses. For the chosen parameters, we obtain a confinement factor in the analyte of $\Gamma_{\text{Analyte}} \approx$ 23%. This will be explained further in this section. Mach–Zehnder interferometer. Here, we discuss the main results in the development of the Y-branch splitter and the sensing area which are the most important parts of the strip-loaded MZI. The Y-branch splitter. Figure 5a shows the normalized transmittance at the output of the splitter as a function of the S-bend length of the splitting part.

At this stage, the sample and the reference arm are similar. Beyond an S-bend length of 150 µm, the transmittance changes insignificantly. Figure 5b shows the electric field profile along the Y-branch splitter. The length of the S-bend is 150 µm and the gap between the two arms is 50 µm. One can observe from the field profile that the input beam splits evenly between the two branches and remains guided without leaking in the slab. The sensing area. The sensing area of the proposed MZI employs a wide opening of the nLOF strip turning the sensor area into a 3-layer asymmetric slab waveguide with the air space as the cladding where the analyte will be further deposited.

This technique is possible for a strip-loaded waveguide since the TiO_2 core region ensures the vertical trapping of light. The modulation of the effective index in the sensing region is greatly influenced by the portion of electric field confined in the cladding (analyte). Figure 6a,b shows the confinement factor as a function of both the TiO_2 thickness and the analyte refractive index. A relatively high confinement of light of about 23% in the analyte can be obtained. The confinement factor, a fraction of the optical energy confined within the sensor's interaction region, is especially crucial because it plays a pivotal role in influencing the sensor's performance. A high confinement factor as in our case not only ensures a considerable amount of optical energy interaction but also extends the interaction length ensuring minute changes in refractive index or analyte concentration are

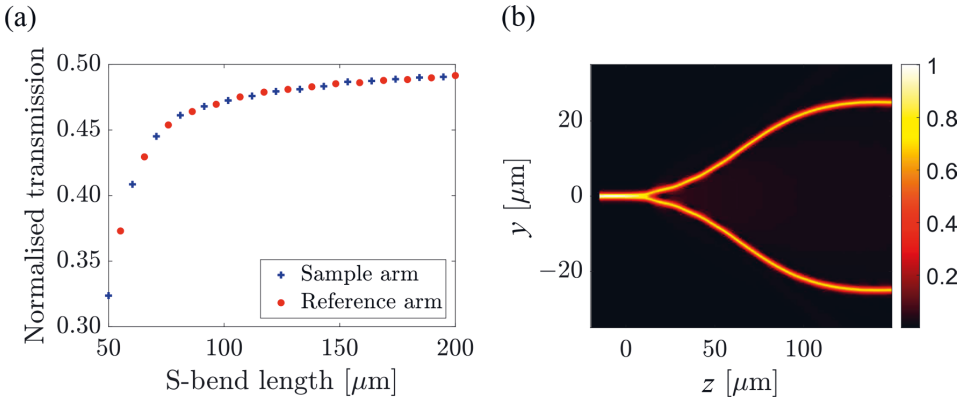

Figure 5. (**a**) Normalized transmittance at Y-branch splitter outputs as a function of the S-bend length for both the sample arm (plus blue curve) and reference arm (dotted red curve). (**b**) Electric field distribution along the Y-branch splitter with an S-bend length of 150 µm. The gap between the two arms is 50 µm.

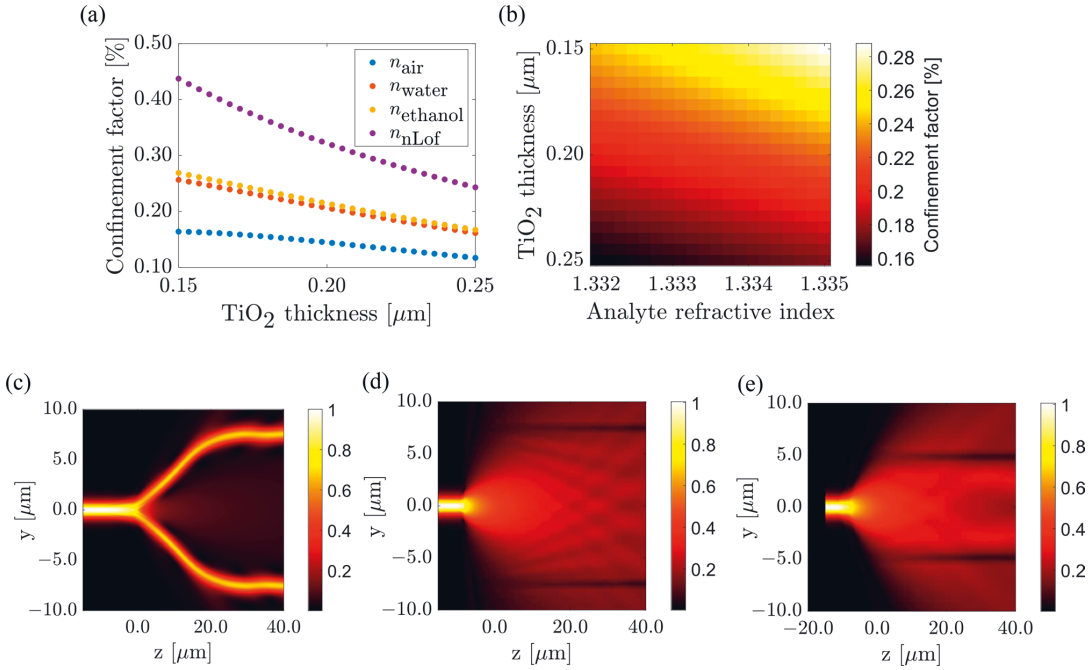

Figure 6. Sensing region simulations (**a**) Confinement factor analysis for different analytes at varying TiO_2 thickness in the sensing region (**b**) Influence of the TiO_2 thickness and the refractive index of interest on the confinement factor in the sensing region. Electric field distribution at the input of the sensing region (top view) for S-bend length of (**c**) 40 μm and (**d**) 5 μm, with 15 μm gap between the arms. (**e**) Corresponds to a bend of 5 μm and a gap of 10 μm.

detected. This prolonged interaction contributes to improved resolution, heightened signal-to-noise ratio, and ultimately enhances overall device efficiency of our MZI.

It can also be seen that the confinement factor is inversely proportional to the thickness of TiO_2, as expected in view of effective index trend, for a fixed refractive index of analyte. Aiming at a high confinement factor in the analyte, for higher sensitivity, the mode will be poorly confined, and the losses will be high and vice versa. Improving the confinement of the mode in the TiO_2 also means low confinement factor in the analyte hence low sensitivity. This means that there is a trade off to be made between the sensitivity and losses when designing evanescent field sensors. The sensing region is similar to a Y-branch but light must not be guided by them. Its design is thus performed using the opposite figure of merit than for the splitters forming the MZI. The objective is to lose the propagation along the S-bend so that the signal can propagate straight and experience the refractive index influence of the analyte. The results from the sensing area with water as cladding material are shown in Fig. 6c–e. Figure 6c,d show the electric field distribution simulated for an S-bend length of 40 μm and 5 μm, respectively. From Fig. 6c, the beam splits and remains guided by the loading strip despite some weak leaking in the slab. On the opposite, from Fig. 6d, the incoming beam is propagating mainly in the open area, ignoring the tight bend and further waveguides. The two arms in the sensing region have the same geometry. The portion of light propagating in the TiO_2 core in the sensing region exhibits a diverging and lateral expansion behavior which causes some light leakage. This can be mitigated by adjusting the width of the sensing area: the smaller the better. One can also note some interference pattern, similar to Talbot effect, coming from the reflection of the mode on the side channel waveguides. This effect is however negligible due to the low effective index contrast and can be further reduced by reducing the gap between the arms to 10 μm, as shown in Fig. 6e. Note that the width of the sensing area is dependent upon the output tapering part at the end of the sensing region. The purpose of the taper is to collect and guide the signals from the sensing region and direct it back to the strip-loaded waveguide. The low index contrast of the waveguide requires that the taper section becomes long as the light modulation inside the guiding layer is small. Consequently, the length of the designed taper was set to 2 mm. This is to ensure that all possible light coming from the sensing region is collected and guided for a nearly adiabatic operation.

Fabricated sample

Figure 7 compiles the scanning electron microscopy (SEM) pictures of the main parts of the fabricated MZI. Figure 7a shows the Y-branch splitter of the MZI with a long S-bend. Figure 7b shows the input of the sensing region with a very short S-bend. Figure 7c shows the end of the sensing region and the beginning of the taper

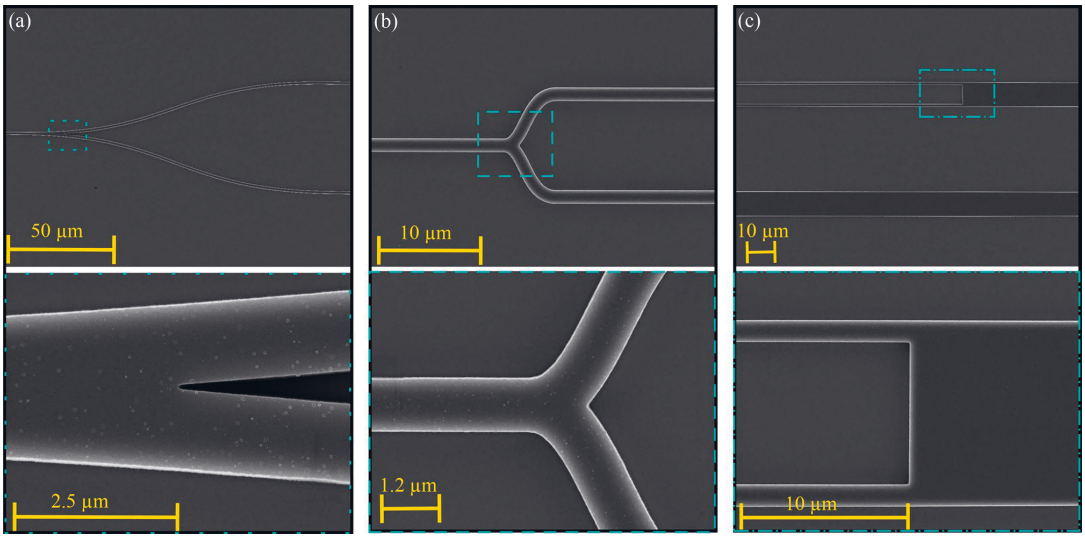

Figure 7. Scanning electron microscope pictures (top views) of different sections of the Mach–Zehnder interferometer. (**a**) Y-branch splitter with a zoomed in on the separation of the two arms. (**b**) Input of the sensing region showing part of the sample arm. (**c**) Latter part of the sensing region and beginning of the taper section.

section. Refer to Fig. 1b for the definition of the different sections of the MZI. In Fig. 7c, one can see the two arms, one with an opening and the other without.

Experimental characterization

A specific sample was fabricated to study the splitters used in the construction of the MZI. Figure 8a shows the output fundamental mode of the Y-branch splitter. As can be seen the input beam splits nicely and the Y-branch splitter works effectively well. This has been verified by measuring the intensity at the output of both waveguides. Figure 8b shows the output mode image of the MZI with air as the cladding material in the sensing region. The reference arm is isolated from any external perturbation and the variation in the sensing arm is only due to the change in concentration of the ethanol-water solution. The measured output power obtained from the power meter is plotted as a function of the analyte refractive index in Fig. 8c. The output power follows a sinusoidal curve as predicted by the theory and such function was used to fit the experimental data. It is to be noted that the length of the sensing zone is a means of setting an offset in terms of path length difference. It has been chosen so that the range of refractive index corresponds to the maximal variation (highest slope in the sine function) of the interference response. The maximum estimated output power when light in the two arms is in phase is about 30 µW. It can also be seen from Fig. 8c that the output power at the output of the MZI interferometer decreases with an increase in refractive index within the range of our analyte which is in line with our simulations. According to the theory ruling the MZI, it shows that the data points are the highest slope of the sine function allowing for the highest sensitivity, as expected. Figure 8d shows a plot of the effective index change as a function of refractive change of the ethanol-water concentration. Note that the data is represented on a semi logarithmic scale to render better the large range of refractive index. The correlation coefficient of the fit is $\rho = 0.9987$, which shows an excellent validity for the model and proves that only the refractive index change of the analyte affects the sensing results. The slope of the fitting curve determined by Eq. 6 gives a homogeneous sensitivity of $S_H = 0.128$, which is excellent according to the state-of-the-art[38]. The operating range in terms of concentration of the studied MZI is of several orders of magnitude ranging from 0.00178 % (w/w) to 0.178 % (w/w) and corresponding to a smallest refractive index change of the analyte of 1.10^{-6}. This is limited by the uncertainty on the measurements that can be greatly reduced by pig-tailing the chip device and building a fluidic cell to ensure an easier application of the solution on the sensing area. Moreover, the range of refractive index, i.e., the range of possible measurable concentration can be tuned by adding an offset to the reference arm. This can be done very accurately by changing slightly the effective index of its fundamental mode, e.g., by varying the width of the loading strip locally, or for a bigger offset, by changing its length. This counts among the advantages of using a strip-loaded waveguide as a platform for sensing.

Conclusion and outlook

A strip-loaded Mach–Zehnder interferometer with a wide opening in the sensing arm has been demonstrated where the variation in the sensing region is detected by measuring the changes in the output power. The MZI features a wide opening in the sensing arm which yields a confinement factor of 23%. The fabrication techniques

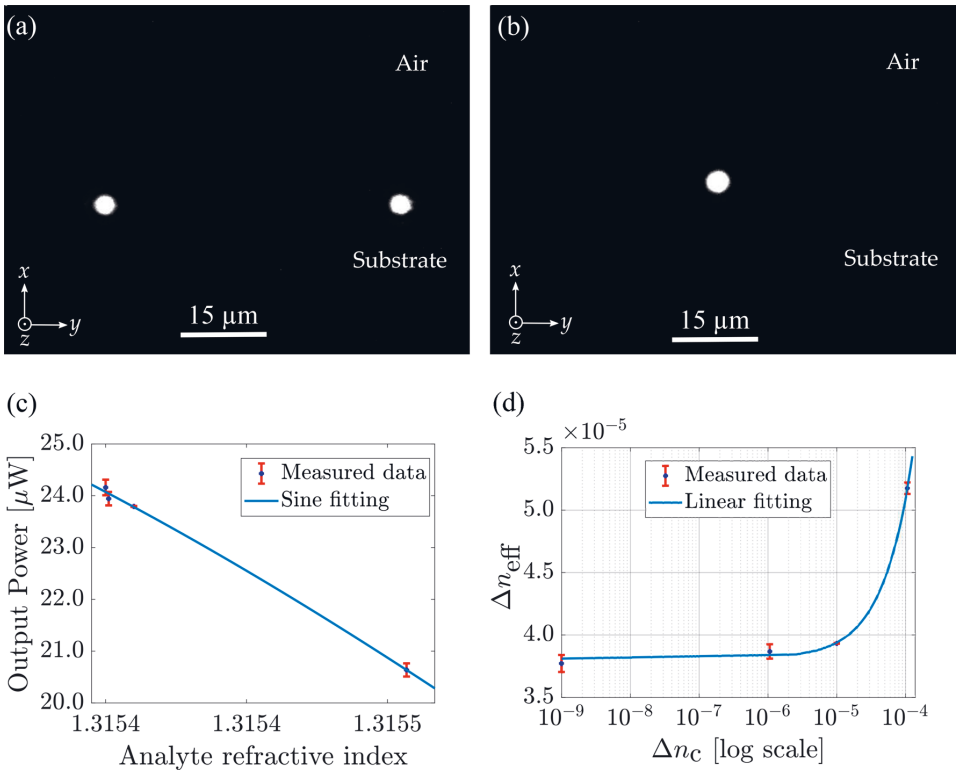

Figure 8. Characterisation results. (**a**) Photograph of the TE fundamental mode at the output of the Y-branch splitter with S-bend length of 150 µm. The separation between the two arms is 50 µm. (**b**) Photograph of the TE fundamental mode at the output of the strip-loaded Mach–Zehnder interferometer with air as cladding material in the sensing region. (**c**) Measured output power as a function of the analyte refractive index with sine fitting. (**d**) Calculated effective index change as a function of refractive index change of the analyte with fitting in microns. The sensing length is 5 mm. The slope represents a homogeneous sensitivity of 0.128.

employed resulted in fine structures allowing interferometry at the most effective response due to the high control of the dimensions. The analyte used was varying concentration of ethanol-water solutions. The homogeneous sensitivity of the sensor was experimentally determined to be 0.128. We demonstrated a minimal variation of refractive index of 1.10^{-6}. for a concentration variation of 0.00178% (w/w). Looking ahead, the development of our strip-loaded Mach–Zehnder interferometer sets the stage for advancing sensor technologies. The precision it offers and its immunity to environmental medium variations due to the buried mode (except for the sensing area) form a key-building block for the design smart sensors using strip-loaded platforms. Integrating this interferometer with neural networks holds the promise of enhancing sensor capabilities, enabling real-time analysis, and providing adaptive responses to dynamic environmental changes. This synergy could usher in a transformative era of intelligent sensor systems across diverse applications.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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References

1. Psarouli, A. *et al.* Monolithically integrated broad-band Mach–Zehnder interferometers for highly sensitive label-free detection of biomolecules through dual polarization optics. *Sci. Rep.* **5**, 17600 (2015).
2. Janik, M., Koba, M., Celebańska, A., Bock, W. J. & Śmietana, M. Live E. coli bacteria label-free sensing using a microcavity in-line Mach–Zehnder interferometer. *Sci. Rep.* **8**, 17176 (2018).
3. Zhao, Z. *et al.* Eight-channel LAN WDM (DE) multiplexer based on cascaded Mach–Zehnder interferometer on SOI for 400 GBe. In *Photonics*, **9**, 252 (MDPI, 2022).
4. Oguma, M. *et al.* Four-channel flat-top and low-loss filter for wide passband WDM access network. *Electron. Lett.* **37**, 1 (2001).

5. Pandey, D. *et al.* A proposal for evading the measurement uncertainty in classical and quantum computing: Application to a resonant tunneling diode and a Mach–Zehnder interferometer. *Appl. Sci.* **9**, 2300 (2019).
6. Taraphdar, C., Chattopadhyay, T. & Roy, J. N. Mach–Zehnder interferometer-based all-optical reversible logic gate. *Opt. Laser Technol.* **42**, 249–259 (2010).
7. Ding, R. *et al.* Low-loss strip-loaded slot waveguides in silicon-on-insulator. *Opt. Express* **18**, 25061–25067 (2010).
8. Furuta, H., Noda, H. & Ihaya, A. Novel optical waveguide for integrated optics. *Appl. Opt.* **13**, 322–326 (1974).
9. Ahmadi, L. *et al.* Strip-loaded waveguide on titanium dioxide thin films by nanoimprint replication. *Opt. Lett.* **42**, 527–530 (2017).
10. Munowitz, M. & Vezzetti, D. J. Lateral confinement in generalized strip-loaded optical waveguides. *J. Appl. Phys.* **68**, 5375–5377 (1990).
11. Munowitz, M. & Vezzetti, D. J. Mode structure and lateral confinement in strip-loaded optical waveguides: Effects of asymmetric cladding. *J. Lightwave Technol.* **10**, 426–431 (1992).
12. Häyrinen, M. *et al.* Low-loss titanium dioxide strip waveguides fabricated by atomic layer deposition. *J. Lightwave Technol.* **32**, 208–212 (2013).
13. Vezzetti, D. J. & Munowitz, M. Design of strip-loaded optical waveguides for low-loss coupling to optical fibers. *J. Lightwave Technol.* **10**, 581–586 (1992).
14. Zhao, K. *et al.* Optimization of light-analyte interaction in Si3N4/polymer hybrid waveguide for sensitive sensing of pyridine vapor with ppb-level detection limit. *Sens. Actuators B Chem.* **377**, 133104 (2023).
15. Hofmann, M. *et al.* Asymmetric Mach–Zehnder interferometers without an interaction window in polymer foils for refractive index sensing. *Appl. Opt.* **55**, 1124–1131 (2016).
16. Pélisset, S. & Roussey, M. Strip-loaded nano-photonics on horizontal slot waveguide. *J. Eur. Opt. Soc. Rapid Pub.* **15**, 1–9 (2019).
17. Pélisset, S., Laukkanen, J., Kuittinen, M., Honkanen, S. & Roussey, M. Modal properties of a strip-loaded horizontal slot waveguide. *J. Eur. Opt. Soc. Rapid Pub.* **13**, 1–8 (2017).
18. Taflove, A., Oskooi, A. & Johnson, S. G. *Advances in FDTD computational electrodynamics: photonics and nanotechnology* (Artech House, 2013).
19. Komatitsch, D. & Tromp, J. A perfectly matched layer absorbing boundary condition for the second-order seismic wave equation. *Geophys. J. Int.* **154**, 146–153 (2003).
20. Shin, W. & Fan, S. Choice of the perfectly matched layer boundary condition for frequency-domain maxwell's equations solvers. *J. Comput. Phys.* **231**, 3406–3431 (2012).
21. Hammer, M. & Ivanova, O. V. Effective index approximations of photonic crystal slabs: A 2-to-1-d assessment. *Opt. Quant. Electron.* **41**, 267–283 (2009).
22. Sharma, B., Kishor, K., Sharma, S. & Makkar, R. Design and simulation of Mach– Zehnder interferometer based directional coupler for optical coherence tomography. In *ICOL-2019: Proceedings of the International Conference on Optics and Electro-Optics, Dehradun, India*, 715–718 (Springer, 2021).
23. Gašo, P. *et al.* 3d polymer based 1x4 beam splitter. *J. Lightwave Technol.* **39**, 154–161 (2020).
24. Das, S. & Pal, S. A simple silica-on-silicon technology-compatible design of 1× 8 optical splitter based on field matching y-branch with s-bend sine taper and arc waveguide. *J. Opt.* **38**, 177–190 (2009).
25. Pal, S. *et al.* Fabrication and performance-analysis of a planar silica-based cascaded symmetric y-branch 1× 8 optical power splitter. *J. Opt.* **38**, 149–159 (2009).
26. Tao, S. *et al.* Cascade wide-angle y-junction 1× 16 optical power splitter based on silicon wire waveguides on silicon-on-insulator. *Opt. Express* **16**, 21456–21461 (2008).
27. Sakamaki, Y., Saida, T., Tamura, M., Hashimoto, T. & Takahashi, H. Low-loss y-branch waveguides designed by wavefront matching method and their application to a compact 1× 32 splitter. *Electron. Lett.* **43**, 1 (2007).
28. Patnaik, N. & C.P, V. A comparative study of litao3 y-branch symmetric and asymmetric optical power splitter with s-bend waveguides with various parameters. *IJITEE***8**, 3280–3282 (2019).
29. Duncan, M. Bézier curves ii. In *Applied Geometry for Computer Graphics and CAD*, 161–186 (Springer, 2005).
30. Choi, H. Y., Kim, M. J. & Lee, B. H. All-fiber Mach–Zehnder type interferometers formed in photonic crystal fiber. *Opt. Express* **15**, 5711–5720 (2007).
31. Villatoro, J., Minkovich, V. P., Pruneri, V. & Badenes, G. Simple all-microstructured-optical-fiber interferometer built via fusion splicing. *Opt. Express* **15**, 1491–1496 (2007).
32. Musa, R. K., Al deen Adnan, S. & Hammadi, A. M. Evanescent field Mach–Zehnder interferometer sensor for concentration measurement. *J. Madenat Alelem Univ. College***10**, 44–55 (2018).
33. Qi, Z.-M., Matsuda, N., Itoh, K., Murabayashi, M. & Lavers, C. A design for improving the sensitivity of a Mach–Zehnder interferometer to chemical and biological measurands. *Sens. Actuators B Chem.* **81**, 254–258 (2002).
34. Ahmadi, L. *et al.* Hybrid layered polymer slot waveguide young interferometer. *Opt. Express* **24**, 10275–10285 (2016).
35. Kedenburg, S., Vieweg, M., Gissibl, T. & Giessen, H. Linear refractive index and absorption measurements of nonlinear optical liquids in the visible and near-infrared spectral region. *Opt. Mater. Express* **2**, 1588–1611 (2012).
36. Lide, D. R. *CRC handbook of chemistry and physics*, **85** (CRC Press, 2004).
37. Doughan, I., Oyemakinwa, K., Ovaskainen, O. & Roussey, M. Low loss vertical tio2/polymer hybrid nano-waveguides. *Nanomaterials* **13**, 469 (2023).
38. Peng, C. *et al.* Optical waveguide refractive index sensor for biochemical sensing. *Appl. Sci.* **13**, 3829 (2023).

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Author contributions

Design, I.D.; Fabrication, O.O. and I.D.; Characterization, I.D. and M.R.; data analysis, I.D. and M.R.; writing, I.D., M.R. and K.O.; conceptualization, and supervision, M.R. All authors reviewed and have given approval to the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Paper **III**

I. Doughan, A. Halder, I. Reduto, M. Koivurova, T. Aalto, M. Roussey, and J. Turunen
"Determination of mode strengths in channel waveguide from the complex electric field"
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Determination of mode strengths in channel waveguide from the complex electric field

Isaac Doughan[1]✉, Atri Halder[1], Igor Reduto[1], Matias Koivurova[1], Timo Aalto[2], Matthieu Roussey[1] & Jari Turunen[1,3]

We show that the mode strengths of a guided field in an arbitrary asymmetric channel waveguide can be uniquely determined from self-referencing interferometric measurements at the exit plane of the waveguide. This requires knowledge of both the amplitude and phase of the complex electric field distribution. Although the amplitude can be obtained from the measured intensity profile easily, the phase retrieval is usually non-trivial. We develop an innovative, alternative and promising technique, where the complex cross-spectral density (CSD) function is measured using a customized wavefront folding interferometer. We then construct the total electric field (complex valued), from which we can determine the strengths of the allowed modes for an asymmetric strip waveguide. Our retrieval algorithm also provides the phase information (intermodal dispersion) associated with each mode, directly from the measured electric field distribution. Moreover, we experimentally demonstrate the developed scheme for different in-coupling (butt-coupling) conditions, resulting in different modal strength distributions.

Applications of integrated photonics in fields such as sensing using fluorescence[1,2], up- or down-frequency light conversion[3,4], photonic computing[5], and other low-photon sources[6] require a precise understanding of photonic circuitry. Specifically, bus waveguides[7] are used as a supporting platform for the functional building blocks, creating the necessary operations along the light path.

While designing a single-mode channel waveguide for a specific wavelength range is relatively easy, it becomes more challenging for very broadband operations, especially when light is generated inside the waveguide for source integration[8,9]. In such cases, several modes may establish, perturbing the signals and modifying the operation of different integrated elements along the circuit. Moreover, since only quasi-polarized modes, and not fully polarized modes, exist in a channel waveguide, signals at the same wavelength can interfere, yielding varied output responses. As a consequence, it is crucial to determine the strength of each mode in such a waveguide. In this article, we propose and present a straightforward yet unique approach to solving this problem.

Modes are ubiquitous across different branches of science. They can be found in many systems described by a partial differential equation supplemented with boundary conditions. A prime example of modal descriptions is usual continuous wave lasers based on stable resonators. Lasers can support multiple transverse modes, depending on the geometry of the laser cavity. Laser modes are said to be uncorrelated because they oscillate at different frequencies[10,11], which leads to reduced spatial coherence. An algorithm to find the modal strengths of partially coherent lasers was introduced in Ref. [12] and experimentally verified in Ref. [13]. An alternative way to determine the modal profiles and their relative strengths was presented in Ref. [14], where an edge-emitting semiconductor diode was chosen for the illustration.

However, modes excited in waveguides by spatially coherent light are correlated, and yield spatially coherent output light. Fortunately, there is also a way to determine the weights of mutually correlated modes. Knowledge of the complex electric field at the exit (or image) plane of the waveguide alone is sufficient to determine the weights of the modes. Mode strength determination of a multi-mode fiber, based on diffractive optical elements [15], SLM based digital holographic techniques [16,17] have been proposed and demonstrated. These techniques are based on overlap integral methods. In this article, we present an experimental procedure based on the measurement of the total electric field at the exit plane of a channel waveguide, enabling the determination of the strength of each mode of a multimode waveguide once coupled to a proper retrieval algorithm.

Several attempts have been made on determining the modal strengths of multimode/few-mode waveguides or fibers [18–24]. In most cases, the modal strengths/weights are estimated from the total intensity at the output plane of the waveguide, while excited with different modal strength distribution. Although these approaches

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provide a fair estimate of the modal strengths, the solutions are not unique. Only complete knowledge of the total (complex-valued) electric field profile at the output can provide a unique solution.

This article deals with the direct retrieval of the strength of each individual mode sustained by a channel waveguide from the total electric field at the exit plane. In section "Concept" we present a conceptual overview of mode strength retrieval process. Section "Modal representations in waveguides" discusses modal representation in waveguides, explaining how modes are represented within the waveguide structure. In Section Complex field and modal strength retrieval, we delve into complex field and modal strength retrieval, presenting equations involving cross-spectral density as the basis of mode strength extraction, accompanied by simulation results. Finally, in Section Experimental results, we demonstrate the technique by presenting and discussing promising experimental results of the retrieved mode strengths.

Concept

A conceptual illustration of the method to retrieve the strength of discrete modes at the output of a channel waveguide is presented in Fig. 1. We consider, as an example, a silicon nitride waveguide, especially designed to sustain several modes in the quasi-TE regime, and only the fundamental quasi-TM mode TM_{00}. The height of the waveguide is 330 nm, its width is 3 μm, and the vacuum wavelength of the light coupled into the waveguide is 520 nm. We inject the light into the waveguide using end butt-coupling method. Immediately after the output, a linear polarizer separates TE from TM contribution. Only the intensity from the superposition of the TE modes is considered. The individual modes are spatially completely coherent, mutually orthogonal, as well as mutually correlated. One gets mutually uncorrelated modes when the individual modes have different frequencies, which is not the case here.

The total light enters a wavefront folding interferometer (WFI), detailed in Refs. [25,26], which measures the CSD by analyzing the interference pattern. The WFI creates two identical replicas of the same incident beam like field first. Next, in the arm-1 (arm-2) the beam wavefront folds along x (y) axis. Finally, from the superposition of the two folded beam like fields and by scanning those along the transverse direction one gets the CSD. The main difference between traditional WFI and our customized WFI is the design. However, we guide the readers to see Refs. [26] for the detailed description and working principle of the WFI, that we have used here. After decoding the output of the WFI using an appropriate algorithm, the weight of each mode can be retrieved. The algorithm

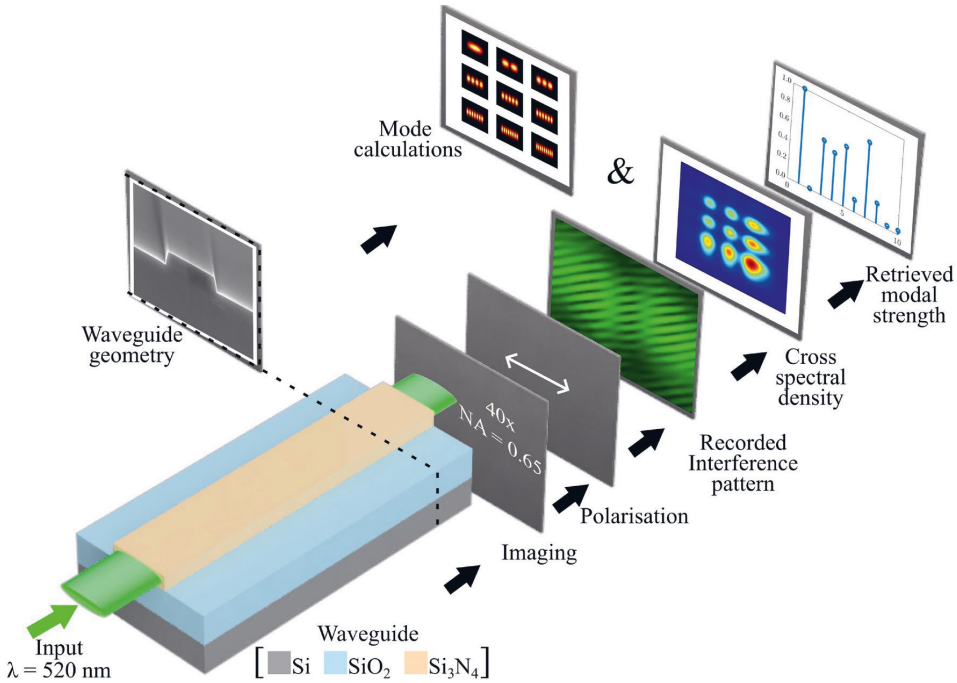

Fig. 1. Conceptual overview of the mode strength retrieval process. A channel waveguide is fed with a 520 nm wavelength light source. The emerging light intensity is captured using a microscope objective, after which the transverse electric (TE) polarization is isolated and the interference pattern is recorded by a wavefront folding interferometer (WFI) for detailed wavefront analysis. From this analysis, the cross-spectral density is determined. The possible modes that the waveguide can support are calculated based on its geometric attributes and refractive index profile. By correlating these theoretically calculated modes with the acquired cross-spectral density, the modal strength of each mode is retrieved.

uses the field distribution simulated for the studied waveguide, i.e., the field distribution of each mode is known a priori. The retrieval provides both the modal strengths as well as the inter-modal dispersion parameters.

Modal representations in waveguides

In the following, we consider the strong hypothesis of a slab waveguide and scalar fields to describe the methodology. The following arguments can be extended to other waveguide designs and vectorial light as well. This is allowed because a polarizer is used at the output of the channel waveguide and because the result concerns only the output plane and not what happens inside the channel. In a waveguide with two-dimensional cross-section, the high and inherent birefringence could completely modify the modal strength distribution at different locations along the propagation.

Due to the orthogonality of modes in a slab waveguide, the amplitude and phase of the guided field E_{out} at the exit (or image) plane of the waveguide may in general be represented as a superposition of the known modal fields $\phi_p(x)$, as in

$$E_{\mathrm{out}} = \sum_{p=0}^{P-1} \alpha_p \phi_p(x), \tag{1}$$

where we have left the frequency dependence implicit for brevity, P is the number of modes, and α_p are the generally complex valued modal weights with $\psi_p = \arg[\alpha_p]$. Since the waveguide modes are mutually orthogonal, they obey the relation

$$\int_{-\infty}^{\infty} \phi_p^*(x)\phi_q(x)\mathrm{d}x = \delta_{pq}, \tag{2}$$

where δ_{pq} is the Kronecker delta. Additionally, the modes $\phi_p(x)$ in slab waveguides are real-valued, although their superposition generally is not. Hence, by multiplying both sides of (1) by $\phi_q^*(x)$, integrating over x, and using (2), we attain the weights from

$$\alpha_p = \int_{-\infty}^{\infty} E_{\mathrm{out}}\phi_p(x)\mathrm{d}x. \tag{3}$$

This gives $\alpha_p \neq 0$ only for modes that propagate in the waveguide. There is also a continuum of radiation modes, but these are not of interest here. Note that (3) gives weights of the amplitudes of the modes, which are complex-valued numbers. The intensity weights

$$c_p = |\alpha_p|^2 \tag{4}$$

are of course real, and correspond to the weights given by the coherence-theoretic algorithm[12,13].

It is also possible to measure the weights from a collimated beam in the far-zone of the waveguide output. The k-space (far-field) expression of the modes is simply given by the Fourier transform – which we denote with a tilde – as in

$$\tilde{\phi}_p(k_x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \phi_p(x)\exp\left(-ik_x x\right)\mathrm{d}x. \tag{5}$$

Similarly, the k-space representation of the total field exiting the waveguide is

$$\tilde{E}_{\mathrm{out}}(k_x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} E_{\mathrm{out}}\exp\left(-ik_x x\right)\mathrm{d}x. \tag{6}$$

Since Fourier transform is a linear operation, we immediately see that

$$\tilde{E}_{\mathrm{out}}(k_x) = \sum_{p=0}^{P-1} \alpha_p \tilde{\phi}_p(k_x), \tag{7}$$

and the orthogonality of the modes is preserved, so that we may write with the use of (1)

$$\alpha_p = \int_{-\infty}^{\infty} \tilde{E}_{\mathrm{out}}(k_x)\tilde{\phi}_p(k_x)\mathrm{d}k_x. \tag{8}$$

Hence, the same weights are obtained by near- and far-field measurements, as one would expect. The present formulation can trivially be extended to 2D modes.

The technique of finding the unknown modal weights of the output field has a remarkable formal similarity to the overlap integral method for input coupling of coherent fields. This method allows one to calculate the weights α_p in end-coupling of a known fully coherent incident field $E_{\rm in}(x)$ into the waveguide, as in

$$\alpha_p = \int_{-\infty}^{\infty} E_{\rm in}(x)\phi_p(x)\mathrm{d}x. \tag{9}$$

The weights given by (9) are the projections of the incident field onto the waveguide-mode basis. These coefficients depend on polarization (TE or TM) of the incident field since the waveguide modes do so. Now the incident field is expanded in the modal basis of the waveguide as

$$E_{\rm in}(x) = \sum_{p=0}^{P-1} \alpha_p\phi_p(x) + E_{\rm r}(x), \tag{10}$$

where $E_r(x)$ contains the uncoupled radiation modes. That is, depending on the input field, a varying fraction of the energy in the incident field is lost in the coupling process. If we denote the coupled field as $E_{\rm c}(x)$, then we can write

$$E_{\rm c}(x) = E_{\rm in}(x) - E_{\rm r}(x). \tag{11}$$

The coupled field then propagates through the waveguide to produce $E_{\rm out}(x)$. However, there is no straightforward general relation between $E_{\rm c}(x)$ and $E_{\rm out}(x)$. Although the modes themselves propagate undisturbed by definition, they can lose or gain energy from other modes, depending on the waveguide geometry, refractive index variations, strain, or other imperfections.

The term overlap integral method arises because (9) can be interpreted as a measure of the overlap between the incident field and a given waveguide mode. If we have perfectly matched incident field and some mode *p*, such that $E_{\rm in}(x) = \phi_p(x)$, we have only one mode for which $\alpha_p = 1$ and all others vanish. Additionally, there is no uncoupled radiation and thus the efficiency of the coupling is unity. It is also obvious that the weights of the excited modes depend on, e.g., defocusing, tilting, or misaligning the incident field. For example, if the amplitude of the incident field matches that of mode *p* but the incident field has phase curvature due to defocusing, then $\alpha_p < 1$ and other propagating waveguide modes (as well as radiation modes) are excited as well.

It should be noted that the overlap integral method is not exact for end-coupling. It ignores reflection from the waveguide end (or end facets) and the evanescent part of the in-coupled field. These aspects have been studied in[27]. However, our analysis is completely based on the total output field, and can be generally applied.

Note that this formulation does not only cover end-butt coupling from free space (focused fields in the waveguide input) but also waveguide-to-waveguide (or fiber-to-fiber) coupling. One just needs to know the incident field at the input plane of the waveguide. However, we are interested in the relative mode strengths of the total output field. In other words, how much energy is in a particular mode relative to other modes. In order to find the answer, in practice, one needs to measure the total (complex) electric field distribution $E_{\rm out}$ at the exit plane. Usually, the phase measurement of total electric field (both the amplitude and phase) requires a reference beam, where the phase profile can be estimated from the interference pattern, e.g., the Twyman-Green interferometer[28]. These types of phase measurement schemes are not trivial while considering the total electric field at the output of a waveguide. Here we present an alternative scheme that utilises a mirror based WFI [25,26] to measure the complex valued two-dimensional cross-spectral density function, from which both the amplitude and phase profiles of the total field at the exit plane of a slab waveguide can be calculated. Notably, this method is self-referencing.

Complex field and modal strength retrieval

The cross spectral density (CSD) for a general scalar field $E(x)$ is defined as

$$W(x_1, x_2) = \langle E^*(x_1)E(x_2)\rangle, \tag{12}$$

where $E(x_i)$ denote the electric field at coordinate x_i, the angle brackets denote ensemble averaging, and asterisk denotes the complex conjugate. The CSD is widely used while dealing with partially coherent fields in the space-frequency domain, see e.g., Ref. [29]. Physically, $W(x_1, x_2)$ represents the correlation between the electric fields at coordinates x_1 and x_2, at a given frequency. However, for a completely coherent field, such as $E_{\rm out}(x)$, the CSD can be written in the factored form[30,31]

$$W(x_1, x_2) = E^*_{\rm out}(x_1)E_{\rm out}(x_2), \tag{13}$$

where the ensemble average vanishes. This is because all realizations of $E_{\rm out}(x)$ are identical, in which case averaging over all realizations is the same as having only one realization. In terms of the amplitudes $E_0(x)$ and phases $\varphi(x)$, this can be written as

$$W(x_1, x_2) = E_0(x_1)E_0(x_2)\exp\left\{-i\left[\varphi(x_2) - \varphi(x_1)\right]\right\}, \tag{14}$$

One can then retrieve the phase information of the total field E_{out} from the CSD, and the amplitude can be calculated from the measured intensity distribution or from the diagonal elements of $W(x_1, x_2)$. Once the complex field has been retrieved, the modal strengths can be retrieved using (3) and (4). To demonstrate the retrieval method numerically, we present two simulated examples in Figs. 2 and 3.

In addition, we note that in the case of two-dimensional (2D) field $E(x, y)$ the CSD is a four-dimensional (4D) function $W(x_1, y_1, x_2, y_2)$ and can be measured using WFI, see Ref. [26]. Therefore, our method can be generalized for such fields, that has both x and y dependency.

In both cases we start by defining the slab waveguide parameters, i.e., the thickness of the guiding layer is $h = 1.5$ μm and the refractive indices of the cladding, guiding layer, and substrate materials are $n_a = 1$, $n_g = 2.04$, and $n_s = 1.46$, respectively, while the wavelength of the illumination is 520 nm. We calculate the possible/allowed modes and their profiles by solving Maxwell's equations within the waveguide structure. For an asymmetric planar waveguide with the above parameters, the modes ϕ_p are of the form [32,33]

$$\phi_{y,p} = \begin{cases} A_p e^{(\rho_p x)}, & \text{for } x \leq 0, \\ A_p \cos(\kappa_p x) + B_p \sin(\kappa_p x), & \text{for } 0 \leq x \leq h, \\ [A_p \cos(\kappa_p h) + B_p \sin(\kappa_p h)]\, e^{-\sigma_p (x-h)}, & \text{for } x \geq h, \end{cases} \tag{15}$$

while the propagation constants of the modes can be found from the following transcendental equation

$$\tan(\kappa_p h) = \frac{\kappa_p(\sigma_p + \rho_p)}{\kappa_p^2 - \sigma_p \rho_p}, \tag{16}$$

where $\sigma_p^2 = \beta_p^2 - k_0^2 n_a^2$, $\kappa_p^2 = k_0^2 n_g^2 - \beta_p^2$, and $\rho_p^2 = \beta_p^2 - k_0^2 n_s^2$. Here, $\phi_{y,p}$ represents the electric field component of the p-th mode polarized along the y-axis, where A_p and B_p denote the amplitude coefficients of the mode. The variable x signifies the spatial coordinate perpendicular to the waveguide layers, with h

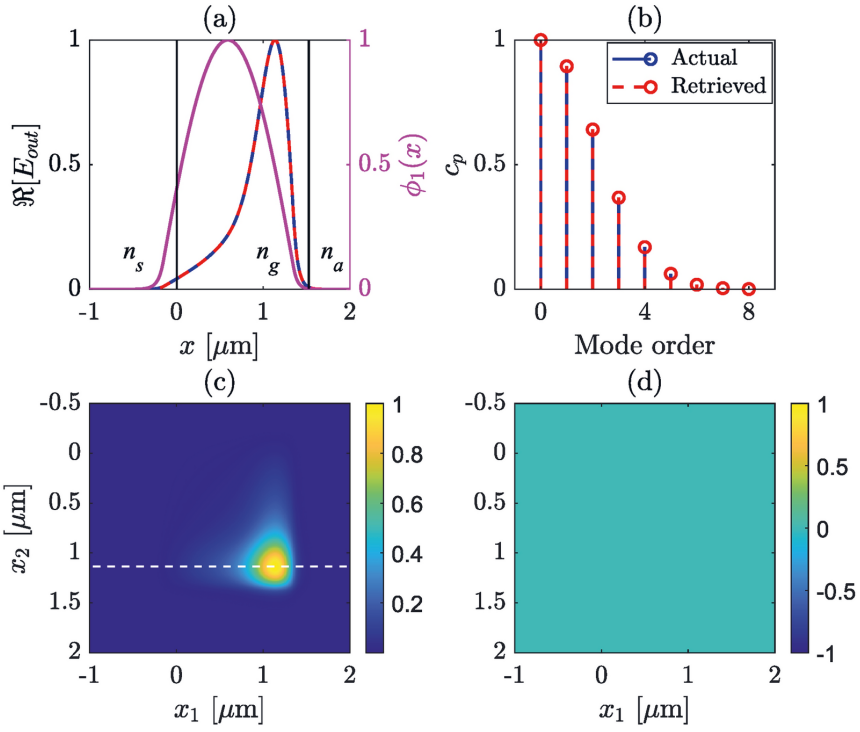

Fig. 2. Retrieved modal strengths and output field profile for an asymmetric slab waveguide, for which we consider only the TE modes. The lowest order mode has maximum strength. In (**a**), magenta line shows the fundamental mode, black lines delimit the waveguide boundaries, red and blue lines show real part of the actual and retrieved total field profiles, respectively. (**b**) actual (blue) and retrieved (red) mode strengths are compared, (**c**) real part of CSD, (**d**) phase of CSD. The white dash line in (**c**) indicate the selected line profile of CSD used for mode strength retrieval.

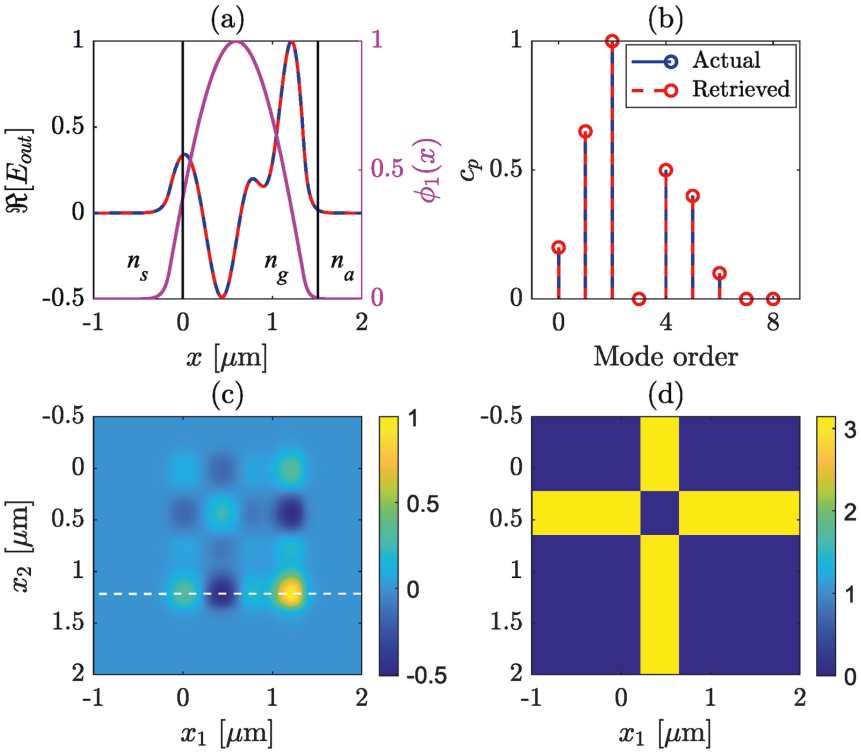

Fig. 3. Retrieved modal strengths and output field profile for an asymmetric slab waveguide, for which we consider only the TE modes. The third order mode has maximum strength. In (**a**), magenta line shows the fundamental mode, black lines mark the waveguide boundaries, red and blue lines show real part of the actual and retrieved total field profiles, respectively. (**b**) actual (blue) and retrieved (red) mode strengths are compared, (**c**) real part of CSD, (**d**) phase of CSD. The white dash line in (**c**) indicate the selected line profile of CSD used for mode strength retrieval.

denoting the guiding layer's thickness. The constants ρ_p and σ_p correspond to the decay rates in the substrate and cladding, respectively, while κ_p represents the transverse wavenumber in the guiding layer. Lastly, β_p is the mode's propagation constant and k_0 the free-space wave number.

Equation (15) represents how the electric field varies along the transverse plane perpendicular to the direction of propagation, chosen here as the z-axis. The solutions of Eqs. (15) and (16) yields the the electric field profiles as well as the propagation constants of the waveguide modes. Once the allowed modes are obtained, their properties such as field distribution, effective refractive index, and mode confinement can be analyzed. The confinement factor here refers to the proportion of optical power that is confined within the various regions of the waveguide.

To demonstrate the retrieval method we present two simulated examples in Figs. 2 and 3. The modal strengths, c_p, are chosen to follow a Gaussian distribution for Fig. 2, i.e., an ideal coupling in the waveguide to excite mainly the fundamental TE_0 mode with all α_p being real valued. The total field (actual) is then calculated following (1), and the real part is shown in panel (a), while the corresponding modal weights are depicted in panel (b) in blue solid line. We calculate the corresponding CSD using (13), the amplitudes and the phases of the CSD are presented in sub figures (c) and (d), respectively. For the Gaussian modal strength distribution the phase is flat, as can be seen from Fig. 2d. Note that, in this particular case, one can construct the total electric field only from the measured intensity across the exit plane of the slab waveguide, since the phase profile is constant. Next, we select a line profile as a function of x_1 from the CSD (at a fixed x_2) where $|W(x_1, x_2)|$ is maximum, indicated as the white dashed line in (c). This one dimensional (1D) cut of $W(x_1, x_2)$ provides the complex valued total electric field. Using (3), we then retrieve all the α_p values and the modal strengths, c_p. The retrieved total field and modal strengths are plotted along with the actual ones for comparison in (a) and (b), respectively, in red dashed lines. In all of the illustrations, the 0th order mode is the same as the fundamental mode.

We follow the same steps for simulating Fig. 3 except we choose random modal strengths. In practice, the random modal strength distribution may arise from several reasons, e.g., defects of the waveguide structure, tilt or miss-aligning the in-coupling field, etc. In this situation the CSD is complex valued and the phase plays a

crucial role. Moreover, since we choose all the α_p values real, the phase of CSD and the total field have a binary nature, which is evident from Eqs. (1) and (14) since all $\phi_p(x)$ values have a binary profile, see also Fig. 3d. In both illustrations the actual and retrieved modal strengths match and so does the electric field profile. This validates our algorithm for the modal strength retrieval from simulated field distributions. The complex valued $W(x_1, x_2)$ can be measured in practice using a WFI or a digital micro-mirror device (DMD) [34]. We note that in both cases we select a 1D cut (as marked by white dashed line in Figs. 2c and 3c) of CSD where $|W(x_1, x_2)|$ is maximum. This particular choice is important from the experimental point of view, which will become clear in Sect. 5, where we show the detailed experimental procedure and results.

Experimental results

In the following experiments, we demonstrate the feasibility of retrieving specific modes of a silicon nitride channel waveguide illustrated in Fig. 1. This waveguide, measuring 3000 nm in width and 330 nm in height, is fabricated on an oxidized silicon wafer. The wafers with LPCVD silicon nitride layer are specially customized and ordered from Silicon Materials (Germany). The waveguides are fabricated by electron-beam lithography of the nLOF resist (AZ2070, Microchemicals, Germany) and followed by reactive ion etching of the silicon nitride layer through the patterned resist. Effective index calculations for such waveguide using Finite Difference Eigenmode (FDE) solver in Ansys Optics indicate the existence of over 20 modes each for transverse electric (TE) and transverse magnetic (TM) polarizations at $\lambda = 520$ nm. However, it is important to note that not all these modes are efficiently guided through the waveguide due to potential losses. In the following, we consider only the TE polarization state selected with a linear polarizer at the output of the waveguide.

A schematic of the experimental setup for the mode measurement is shown in Fig. 4. We use a fiber coupled laser source (MCLS1 controller together with a custom laser diode from ThorLabs). We inject light from a finite bandwidth (1.2 nm) continuous wave laser having a central wavelength of 520 nm into the channel waveguide using a tapered lens fiber (focused spot size of 2 ± 0.5 μm). This butt-coupling involves aligning the input end facet of the waveguide with the optical fiber in such a way that light can be transferred with minimal loss, despite a modal mismatch between the fiber and the channel waveguide. The butt-coupling technique also allows us to laterally shift the fiber or tilt it with high precision, thus altering/adjusting the modal strength distribution, i.e., exciting preferentially the fundamental or the higher order modes. Once the in-coupling is done, the output plane of the channel waveguide is imaged (with sufficient magnification) onto the WFI camera. We use a microscope objective (Olympus Plan N, $\mathrm{NA} = 0.65, 40\times$) for the imaging purpose. We place a high quality linear polarizer (Glan-Thompson GTH10M polarizer from Thorlabs) to allow only the TE modes.

Next, a WFI is used to capture an interference pattern while keeping all the other components in the setup fixed. The WFI measures the complex valued CSD in average and difference coordinate: $(\bar{x}, \Delta x)$, which is then mapped to absolute coordinates (x_1, x_2). See, Refs. [25,26] for details on the measurement. Any 1D cut from the measured $W(x_1, x_2)$ for a fixed value of x_2 (or x_1) can be used for retrieving the modal strength distribution and the modal dispersion. However, we choose the 1D cut such that it passes through a point where $|W(x_1, x_2)|$ is maximum to maximize the signal-to-noise ratio. The field retrieved in this way represents the output field with at most a uniform pi phase shift ambiguity. However, this does not affect mode retrieval.

Our experimental setup, as illustrated in Fig. 4, reveals some limitations in the mode detection. Specifically, the number of observed and measured modes is constrained by the NA of the microscope objective used to image the waveguide's output plane onto the WFI. The finite numerical aperture of the microscope objective (and overall imaging system) limits the number of modes to be collected. This comes from the simple fact that the divergence of each mode increases with the mode order. Modes with propagation angles exceeding the microscope objective's convergence angle are truncated, making the capture of all propagating modes highly

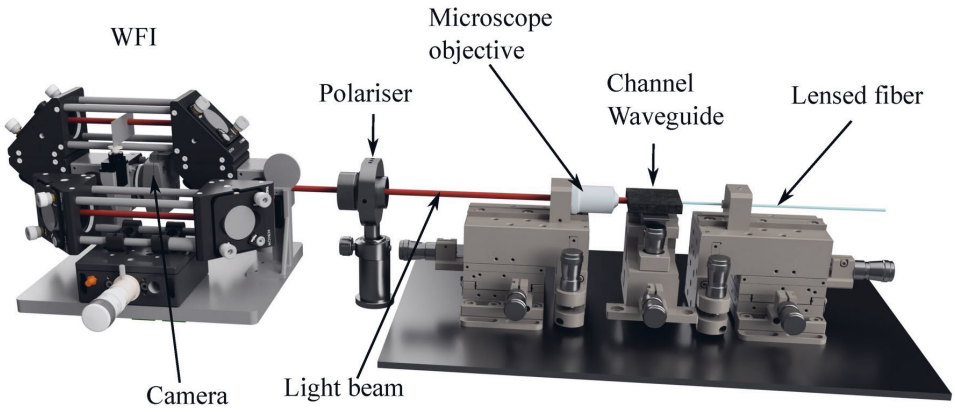

Fig. 4. Schematic of the experimental setup showing the WFI on the left, the polariser and the waveguide stage. The lensed fiber is connected to a fiber-coupled laser source from ThorLabs, which is not shown here.

challenging, if not impossible. More developed imaging system can be used in order to increase the collection angle. However, this is not required to validate our method.

An interesting aspect of our experiment is the ability to adjust the mode order with the highest weight simply by shifting the lensed fiber laterally or by altering the injection angle. This versatility is demonstrated in Figs. 5, 6, and 7, which represent different excitation conditions for TE modes. For each case, the mode strengths have been measured and quantified in the same Si_3N_4 waveguide as previously discussed. It is important to highlight that although more than 20 modes were used for calculating mode strengths, only the first 15 modes are presented, as the contributions of higher-order modes are either zero or negligible.

In Fig. 5a both the measured (in blue) and the retrieved (in red) electric field profile (real part) are presented. The retrieved complex valued electric field is constructed using the normalized strength of modes as shown in Fig. 5b, where the normalized modal weight (in blue circles) and associated phases (in red circles) of each of the first 15 quasi-TE modes is illustrated. The agreement between the measured and retrieved field profiles indicates the robustness of the algorithm. Figure 5c showcases the measured CSD, while Fig. 5d displays the retrieved CSD at the output plane of the channel waveguide utilizing the retrieved mode strengths.

A consistent display arrangement is maintained for various cases, as demonstrated in Figs. 6 and 7 for the same channel waveguide with different injection condition. A distinction can be made between Figs. 5c, 6c, and 7c, which is due to the variations in the modal strength of individual modes during measurements. The modal strength distribution is altered by shifting the lens fiber along the transverse plane slightly. In Fig. 5b, the fundamental mode exhibits the highest strength, followed by the first-order mode. Contributions from other modes (3rd–5th) are minimal, while others are approaching zero. Similarly, in Fig. 6, the fundamental mode dominates in strength, although the first-order mode vanishes this time. Several other higher-order modes have weights greater than, or equal to 0.5. The strength of higher-order modes in Fig. 6b is significant and adds more features in the total electric field, unlike the scenario in Fig. 5b.

In Fig. 7b, the fourth-order mode is dominating followed by the 5th order, whereas the fundamental, 2nd, and 3rd-order modes have similar strengths. Notably, the contribution of higher-order modes is substantial in the last two illustrations. In all the cases, we find good agreement between the retrieved and the measured electric

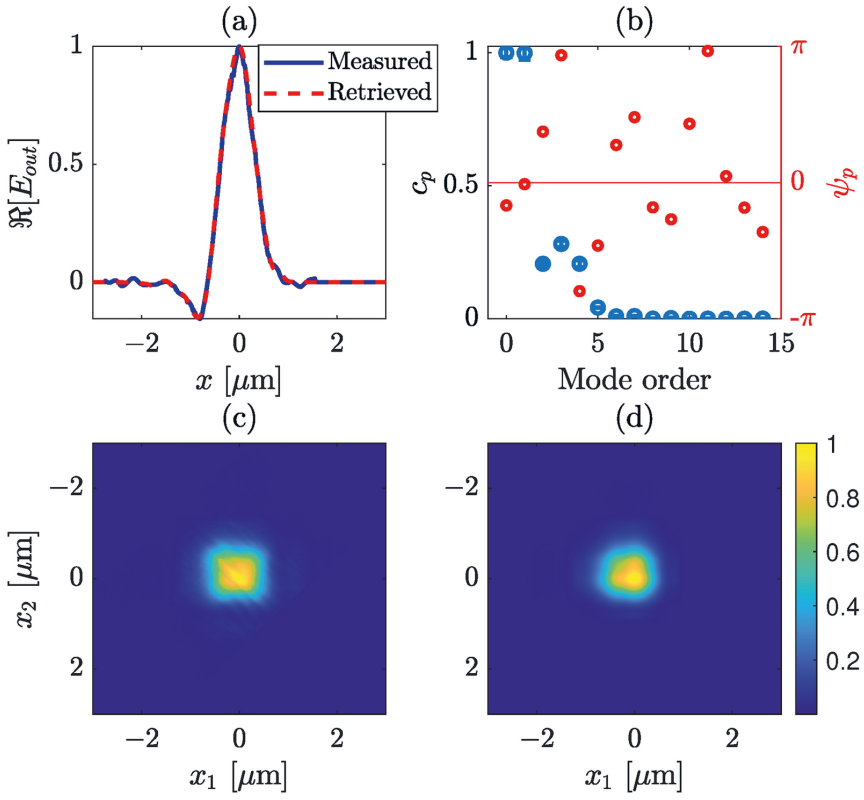

Fig. 5. Retrieved modal strengths and output field profile from a silicon nitride waveguide, where we consider only TE modes. (**a**) Measured (blue) and retrieved (red) intensity profiles of the total field at the output plane of the waveguide, (**b**) retrieved mode strengths (blue) and phases (red), (**c**) measured, and (**d**) retrieved absolute CSD. The first two modes have the highest contribution. The error bars correspond to one standard deviation.

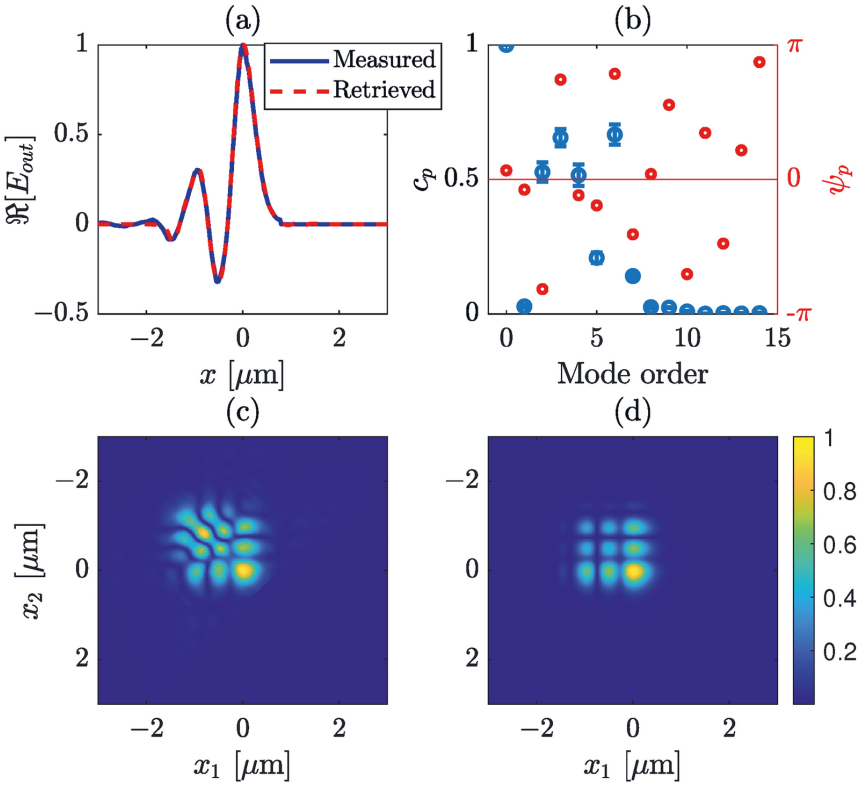

Fig. 6. Retrieved modal strengths and output field profile from a silicon nitride waveguide, where we consider only TE modes. (**a**) Measured (blue) and retrieved (red) intensity profiles of the total field at the output plane of the waveguide, (**b**) retrieved mode strengths (blue) and phases (red), (**c**) measured, and (**d**) retrieved absolute CSD. The zeroth mode has the largest contribution, while the 2nd, 3rd,4th, and 6th have weights higher than 0.5. The error bars correspond to one standard deviation.

field profiles, as illustrated in Figs. 5a, 6a, and 7a. However, a slight disagreement can be observed between the retrieved and measured CSDs. We note that although in the retrieval algorithm we have considered a strictly monochromatic field, in practice the laser we used has (Lorentzian shaped) finite bandwidth, a full width at half maximum of 1.2 nm. In addition, although in our analysis we have considered that a perfect image of the exit plane of the waveguide is captured by the WFI there might be a couple of micron error in the object plane. In practice, this error could arise from the depth of focus of the microscope objective or/and from the manual adjustment of mechanical stage that holds the objective. This mean that the measured CSD is not perfect, rather a slightly propagated CSD. This effect is illustrated in Fig. 8, where in (a) we show the CSD at the exit plane (at $z = 0$), and in (b) at $z = 2$ µm. A slight stretching in the propagated CSD can be observed, which we believe also describes the discrepancies between the retrieved and measured CSDs in Figs. 5, 6, and 7.

In all of the presented cases, we calculate the error bars associated with the modal strengths by numerically adding random number distribution - with the range of $\pm 12.5\%$ of peak amplitude value - with the amplitude bins of the measured electric field. Note, however, that our measured CSDs do not feature such large noise and the error bars computed in this way represent a worst case scenario.

The error bars represent one standard deviation from the mean, calculated from 20 independent realizations of numerically calculated electric field E_{out}, in each case. In practice, these may arise from low signal conditions. The error bars show the robustness of the presented scheme. Finally, from the retrieved phase distribution associated with each modal strength (since α_p values are complex), the dispersion between the different modes can be estimated. This dispersion is due to both the in-coupling condition and different propagation constants of different mode orders.

Conclusion

In summary, this paper introduces an innovative method for determining mode strengths in arbitrary asymmetric channel waveguides by measuring the complex electric field distribution at the exit plane using WFI. The proposed technique, demonstrated experimentally with a silicon nitride channel waveguide, provides

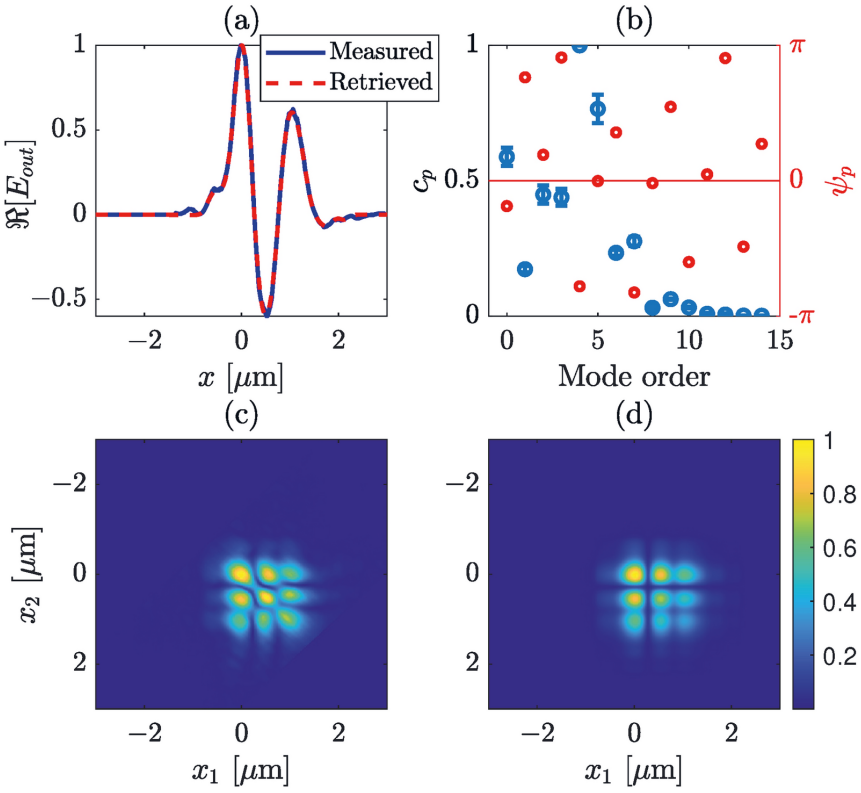

Fig. 7. Retrieved modal strengths and output field profile from a silicon nitride waveguide, where we consider only TE modes. (**a**) Measured (blue) and retrieved (red) intensity profiles of the total field at the output plane of the waveguide, (**b**) retrieved mode strengths (blue) and phases (red), (**c**) measured, and (**d**) retrieved absolute CSD. The 4th order mode has the largest contribution, while the fundamental, 2nd, 3rd and 5th order modes have weights around or over 0.5. The error bars correspond to one standard deviation.

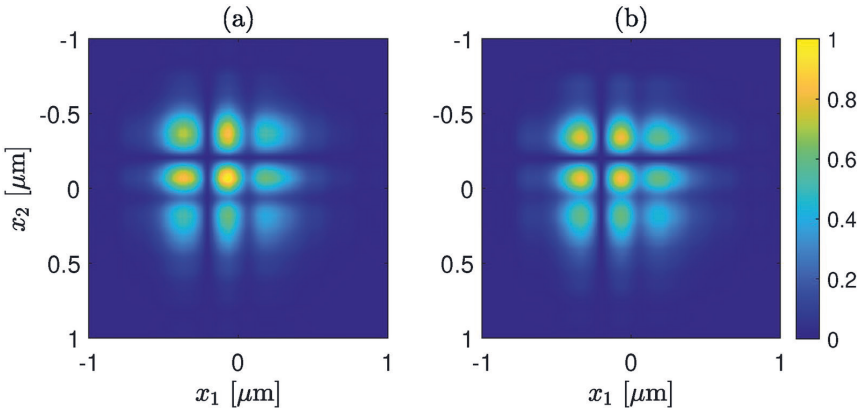

Fig. 8. Illustration of the effect of propagation CSD, (**a**) at $z = 0$, and (**b**) at $z = 2$ µm.

a robust approach for estimating modal strengths in spatially coherent light, where guided modes exhibit correlation, resulting in spatially completely coherent output. Our approach showcases adaptability, allowing for the adjustment of mode order with the highest intensity by manipulating the injection angle. However, acknowledging limitations, such as mode truncation due to microscope objective constraints, opens avenues for future experimental setups to comprehensively observe all propagating modes. Furthermore, these investigations can be extended to the incoherent case, broadening the applicability of the proposed method and providing deeper insights into waveguide characteristics. Our study contributes to the understanding and characterization of guided modes in channel waveguide design. The presented results and methodology offer valuable insights for researchers and engineers in the field of integrated optics and photonics, showcasing promise for optimizing device performance in applications such as optical communication and sensing.

Finally, we also emphasize that the mode strength retrieval based on measured intensity distribution is not unique. However, here we demonstrate an alternative approach where the retrieval is based on the measured complex valued CSD. Since we can get complete information on both the amplitude and phase profile of the electric field from the complex-valued CSD, our retrieval relies on the measured complex-valued electric field. Therefore, this method provides a unique solution. Moreover, the inter-modal dispersion properties of a waveguide can also be estimated via our measurement scheme. The presented error analysis also shows the robustness of our technique. Such a technique can easily be implemented in any integrated optics characterization system, even at the industrial level.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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References

1. Vetrone, F. et al. Temperature sensing using fluorescent nanothermometers. *ACS Nano* **4**, 3254–3258 (2010).
2. Doughan, I., Oyemakinwa, K., Ovaskainen, O. & Roussey, M. Strip-loaded Mach-Zehnder interferometer for absolute refractive index sensing. *Sci. Rep.* **14**, 3064 (2024).
3. Chen, G., Yang, C. & Prasad, P. N. Nanophotonics and nanochemistry: Controlling the excitation dynamics for frequency up-and down-conversion in lanthanide-doped nanoparticles. *Accounts Chem. Res.* **46**, 1474–1486 (2013).
4. Zeng, H., Li, R. & Li, W. Simultaneous frequency up/down converting interface based on a single hardware incorporating two phase-correlated photonic mixers. *Opt. Express* **30**, 9643–9654 (2022).
5. Shastri, B. J. et al. Photonics for artificial intelligence and neuromorphic computing. *Nat. Photon.* **15**, 102–114 (2021).
6. Oxborrow, M. & Sinclair, A. G. Single-photon sources. *Contemp. Phys.* **46**, 173–206 (2005).
7. Doughan, I., Oyemakinwa, K., Ovaskainen, O. & Roussey, M. Low loss vertical tio2/polymer hybrid nano-waveguides. *Nanomaterials* **13**, 469 (2023).
8. Lau, R. K. et al. Octave-spanning mid-infrared supercontinuum generation in silicon nanowaveguides. *Opt. Lett.* **39**, 4518–4521 (2014).
9. Karim, M. & Rahman, B. Ultra-broadband mid-infrared supercontinuum generation using chalcogenide rib waveguide. *Opt. Quant. Electron.* **48**, 1–10 (2016).
10. Svelto, O. *Principles of Lasers*. (Springer, 2009).
11. Wolf, E. & Agarwal, G. S. Coherence theory of laser resonator modes. *J. Opt. Soc. Am. A* **1**, 541–546 (1984).
12. Turunen, J., Tervonen, E. & Friberg, A. T. Coherence theoretic algorithm to determine the transverse-mode structure of lasers. *Opt. Lett.* **14**, 627–629 (1989).
13. Tervonen, E., Turunen, J. & Friberg, A. T. Transverse laser-mode structure determination from spatial coherence measurements: Experimental results. *Appl. Phys. B* **49**, 409–414 (1989).
14. Warnky, C. M., Anderson, B. L. & Klein, C. A. Determining spatial modes of lasers with spatial coherence measurements. *Appl. Opt.* **39**, 6109–6117 (2000).
15. Khonina, S. N., Skidanov, R. V., Kotlyar, V. V., Jefimovs, K. & Turunen, J. Phase diffractive filter to analyze an output step-index fiber beam. In Wyrowski, F. (ed.) *Wave-Optical Systems Engineering II*, vol. 5182, 251 – 259. (International Society for Optics and Photonics SPIE, 2003).
16. Flamm, D. et al. All-digital holographic tool for mode excitation and analysis in optical fibers. *J. Lightwave Technol.* **31**, 1023–1032 (2013).
17. Gervaziev, M. D. et al. Mode decomposition of multimode optical fiber beams by phase-only spatial light modulator. *Laser Phys. Lett.* **18**, 015101 (2020).
18. Nicholson, J. W., Yablon, A. D., Fini, J. M. & Mermelstein, M. D. Measuring the modal content of large-mode-area fibers. *IEEE J. Sel. Top. Quant. Electron.* **15**, 61–70 (2009).
19. Yan, L., Barankov, R., Steinvurzel, P. & Ramachandran, S. Modal-weight measurements with fiber gratings. *J. Lightwave Technol.* **33**, 2784–2790 (2015).
20. Bolshakov, M. V., Kundikova, N. D. & Vlazneva, M. A. Modal power decomposition of light propagating through multimode optical fiber. *Opt. Commun.* **365**, 1–6 (2016).
21. Liu, A. et al. Analyzing modal power in multi-mode waveguide via machine learning. *Opt. Express* **26**, 22100–22109 (2018).
22. Zhou, H. et al. Mode measurement of few-mode fibers by mode-frequency mapping. *Opt. Lett.* **43**, 1435–1438 (2018).
23. Yan, W., Xu, X. & Wang, J. Modal decomposition for few mode fibers using the fractional fourier system. *Opt. Express* **27**, 13871–13883 (2019).
24. Manuylovich, E. S., Dvoyrin, V. V. & Turitsyn, S. K. Fast mode decomposition in few-mode fibers. *Nat. Commun.* **11**, 5507 (2020).
25. Koivurova, M., Partanen, H., Lahyani, J., Cariou, N. & Turunen, J. Scanning wavefront folding interferometers. *Opt. Express* **27**, 7738–7750 (2019).
26. Halder, A. et al. Mirror-based scanning wavefront-folding interferometer for coherence measurements. *Opt. Lett.* **45**, 4260–4263 (2020).
27. Saastamoinen, T., Kuittinen, M., Vahimaa, P., Turunen, J. & Tervo, J. Focusing of partially coherent light into planar waveguides. *Opt. Express* **12**, 4511–4522 (2004).
28. Malacara, D. *Optical Shop Testing*. (Wiley, 2007).
29. Mandel, L. & Wolf, E. *Optical Coherence and Quantum Optics*. (Cambrigde University, 1995).

30. Lakki, M. A., Friberg, A. T. & Setälä, T. Complete coherence of random, nonstationary electromagnetic fields. *Opt. Lett.* **46**, 1756–1759 (2021).
31. Halder, A. Intensity-induced phase in linear optics. *J. Opt. Soc. Am. A* **40**, 1583–1586 (2023).
32. Partanen, H., Tervo, J. & Turunen, J. Coupling of spatially partially coherent beams into planar waveguides. *Opt. Express* **23**, 7879–7893 (2015).
33. Saleh, B. E. A. & Teich, M. C. *Fundamentals of Photonics*. (Wiley, 1991).
34. Partanen, H., Turunen, J. & Tervo, J. Coherence measurement with digital micromirror device. *Opt. Lett.* **39**, 1034–1037 (2014).

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Author contributions
I.D. and A.H. performed the numerical calculations and simulations. I.D. built the experimental setup. I.D. and A.H. run the experiment and did the data analysis. I.R. fabricated the waveguide. M.K., T.A., M.R., and J.T. contributed to the scientific discussions and manuscript preparation. M.R. and J.T. supervised the work, J.T. developed the theory.

Declarations

Competing interests
The authors declare no competing interests

Additional information
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ISAAC DOUGHAN

This thesis focuses on the development of polymer-titania hybrid waveguides for photonic sensing applications. It investigates waveguide mode behavior and presents the design, fabrication, and characterization of a low loss TiO2-polymer hybrid waveguide optimized for sensing. Further, the sensing capabilities of strip-loaded hybrid waveguides are demonstrated through the design, fabrication, and characterization of a sensitive Mach-Zehnder interferometer for refractive index sensing with improved performance.

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